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Título/Title Analysis of CFC shear webs with lightening holes																	
Palabras clave/Key words CARBON FIBER COMPOSITES, FINITE ELEMENT METHOD, BUCKLING, SHEAR STRENGTH, WEBS			Clasificación acceso Access class P - 2														
Resumen/Summary The objective of this Report is to keep all the information resulting from finite element models developed in the analysis of shear web panels, manufactured from CFC plies, with single centred lightening hole, and submitted to pure shear load that were derived from EDAVCOS program. Additional finite element models, based on EDAVCO's ones, have been developed in order to cover a wider range of parameters and configurations. As a result of this Report a design criteria will be derived for this kind of structural elements. Related FEM models were developed for two different materials and five lay-ups as follows (quasi-orthotropic behaviour of the material is assumed): <table border="0" style="width: 100%;"> <tr> <td style="width: 30%;"><u>Material:</u></td> <td><u>Laminate:</u> (%₀, %_{±45}, %₉₀)</td> </tr> <tr> <td>AS4/8552</td> <td>(30/60/10)_s</td> </tr> <tr> <td>AS4/8552</td> <td>(10/80/10)_s</td> </tr> <tr> <td>AS4/8552</td> <td>(0/40/60)_s</td> </tr> <tr> <td>AS4/8552</td> <td>(0/80/20)_s</td> </tr> <tr> <td><u>Material:</u></td> <td><u>Laminate:</u> (%_{0/90}, %_{±45})</td> </tr> <tr> <td>AGP193-PW/8552</td> <td>(40/60)_s</td> </tr> </table> Analized shear webs are square or rectangular, with a centred lightening hole. Some different hole diameters and kinds of reinforcement methods around the hole were considered. Requested output data for all FEM models were related to Linear Strength Analysis and Linear Buckling Analysis, respectively MSC/NASTRAN solutions Nos. 101 and 105. PROPIEDAD DE CONSTRUCCIONES AERONAUTICAS, S.A. Este documento no puede ser utilizado ni reproducido total o parcialmente sin la previa autorización escrita de la Dirección de Proyectos y Sistemas de C.A.S.A. CONSTRUCCIONES AERONAUTICAS, S.A. PROPERTY. This document shall neither be used nor completely or partially reproduced without previous written authorization by C.A.S.A. Engineering and System Direction.				<u>Material:</u>	<u>Laminate:</u> (% ₀ , % _{±45} , % ₉₀)	AS4/8552	(30/60/10) _s	AS4/8552	(10/80/10) _s	AS4/8552	(0/40/60) _s	AS4/8552	(0/80/20) _s	<u>Material:</u>	<u>Laminate:</u> (% _{0/90} , % _{±45})	AGP193-PW/8552	(40/60) _s
<u>Material:</u>	<u>Laminate:</u> (% ₀ , % _{±45} , % ₉₀)																
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AGP193-PW/8552	(40/60) _s																
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	G. López Martínez																
	Comprobado Ckecked	J. C. Gómez López															
	Aprobado Approved	M. Huertas García															

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1 Introduction

1.1 Objectives

Objective of this Report is the development of an analysis method, as simple as possible, for CFC shear webs having centred lightening holes. This kind of structural element is very common in aeronautical structures, (v. g.: ribs and spars).

Analysed elements are empty holes, (without any kind of cover or filling), machined-out from rectangular or square shear webs. Capabilities and limitations of the selected lightening holes and basic webs used are summarised below:

- ❑ Lightening hole must have circular or hippodromic shape,
- ❑ Only pure shear flow over the holed panel has been considered,
- ❑ Shear webs are simply supported in their entire external perimeter, and the hole edge is free,
- ❑ Only solid laminates in Carbon Fiber Tape or Fabric have been considered,
- ❑ Buckling results shown on this Report have been obtained employing "B" values for material properties, average between tension and compression ones. Environmental conditions considered in all cases have been [H/W 2], according to AIRBUS standards.
- ❑ Maximum deformation results shown on this Report have been calculated employing "Mean" values, average from tension and compression ones. As for buckling analysis, environmental conditions considered have been [H/W 2].

All the finite element models were stored in Ref. 16, together with this Report back-up.

Somme additional FEM models were developed exclusively for this Report, but they are obtained on the whole of cases by addition/deletion of part of elements from the EDAVCOS FEM models or by substitution of the plate property entries on the models Bulk Data. (i.e.: in order to change laminate properties and/or ply material properties). All new FEM models were made with same mesh criteria, loading and support conditions than the original ones developed for EDAVCOS Tasks.

Additional pursued objectives for these kind of structural elements are,

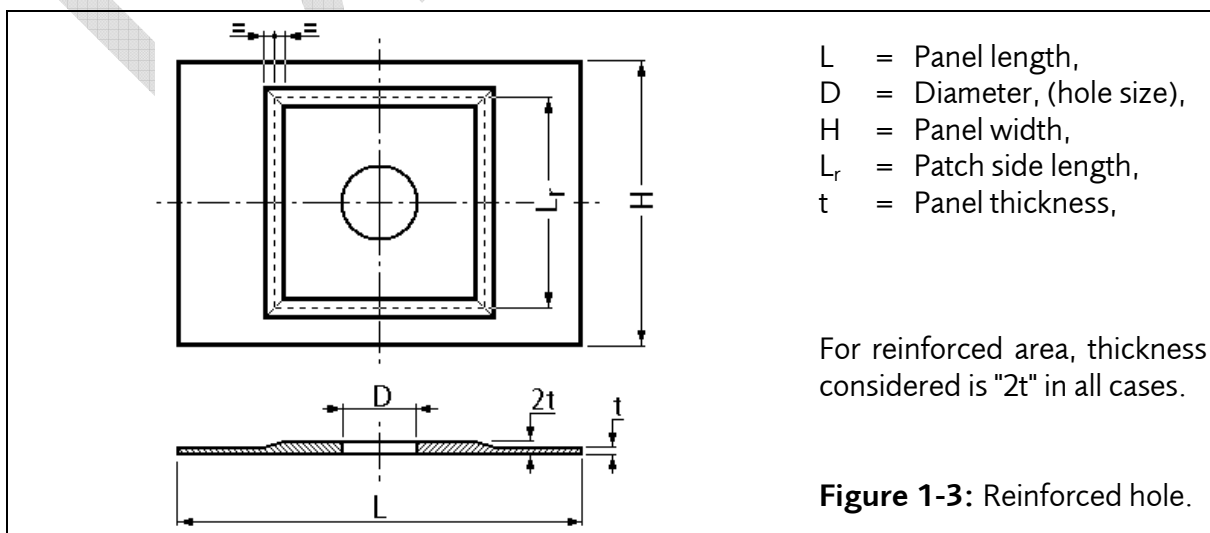
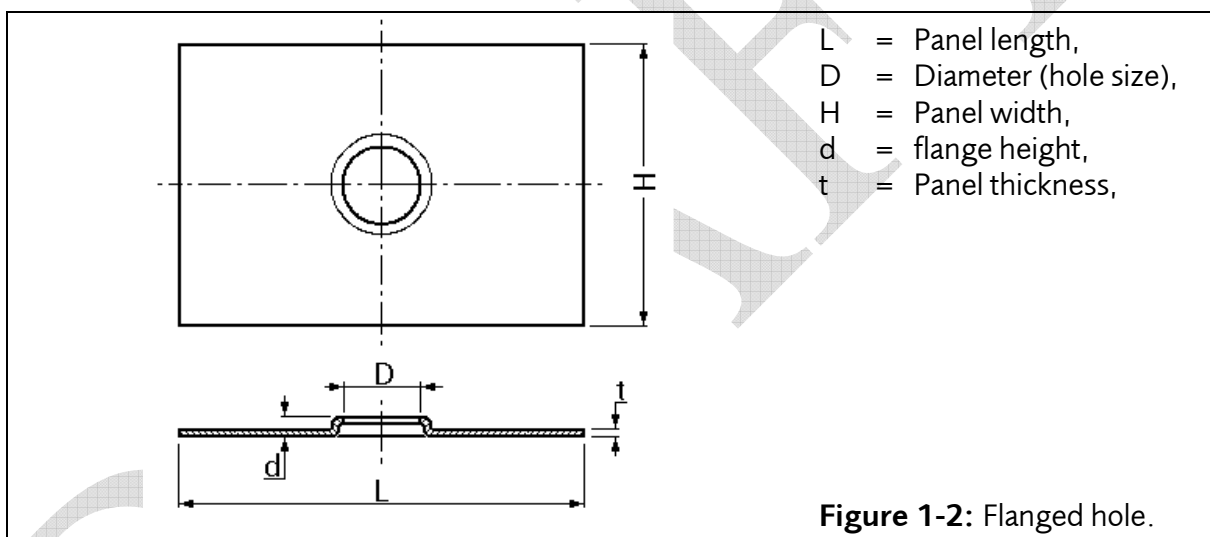
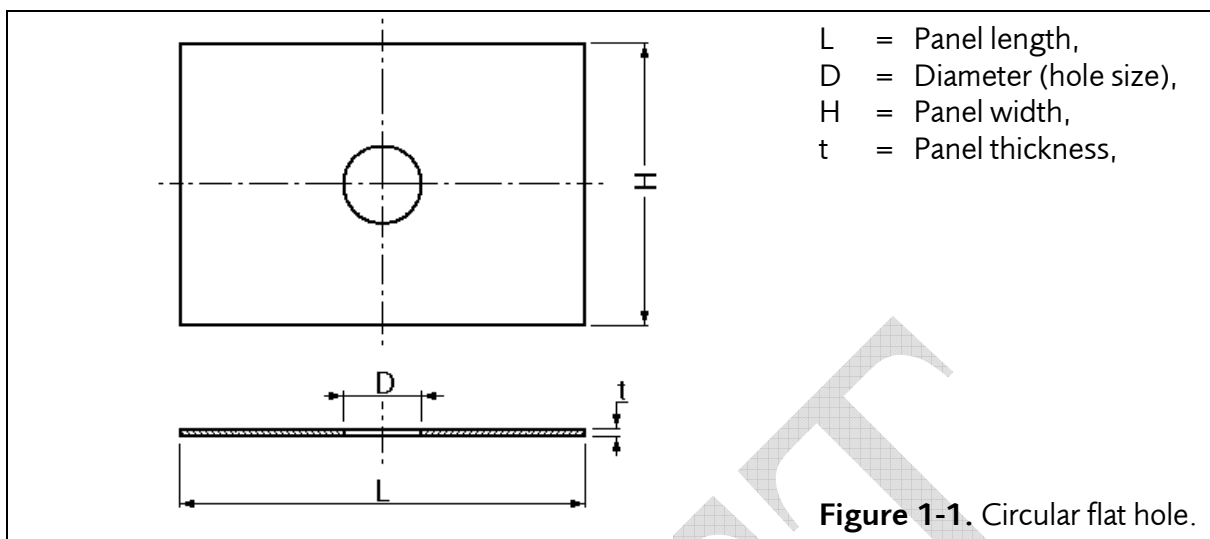
- ❑ Show how to perform a new FEM model having different geometry and/or material, than the ones shown herein,
- ❑ Organise all the information obtained from FEM models analysis in order to develop some design and strength justification guidelines,
- ❑ Obtain Tables and/or Charts showing the web overall shear allowable loads as a function of the selected design parameters.

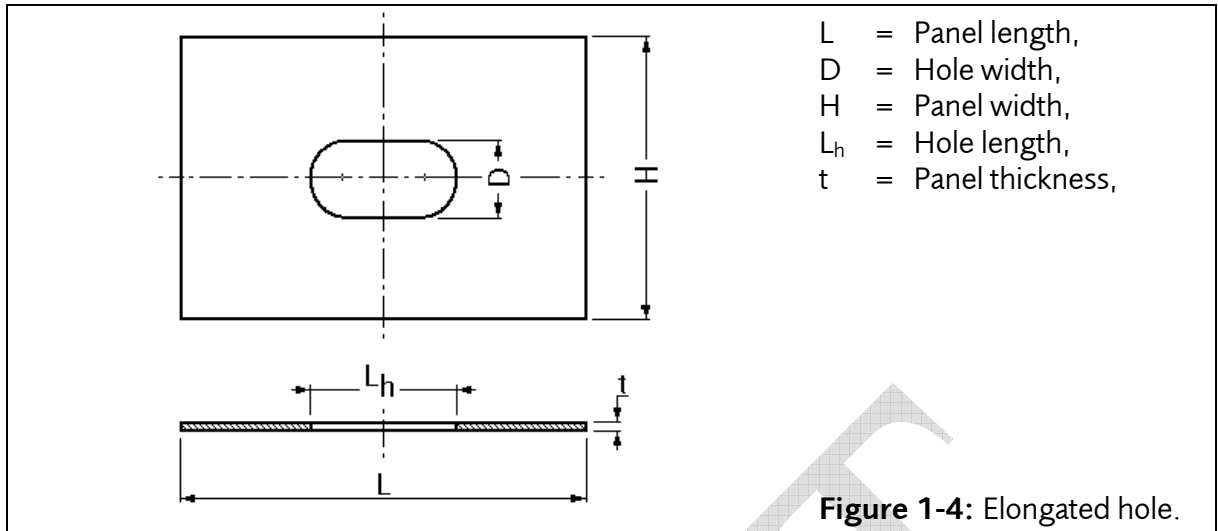
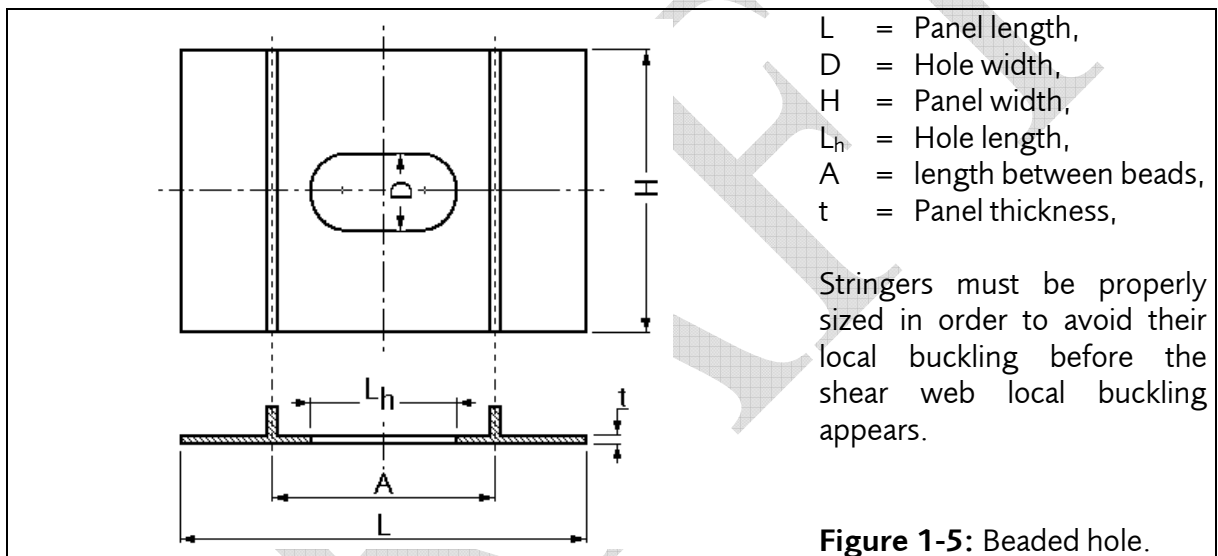
No Drawings were made in order to show the modelled structural details except those included in Ref. 1, (EDAVCOS Program). However, geometries used for FEM models will be sketched hereafter.

The analysed shear webs were grouped by a geometrical criterion. The cut-out shape and the local reinforcement method selected compose the classification criteria:

1. - Circular flat holes: shear webs having a centred circular hole without any kind of reinforcement around it. Several sizes of holes were modelled. (See Figure 1-1).
2. - Flanged holes: the hole edge is reinforced by a flange. The size of the flange is a variable parameter. Thickness is kept uniform through the whole panel, including the flange. (See Figure 1-2).
3. - Reinforced holes: shear web area around the hole is spliced by a square shaped CFC solid laminate patch, having the same lay-up than the basic shear web. Some different patch sizes have been used. (See Figure 1-3).
4. - Elongated holes: flat shear webs having a centred elongated hole, (also called hippodromic holes). Some elongation ratios and hole sizes have been considered. These kinds of FEM models were developed in order to analyse the effect of the hole shape. (See Figure 1-4).
5. - Beaded holes: flat shear webs having an elongated centred hole reinforced by two beads (or stringers), near the hole. Full support from the beads to the panel is considered. (See Figure 1-5).

In all these groups, some different laminates, thicknesses and hole sizes were considered. Figures 1-1 to 1-5 shows the general arrangement considered for lightening shear webs, and the geometrical variables considered for the analysis.




Figure 1-4: Elongated hole.

Figure 1-5: Beaded hole.

In all FEM models, the thickness "t" depends on the plies material and their stacking sequence.

Results obtained from FEM models are compared against results from the flat shear web without the hole, in order to perform a parametric analysis. Data obtained from the analysis will be organised in Tables and Charts as follows,

- Lightening shear web buckling flow, as a function of the buckling shear flow for the flat full shear web and the relative size of the lightening hole. It gives the structural element failure load level due to buckling, i.e.

$$q_{BK}/q_0 = f(D/H),$$
- Maximum microstrain level reached by the lightening shear web as a function of the average micro-strains level reached by the full shear web and the relative size of the hole. This kind of charts give the structural element failure load level due to material strength,

$$|\epsilon_{MAX}|/\epsilon_{AVG} = f(D/H),$$

Results obtained for flat shear web without lightening hole –used in comparisons listed above– were obtained from References 17 and 18, which are under AIRBUS standards for CFC solid laminate structural elements. All these data were also stored in Reference 12 and will be summarized later on this Report.

1.2 References

- Ref. 1 CASA Document No.: SP-T-ADP-99001, Issue B, Oct/99, "Test specification of flat panels with centred access hole loaded in shear",
- Ref. 2 E.S.D.U. Standards. Items Nos.: 71005, 75034, 75035.
- Ref. 3 EDAVCOS Deliverable Document D1.2-4, Appendix M, Aug/00, "Predict buckling strength of shear webs with access hole", [Report No. CASA.T1.TR.8],
- Ref. 4 EDAVCOS Deliverable Document D2.2-1, Feb/00, "Predict buckling strength of shear webs with access hole", [Report No. CASA.T2.TR.4],
- Ref. 5 EDAVCOS Deliverable Document D1.2-2, Aug/00, "Description of local analysis methods for flat panels with access holes", [Report No. CASA.T1.TR.6],
- Ref. 6 EDAVCOS Deliverable Document D2.2-2, Sep/00, "Final numerical analysis: Buckling strength of shear webs with access hole", [Report No. CASA.T2.TR.6],
- Ref. 7 EDAVCOS Deliverable Document D2.3-1, Jan/01, "Analysis-Test correlation: Failure of shear webs with access hole", [Report No. CASA.T2.TR.10],
- Ref. 8 MSC/NASTRAN Version 70.5 User's Manuals,
- Ref. 9 MSC/PATRAN Versions 8.5 and 9.0 User's Manuals,
- Ref. 10 CASA Structural Design Manual, Vol. XI,
- Ref. 11 DERA contribution to EDAVCOS program, Task 3.4, "Failure analysis of shear web specimens",
- Ref. 12 CASA Technical Document No. 35D-980-80161: "Spar web tests analysis of results", Jul. 89.
- Ref. 13 NT-AO-ECE-063 B: "Criterio de fallo de larguero de timón", Sept. 93
- Ref. 14 NT-F-ADS-00510: "Criterio de fallo de los largueros del cajón del estabilizador horizontal del avión A340-600", Mar. 01.
- Ref. 15 NT-F-ADS-00508: "Criterio de fallo del larguero del timón del avión A340-600", Nov. 00.
- Ref. 16 OAAA-NT-T-ADP-01006, EADS-AIRBUS backup,
- Ref. 17 ANCOLAB Program, Version 3.0
- Ref. 18 PANPA Program, Version 4.0
- Ref. 19 FEA Program, Version 1.0
- Ref. 20 EH Program, Version 2.1

2 Summary of analysed laminates

This section shows mechanical properties per ply and the full lay-up properties used in the analysis of shear web, with and without the lightening holes. In both cases, the laminate properties were obtained from ply properties and classical laminates theory. (Ref. 17).

For lightening shear web FEM models, NASTRAN entry cards used in order to simulate material properties have been PSHELL cards combined with MAT2 cards, including both membrane and bending properties.

Plies and laminates analysed on this Report are shown below. Environmental conditions considered have been [H/W 2].

Ply material Name:	Ply thickness, (mm/ply)	Stacking sequence: [% of plies 0°, ±45°, 90°]	Laminate, (basic) thickness, (mm)
Z-19.775 AS4/8552	0.184	(+/-/0/90/-/+0/+/-/0) _s [30/60/10]	3.68 20 plies
Z-19.775 AS4/8552	0.184	(+/-/0/-/+/-/90/-/+) _s [10/80/10]	3.68 20 plies
Z-19.775 AS4/8552	0.184	Orthotropic laminate [0/40/60]	3.68 20 plies
Z-19.775 AS4/8552	0.184	Orthotropic laminate [0/80/20]	3.68 20 plies
Z-19.760 AGP193-PW/8552	0.205	(±, 0/90, ±, 0/90, ±) _s [40/60]	2.05 10 plies

Table 2-1: CFC Materials and lay-ups used.

Properties per ply were taken from Ref. 10. Note that all shown values are average values between tension and compression ones, -i.e. $E_{11} = (E_{1T} + E_{1C})/2$, where E_{11} is the Young's modulus used in the analysis and E_{1T} , E_{1C} are tension/compression values respectively-.

Material = CFC Tape, AS4/8552 (CASA Group V)			Conditioning: [H/W 2]	
Thickness = 0.184 (mm/ply)				
Properties	E ₁₁ (Mpa)	E ₂₂ (Mpa)	G ₁₂ (Mpa)	μ ₁₂
B values	114500.	7150.	3300.	0.30
Average values	131000.	8850.	3950.	0.30

Material = CFC Fabric, AGP193-PW/8552 (CASA Group V)			Conditioning: [H/W 2]	
Thickness = 0.205 (mm/ply)				
Properties	E ₁₁ (Mpa)	E ₂₂ (Mpa)	G ₁₂ (Mpa)	μ ₁₂
B values	55000	55000.	2900.	0.05
Average values	58000.	58000.	3400.	0.05

Table 2-2: CFC plies used. Mechanical properties per ply.

Laminate average properties and MAT2 NASTRAN entry cards on Tables 2-4 and 2-5 were obtained from Ref. 17, (ANCOLAB program), the basic laminate thickness, the stacking sequences shown in Table 2-1, and the properties per ply as shown in Table 2-2.

Reference 16 shows the files used in the analysis of laminate properties. These files are in directory called "material" of this Reference, and they are named as Table 2-3 shows.

Ply, Laminate, Thickness	Env. Cond.:	Ply Prop.	Analysis File Names:
[30/60/10] 3.68 mm	H/W 2	B Avg.	material/30_60_10/hw2/valorB/anco306010.res material/30_60_10/hw2/average/anco306010.res
[10/80/10] 3.68 mm	H/W 2	B Avg.	material/10_80_10/hw2/valorB/anco108010.res material/10_80_10/hw2/average/anco108010.res
[0/40/60] 3.68 mm	H/W 2	B Avg.	material/0_40_60/hw2/valorB/anco04060.res material/0_40_60/hw2/average/anco04060.res
[0/80/20] 3.68 mm	H/W 2	B Avg.	material/0_80_20/hw2/valorB/anco08020.res material/0_80_20/hw2/average/anco08020.res
[40/60] 2.05 mm	H/W 2	B Avg.	material/40_60/hw2/valorB/anco4060.res material/40_60/hw2/average/anco4060.res

Table 2-3: Laminate Properties Analysis Files.

Ply: Z-19.775			Laminate Average Properties				
Laminate, Thickness	Env. Cond.:	Ply Prop.	E_{xx}	E_{yy}	G_{xy}	μ_{xy}	μ_{yx}
			N/mm ²			—	
[30/60/10] 3.68 mm	H/W 2	B	47013.816	28964.840	19023.490	0.5160	0.3179
		Avg.	54127.156	33566.871	21884.449	0.5135	0.3184
[10/80/10] 3.68 mm		B	26906.543	26906.543	24264.654	0.5751	0.5751
		Avg.	31222.688	31222.686	27862.598	0.5724	0.5724
[0/40/60] 3.68 mm		B	16376.354	74343.375	13782.326	0.1521	0.6907
		Avg.	19281.301	85326.359	15906.303	0.1535	0.6795
[0/80/20] 3.68 mm		B	18925.709	32820.754	24264.656	0.4534	0.7862
		Avg.	22117.355	38036.105	27862.598	0.4526	0.7784

Ply: Z-19.760			Laminate Average Properties				
Laminate, Thickness	Env. Cond.:	Ply Prop.	E_{xx}	E_{yy}	G_{xy}	μ_{xy}	μ_{yx}
			N/mm ²			—	
[40/60] 2.05 mm	H/W 2	B	34363.066	34363.066	16874.281	0.4065	0.4065
		Avg.	36641.195	36641.195	17931.426	0.3998	0.3998

Table 2-4: Summary of Analysed Laminates Average Properties.

For the analysis of other laminate thicknesses different from the ones shown herein, same stacking sequence used for the basic laminates has been considered, but repeating each ply at each position so as to complete the desired laminate thickness.

This procedures makes that only PSHELL NASTRAN cards needs to be changed, —the fields referred to the laminate thickness—, because the MAT2 entries depend only on the laminate

stiffness matrix, $[A_{ij}]/t$ and $12[D_{ij}]/t^3$, which need no change if the laminate is constructed as it was explained before.

Ply, Laminate, Thickness	Env. Cond.:	Ply Prop.	MAT2 No.:		Composite Laminate NASTRAN MAT2 card entries:					
					Non-blank NASTRAN Field ID-No.					
					Terms of material Prop. Matrix:					
					3 G11	4 G12	5 G13	6 G22	7 G23	8 G33
Tape, z-19.775 [30/60/10] 3.68 mm	H/W 2	B	1000	M	56240.9	17880.6	0.0	34649.6	0.0	19023.5
			1100	B	48034.0	19610.2	971.6	39397.4	971.6	20753.1
		Avg.	1000	M	64708.0	20605.7	0.0	40128.6	0.0	21884.4
			1100	B	55361.4	22578.5	1106.1	45529.6	1106.1	23857.2
Tape, z-19.775 [10/80/10] 3.68 mm	H/W 2	B	1000	M	40204.1	23121.8	0.0	40204.1	0.0	24264.7
			1100	B	47986.4	23436.2	971.6	31792.9	971.6	24579.1
		Avg.	1000	M	46440.1	26583.8	0.0	46440.1	0.0	27862.6
			1100	B	55298.8	26942.5	1106.1	36864.2	1106.1	28221.3
Tape, z-19.775 [0/40/60] 3.68 mm	H/W 2	B	1000	M	18299.4	12639.5	0.0	83073.5	0.0	13782.3
			1100	B	18299.4	12639.5	0.0	83073.5	0.0	13782.3
		Avg.	1000	M	21527.3	14627.5	0.0	95265.6	0.0	15906.3
			1100	B	21527.3	14627.5	0.0	95265.6	0.0	15906.3
Tape, z-19.775 [0/80/20] 3.68 mm	H/W 2	B	1000	M	29408.4	23121.8	0.0	50999.8	0.0	24264.7
			1100	B	29408.4	23121.8	0.0	50999.8	0.0	24264.7
		Avg.	1000	M	34150.4	26583.8	0.0	58729.9	0.0	27862.6
			1100	B	34150.4	26583.8	0.0	58729.9	0.0	27862.6
Fabric, z-19.760 [40/60] 2.05 mm	H/W 2	B	1000	M	41163.6	16731.2	0.0	41163.6	0.0	16874.3
			1100	B	40045.6	17849.1	0.0	40045.6	0.0	17992.2
		Avg.	1000	M	43613.9	17438.7	0.0	43613.9	0.0	17931.4
			1100	B	42451.4	18601.2	0.0	42451.4	0.0	19093.9

Table 2-5: Summary of Analysed Laminates MAT2 NASTRAN entry cards.

In FEM models of lightening shear webs used on this Report, the buckling analysis is performed by considering "B" values for laminate properties, (elastic modulus), whereas the strength analysis is made with "Average" laminate properties. Table 2-5 shows laminate properties for both values.

Analyses of shear webs without lightening hole have been carried out with Ref. 18, (PANPA program). This analysis method calculates the buckling shear flow for a flat web, being the data entry the geometry and the terms of the laminate stiffness matrix, as follows,

...					
$A_{11},$	$A_{12},$	$A_{13},$	$A_{22},$	$A_{23},$	A_{33}
$B_{11},$	$B_{12},$	$B_{13},$	$B_{22},$	$B_{23},$	B_{33}
$D_{11},$	$D_{12},$	$D_{13},$	$D_{22},$	$D_{23},$	D_{33}
...					

Table 2-6: (Ref. 18) Format of Laminate Properties Entry.

Relationships between NASTRAN fields shown in Table 2-5 and terms of laminate stiffness matrix as shown in Table 2-6 are as follows,

$$A_{ij} = [G_{ij}]_M \times t, \quad \text{eq. [2.1-1]}$$

$$D_{ij} = ([G_{ij}]_B \times t^3)/12, \quad \text{eq. [2.1-3]}$$

in this case, and due to the symmetry of all composite laminates used on this Report,

$$B_{ij} = 0.0 \quad \text{eq. [2.1-2]}$$

In previous equations,

t is the composite laminate thickness,

$[G_{ij}]_M$ are the terms of the stiffness matrix corresponding to the laminate in-plane behaviour, (MAT2 1000 terms in Table 2-5), and

$[G_{ij}]_B$ are the terms of the stiffness matrix corresponding to the laminate bending behaviour, (MAT2 1100 terms in Table 2-5).

Equations 2.1-1 thru 2.1-3 allow the analysis of any shear web thickness using the laminate properties summarised in Tables 2-4 and 2-5. If orthotropic behaviour is assumed $[G_{ij}]_M$ and $[G_{ij}]_B$ are the same matrix.

3 Lightning shear web FEM models

Even though the lightning shear web FEM models have been described in Refs. 4 and 5, a small review of this description will be included here in order to clarify the results obtained from them.

All FEM models used on this Report were pre- and post-processed with MSC/PATRAN program, (Version 9.0), that was also the tool used in order to produce the FEM model images included on this Report. Solver utility was MSC/NASTRAN, version 70.5.

Both programs are run on a SUN UltraSPARC/336 machine, and the installed operative system is SunOS 8.0, (Unix).

All FEM models shown herein were stored in **Reference 16**. Files used in calculation of lightning shear webs, (FEM model files), were,

- Bulk data file,

Model_file.bdf ,
- Output files produced by MSC/NASTRAN and MSC/PATRAN,

Model_file./log/F06/F04/db .

Table 3-1 lists all lightning shear web FEM models produced from geometry point of view. Calculations of each FEM model have been carried out for the five different materials summarised in Tables 2-4 and 2-5.

Two NASTRAN runs per geometry and material were made. The former was a linear buckling solution, which was made with "B" laminate values. The second one was a linear static solution, for a shear applied flow of 100.0 N/mm, which was made in order to compare maximum strains levels reached between different FEM models. This run was carried out considering average laminate properties.

All the analyses considered two load cases, which corresponds to positive and negative applied shear flow, respectively.

Geometrical parameters on Table 3-1 are in accordance with sketches in Figures 1-1 through 1-5.

Shear web FEM models designation criteria is tabulated below. Practical application of designation criteria is shown on last column of Table 3-1, (FEM model file name).

Type of Hole	L (mm)	H (mm)	D (mm)	d (mm)	Lr (mm)	Lh (mm)	A (mm)
Plain [c]	133. [l00]	133 [h0]	100. [d2]	5. [s1]	150. [r1]	150. [p2]	200. [a2]
Flanged [f]	200. [l1]	200. [h1]		10. [s2]	200. [r2]	200. [p3]	400. [a3]
Reinforced [r]	266. [l01]	400. [h4]		20. [s3]	400. [r3]		1000. [a4]
Elongated [e]	400. [l3]	1000. [h5]					
Beaded [b]	800. [l5]						
	1000. [l6]						
	2000. [l7]						

Type of hole:	L (mm)	H (mm)	D (mm)	d (mm)	L _r (mm)	L _h (mm)	A (mm)	FEM model file name:
Plain circular (See Figure 1-1)	133.	133.	100.	—	—	—	—	CL00H0D2
	266.							CL01H0D2
	200.	200.						CL0H1D2
	400.							CL3H1D2
	400.	400.						CL3H4D2
	800.							CL5H4D2
	1000.	1000.						CL6H5D2
	2000.							CL7H5D2
	Flanged circular (See Figure 1-2)	200.						200.
10.			FL0H1D2S2					
20.			FL0H1D2S3					
5.			FL3H1D2S1					
10.			FL3H1D2S2					
20.			FL3H1D2S3					
400.		400.	5.	FL3H4D2S1				
			10.	FL3H4D2S2				
			20.	FL3H4D2S3				
			5.	FL5H4D2S1				
			10.	FL5H4D2S2				
			20.	FL5H4D2S3				
800.		400.	5.	FL6H5D2S1				
			10.	FL6H5D2S2				
			20.	FL6H5D2S3				
			5.	FL7H5D2S1				
			10.	FL7H5D2S2				
			20.	FL7H5D2S3				
1000.		1000.	5.	FL7H5D2S1				
			10.	FL7H5D2S2				
			20.	FL7H5D2S3				
			5.	FL7H5D2S1				
			10.	FL7H5D2S2				
			20.	FL7H5D2S3				
2000.	1000.	5.	FL7H5D2S1					
		10.	FL7H5D2S2					
		20.	FL7H5D2S3					
		5.	FL7H5D2S1					
		10.	FL7H5D2S2					
		20.	FL7H5D2S3					
		5.	FL7H5D2S1					
		10.	FL7H5D2S2					
		20.	FL7H5D2S3					
		5.	FL7H5D2S1					
		10.	FL7H5D2S2					
		20.	FL7H5D2S3					
Reinforced circular (See Figure 1-3)	200.	200.	100.	—	—	—	RL0H1D2R1	
	400.						RL3H1D2R1	
	400.						RL3H1D2R2	
	400.	400.					150.	RL3H4D2R1
							200.	RL3H4D2R2
							150.	RL5H4D2R1
	800.	400.					200.	RL5H4D2R2
							400.	RL5H4D2R3
							150.	RL6H5D2R1
	1000.	1000.					200.	RL6H5D2R2
							400.	RL6H5D2R3
							150.	RL7H5D2R1
	2000.	1000.					200.	RL7H5D2R2
							400.	RL7H5D2R3
							400.	RL7H5D2R3

Type of hole:	L (mm)	H (mm)	D (mm)	d (mm)	L _r (mm)	L _h (mm)	A (mm)	FEM model File name:
Plain elongated (See Figure 1-4)	200.	200.	100.	—	—	150.	—	EL0H1D2P2
	400.					150.		EL3H1D2P2
						200.		EL3H1D2P3
	400.	400.				120.		EL3H4D2P2
						200.		EL3H4D2P3
	800.					150.		EL5H4D2P2
						200.		EL5H4D2P3
	1000.	1000.				150.		EL6H5D2P2
						200.		EL6H5D2P3
						150.		EL7H5D2P2
2000.		200.	EL7H5D2P3					
Beaded (See Figure 1-5)	400.	200.	100.	—	—	150.	200.	BL3H1D2P2A1
	800.	400.				150.	400.	BL5H4D2P2A4
						200.		BL5H4D2P3A4
	2000.	1000.				150.	1000.	BL7H5D2P2A5
						200.		BL7H5D2P3A5

Table 3-1: CFC lightening webs submitted to shear load. List of FEM models.

As it can be seen from the previous Table, the file designation criteria do not take into account the laminate properties, nor the shear web thickness. This information can be found in the folder tree where the files have been stored (see [Ref. 16](#) and Figure 3-1).

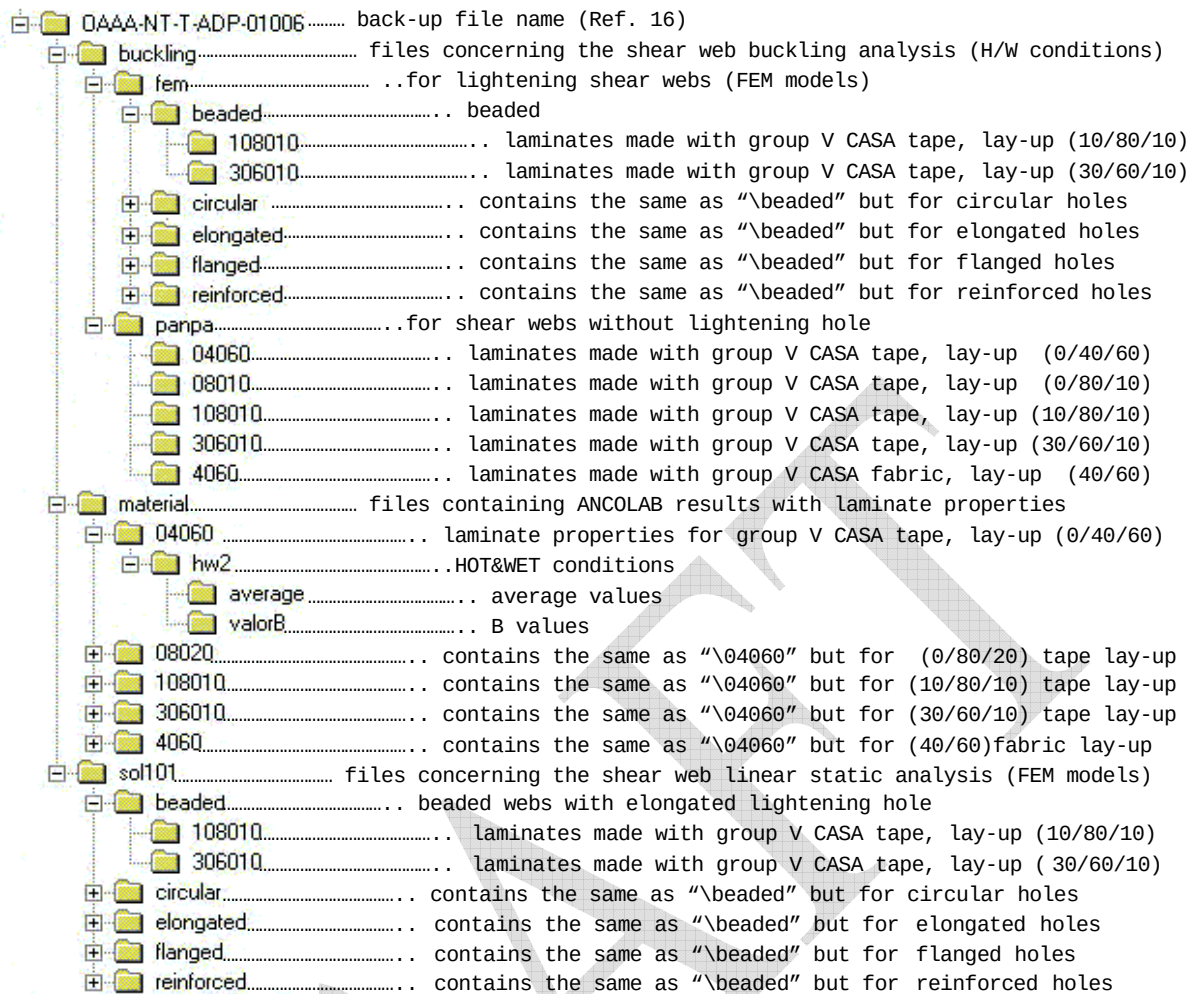


Figure 3-1: Chart of analysis files used. -

3.1 Reference Axes system

All FEM models have been referred to the following basic axis reference system, called the global system. (In all cases, the flat shear web is contained in the OXY plane):

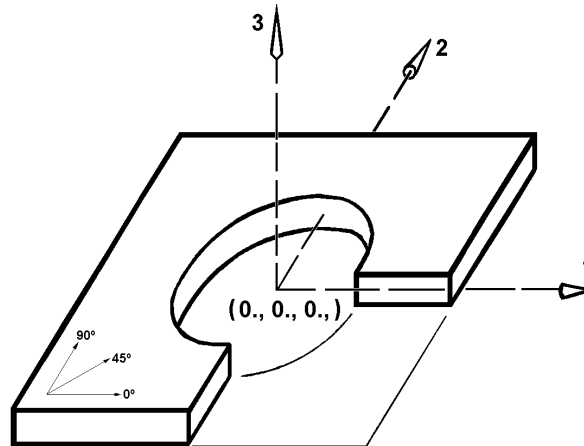


Figure 3-2: Global FEM model reference coordinate system.

Origin: (0., 0., 0.) It is placed within the basic laminate middle plane, and matching with the symmetry centre of each FEM model.

Axis "1", (OX) It matches with the 0° lay-up direction,

Axis "2", (OY) It matches with the 90° lay-up direction,

Axis "3", (OZ) It is normal to the shear web middle plane.

Figure 3-2 shows an example of the global coordinate system for a generic FEM model. All the GRID points corresponding to each FEM model are referred to this global coordinate system.

As it can be seen from Figure 3-2, the 0° direction for all laminates used, (material Reference Axis System), must be in accordance with the "1" direction of global coordinate system.

The elements local coordinate system depends on the MSC/NASTRAN connectivity cards used, which follow these general rules:

- Plane elements: (CQUAD4), 1- axis is radial respect to the hole, in the laminate middle plane and oriented outside the hole. 2- Axis is tangential to the hole, and 3- axis is parallel to the global 3- axis.
- Linear elements: (CROD). Their 1- axis is tangential to the hole and has the same orientation than the 2- axis of surrounding plane elements. 2- Axis is radial, oriented to the center of the hole, and 3- axis is parallel to the same axis in the global coordinate system.

An auxiliary reference coordinate system is used in order to define the web support conditions. It is equal to the one shown in figure 3-2, but rotated around 3- axis until 1- axis crosses the panel corner. This reference system is needed to constrain the rigid solid degrees of freedom without interfering the panel behaviour.

3.2 FEM models mesh description

3.2.1 Grids and connectivity cards used

For FEM models having circular lightening hole, the whole mesh around the hole has been issued in a radial way respect to the lightening hole, containing the lay-up middle plane.

Figure 3-3 shows an example. Rectangular plates have been meshed in the same way, by considering a square web having the rectangular minimum side length and adding later rectangular portions of mesh, just to fill the desired dimensions, (see Figure 3-4).

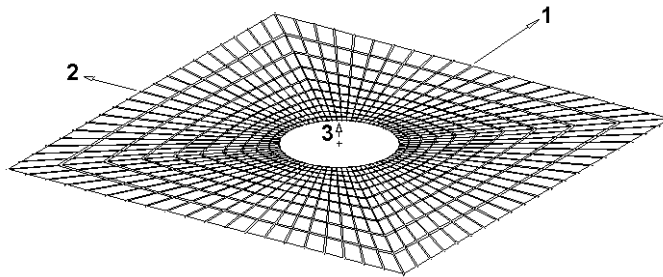


Figure 3-3: Lightening flat web FEM model mesh.

FEM models corresponding to the flanged holes were issued in the same way, but taken into account the flange geometry. Flanges considered on this Report are not symmetric respect to the shear web middle plane.

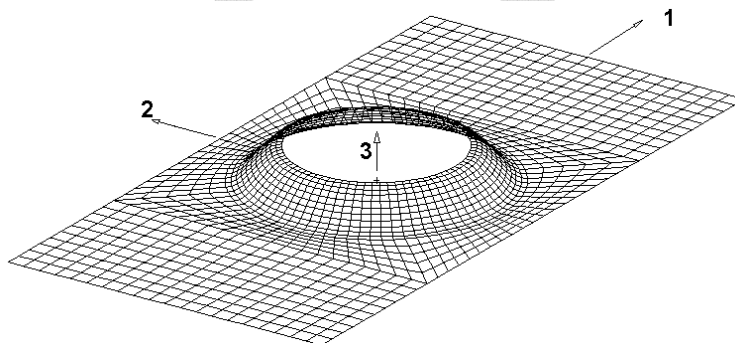


Figure 3-4: Lightening flanged web FEM model mesh.

Flange geometry depends on its height. Inner diameter is kept constant for all the configurations and has a value of 100 mm. Flange slope and outer diameter vary with the height of the flange in the following way:

Flange height	d (mm)	5.0	10.0	20.0
Outer diameter	Φ_{out} (mm)	129.3	139.9	159.9
Flange slope	α (°)	31.3	45.0	45.0

FEM models for reinforced shear webs were obtained from the circular or elongated hole FEM models, by modifying thickness and location of shear web middle plane for the reinforced zones. Obtained FEM models are not symmetric respect to the middle plane of basic lay-up. Reinforced area around the hole was modeled first, after this was meshed the web area corresponding to the basic lay-up thickness and, finally, the ply-drop-off zones were filled. Figure 3-5 sketches an example. (Reinforcement size definition does not include the ply-drop-off zone).

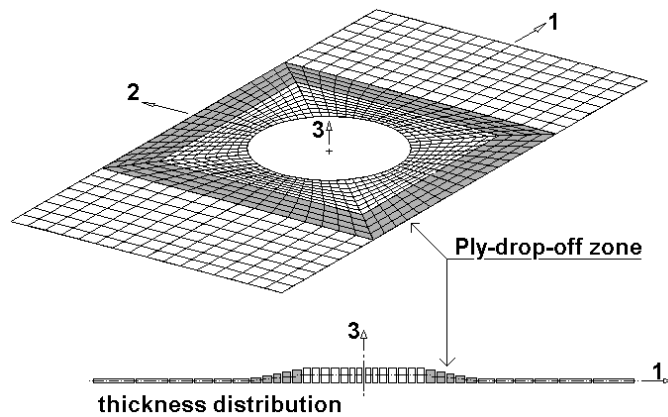


Figure 3-5: Lightening reinforced web FEM model mesh.

FEM models having an elongated hole were constructed from the geometry corresponding to a shear web having a circular centred hole by placing rectangular mesh portions, as it is shown in Figure 3-6.

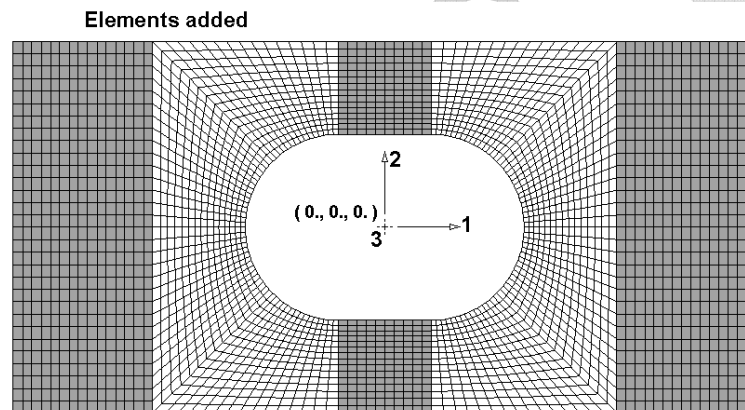


Figure 3-6: Elongated hole web FEM model mesh.

Beaded hole FEM models were made from an elongated hole FEM model, by adding the beads besides the hole.

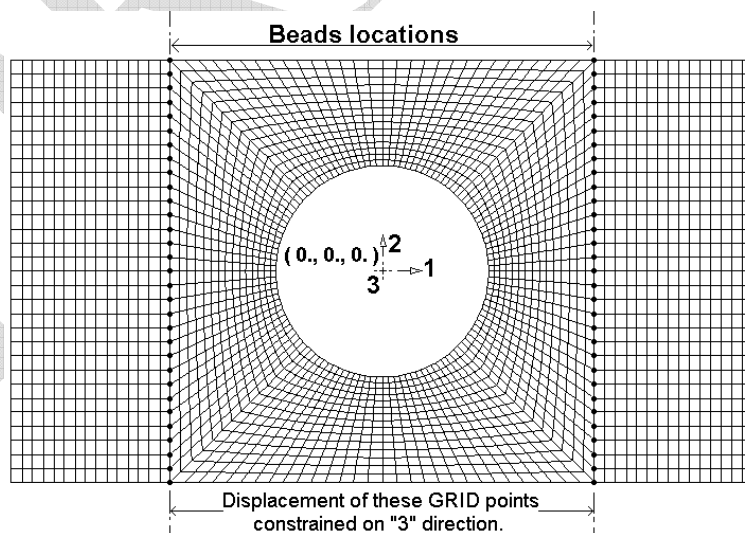


Figure 3-7: Beaded panel with lightening hole. FEM model mesh.

Only support conditions produced by the beads were simulated. Therefore, GRID points placed at the bead location have their displacement constrained in "3" direction. (I.e., it is assumed that the first local buckling mode corresponds to the shear web, -see figure 3-7)

The following element cards have been used in all FEM models:

- CQUAD4 cards, in order to simulate the shear web. They are related to PSHELL and MAT2 entries representing the basic laminated used.
- CROD cards simulating the hole edge. They have a negligible cross sectional area, because they are only used in order to obtain the maximum strain level at this zone. They are attached to the MAT1 entry, which corresponds to a standard aluminium alloy.

Due to the fact that all FEM models are small, no special identification rules have been followed for the MSC/NASTRAN GRID cards nor for the connectivity cards. Only the next format rules have been established in order to distinguish between GRIDS and connectivities:

Card Types:	Identification Nos.:
GRID	9XXXX
CQUAD4	4XXXX
CBEAM	2XXXX
CROD	1XXXX

3.2.2 Materials and mechanical properties

1. Properties for ROD Elements:

For linear elements placed in the hole edge standard aluminium alloy properties are assumed, ($E_{xx} = 71020 \text{ N/mm}^2$)

Negligible cross sectional area has been considered for those elements: $A = 0.001 \text{ mm}^2$.

The objective of these data was to obtain the peak tangential strains nearest as possible to the hole edge.

Thus, the PROD and MAT1 cards used for this kind of element on all FEM models are the ones shown below.

1	2	3	4	5	6	7	8	9	10
PROD	100	1000	0.001						

1	2	3	4	5	6	7	8	9	10
MAT1	1000	71020.		0.33					

Those entries are common for all the produced lightening shear web FEM models.

2. Properties for SHELL Elements:

They depend on the lay-ups used, which were summarised in Table 2-1.

All of them have been calculated considering "H/W 2" standard environmental conditions and for the different thickness considered.

MSC/NASTRAN cards used for each composite laminate are PSHELL and MAT2 entries, (example below shows Tape 30/60/10 laminate, having a total thickness of 3.68 mm in H/W2 conditions and considering "B" values for ply properties),

1	2	3	4	5	6	7	8	9	10
PSHELL	200	1000	3.68	1100	1.0		1.0	0.	
	-1.84	1.84	0						

1	2	3	4	5	6	7	8	9	10
MAT2	1000	56240.9	17880.6	0.0	34649.6	0.0	19023.5		
MAT2	1100	48034.0	19610.2	971.6	39397.4	971.6	20753.1		

As it can be seen, the MAT2 card entry fields are those given from Table 2-5 of this Report.

3.2.3 Supports and loading conditions

All the lightening shear webs are supported in the same way, close to the commonly called "simply supported" edges. Figure 3-9 sketches an example.

In terms of the auxiliary co-ordinate system, (see Figure 3-9), the selected support is done through SPC and SPCD MSC/NASTRAN entry cards, and their effect is:

1. All GRID points at the shear web external contour are fixed in 3- axis; that means that the vertical displacement outside the web plane of the web edges is not allowed,
2. A GRID point near to the down-left web corner is fixed in 1- and 2-axis; this means that no in-plane displacement of this point is allowed,
3. The nearest GRID point to the upper-right web corner is fixed in 2-axis. Together with the previous support condition, this means that the shear web can not rotate as a solid around the 3- axis.

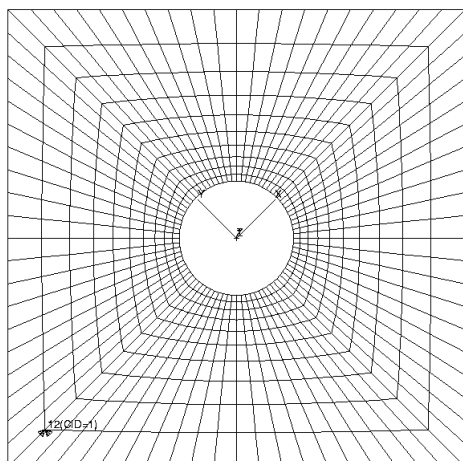
Since the applied loads are in the web plane and form an equilibrated load system (its resultant is equal to zero), the reaction forces in those SPC's must be null. This item is controlled for each analysed model.

MSC/NASTRAN card entries are shown below; GRID point identification numbers can be seen from Figure 3-9, for the example shown.

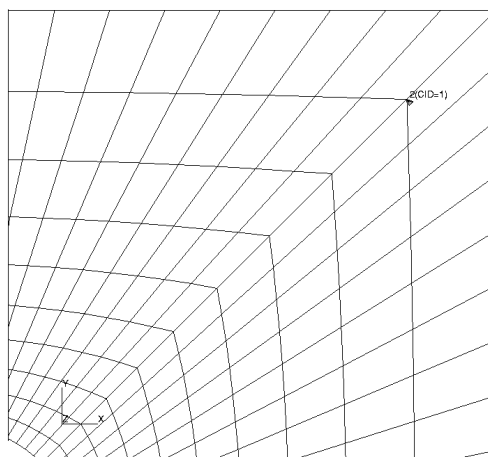
1	2	3	4	5	6	7	8	9	10
SPCADD	2	1	3	4					
\$									
SPC1	1	3	SIDE1 1	THRU	SIDE1 n				
SPC1	1	3	SIDE2 1	THRU	SIDE2 m				
SPC1	1	3	SIDE3 1	THRU	SIDE3 n				
SPC1	1	3	SIDE4 1	THRU	SIDE4 m				
\$									
SPC1	3	12	GRID A						
SPC1	4	2	GRID B						

Note that in the example the GRID point Identification numbers for shear web corners are fictitious, but they must match with the sketch details included below in figure 3-9.

SPC1 Id-No.: 3



SPC1 ID-No.: 4



SPC1 ID-No.: 1

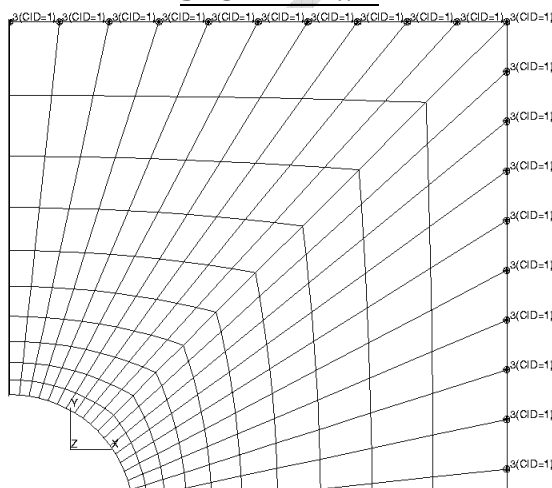


Figure 3-9: Lightning Shear Webs Support Conditions.

In reference to the applied loads, all the lightning shear webs are loaded in the same way, by the means of a positive/negative pure shear flow applied at the GRID points placed over the external contour of the shear web. Figure 3-10 shows an example.

The selected way to load the shear web fulfils that the resultant load over the full shear web is null. The nodal load at each grid is obtained multiplying the applied shear flow and the side length of both elements connected to the grid.

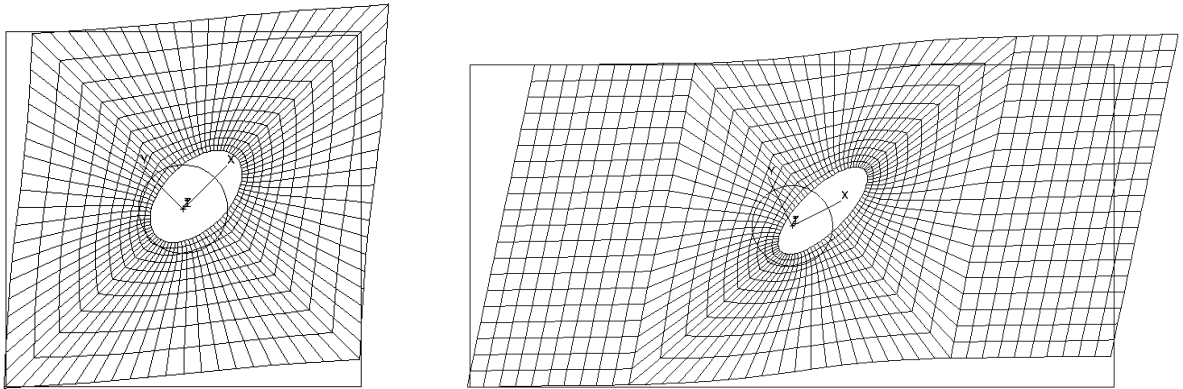


Figure 3-11: Size of the hole and deformation of shear web edges.

1	2	3	4	5	6	7	8	9	10
\$ LOAD OVER SHEAR WEB SIDE "1":									
\$									
FORCE	1000	SIDE1 1	0	1000.	0.	1.	0.		
FORCE	1000	SIDE1 2	0	1000.	0.	1.	0.		
FORCE	1000	SIDE1 2	0	1000.	0.	1.	0.		
...									
FORCE	1000	SIDE1 n-1	0	1000.	0.	1.	0.		
FORCE	1000	SIDE1 n-1	0	1000.	0.	1.	0.		
FORCE	1000	SIDE1 n	0	1000.	0.	1.	0.		
\$									
\$ LOAD OVER SHEAR WEB SIDE "2":									
\$									
FORCE	2000	SIDE2 1	0	1000.	-1.	0.	0.		
FORCE	2000	SIDE2 2	0	1000.	-1.	0.	0.		
FORCE	2000	SIDE2 2	0	1000.	-1.	0.	0.		
...									
FORCE	2000	SIDE2 m-1	0	1000.	-1.	0.	0.		
FORCE	2000	SIDE2 m-1	0	1000.	-1.	0.	0.		
FORCE	2000	SIDE2 m	0	1000.	-1.	0.	0.		
\$									
\$ LOAD OVER SHEAR WEB SIDE "3":									
\$									
FORCE	3000	SIDE3 1	0	1000.	0.	-1.	0.		
FORCE	3000	SIDE3 2	0	1000.	0.	-1.	0.		
FORCE	3000	SIDE3 2	0	1000.	0.	-1.	0.		
...									
FORCE	3000	SIDE3 n-1	0	1000.	0.	-1.	0.		
FORCE	3000	SIDE3 n-1	0	1000.	0.	-1.	0.		
FORCE	3000	SIDE3 n	0	1000.	0.	-1.	0.		
\$									
\$ LOAD OVER SHEAR WEB SIDE "4":									
\$									
FORCE	4000	SIDE4 1	0	1000.	1.	0.	0.		
FORCE	4000	SIDE4 2	0	1000.	1.	0.	0.		
FORCE	4000	SIDE4 2	0	1000.	1.	0.	0.		
...									
FORCE	4000	SIDE4 m-1	0	1000.	1.	0.	0.		
FORCE	4000	SIDE4 m-1	0	1000.	1.	0.	0.		
FORCE	4000	SIDE4 m	0	1000.	1.	0.	0.		
\$									

Table 3-2: Example of shear web loading cards.

4 FEM model results

Hereafter will be summarised the results obtained from lightening shear web FEM models.

Files corresponding to the MSC/NASTRAN linear buckling solution, (SOL 105), and the linear static one, (SOL 101), for all the analysed geometries were fully stored in Reference 16. This includes the bulk data files, and the corresponding output files, (F06, F04, LOG, XDB and/or OP2 extensions).

Buckling shear allowable flows for all the analysed geometries showed on Table 3-1 have been tabulated below.

NASTRAN linear static runs, (SOL 101), have been reproduced from buckling ones by making the following changes over the bulk data files:

- Change line "SOL 105" by "SOL 101" in the NASTRAN Executive Control Section,
- Delete line "METHOD=10", in the NASTRAN Case Control Section; changes must also be done in lines affecting to the output data requested,
- Delete line "EIGRL,10,,,5" in the NASTRAN Bulk Data Section.
- Change MAT2 cards, since Average properties are used for linear static runs instead of B values.

Related to the NASTRAN linear static analyses, following data are requested:

1. Maximum and minimum stress levels at linear elements placed at the hole. Strain level at the hole is obtained from these data by taking into account that the elastic modulus for all these elements is 71020 N/mm^2 .
2. Maximum and minimum strains level reached, (principal strains), for all de CQUAD4 elements, and locations where they take place.
3. Maximum displacement, (it always takes place at the extreme diagonal nodes), and
4. Average shear stress/strains level reached

From 1 and 2, it is obtained that principal strains over CQUAD4 elements are less critical than for linear elements. This assessment is true for all the considered configurations except for the flanged circular holes. In this last case, maximum strain does not appear at the hole edge but at the flange base, due to the bending it induces. For large holes and large flange heights, a second strain concentration area appears nearby the web edge. This is due to the reduced distance existing between the flange base and the web edge.

In all cases, critical elements are located along the shear web diagonal, working nearby pure compression load. The output strain values for critical elements in tension are very close (in absolute value) to those elements found critical under compression flow.

4.1 Buckling FEM model results

4.1.1 Flat webs with circular lightening hole

Obtained results are shown in tabular form. There is one table per material considered including the following data:

1. FEM model considered,
2. Geometrical parameter for the FEM considered, in accordance with Figure 1-1,
3. First buckling shear flow (q_{BK})

Table 4-1: Results for [30/60/10] % tape Group V solid laminate (AS4/8552)

FEM model	L	H	D	t	1 st (q_{BK})
ID. Name:	mm	mm	mm	mm	N/mm
CL00H0D2	133	133	100	3.68	171.83
CL01H0D2	266	133	100	3.68	178.49
CL0H1D2	200	200	100	3.68	159.25
CL3H1D2	400	200	100	3.68	144.39
CL3H4D2	400	400	100	3.68	80.24
CL5H4D2	800	400	100	3.68	60.12
CL6H5D2	1000	1000	100	3.68	17.62
CL7H5D2	2000	1000	100	3.68	11.85

Table 4-2: Results for [10/80/10] % tape Group V solid laminate (AS4/8552)

FEM model	L	H	D	t	1 st (q_{BK})
ID. Name:	mm	mm	mm	mm	N/mm
CL00H0D2	133	133	100	3.68	186.67
CL01H0D2	266	133	100	3.68	173.88
CL0H1D2	200	200	100	3.68	170.66
CL3H1D2	400	200	100	3.68	142.20
CL3H4D2	400	400	100	3.68	85.48
CL5H4D2	800	400	100	3.68	59.49
CL6H5D2	1000	1000	100	3.68	18.71
CL7H5D2	2000	1000	100	3.68	11.70

Table 4-3: Results for [0/40/60] % tape Group V solid laminate (AS4/8552)

FEM model	L	H	D	t	1 st (q_{BK})
ID. Name:	mm	mm	mm	mm	N/mm
CL00H0D2	133	133	100	3.68	129.61
CL01H0D2	266	133	100	3.68	148.45
CL0H1D2	200	200	100	3.68	126.76
CL3H1D2	400	200	100	3.68	131.13
CL3H4D2	400	400	100	3.68	68.20
CL5H4D2	800	400	100	3.68	64.44
CL6H5D2	1000	1000	100	3.68	15.38
CL7H5D2	2000	1000	100	3.68	13.36

Table 4-4: Results for [0/80/20] % tape Group V solid laminate (AS4/8552)

FEM model	L	H	D	t	1 st (q _{BK})
ID. Name:	mm	mm	mm	mm	N/mm
CL00H0D2	133	133	100	3.68	186.45
CL01H0D2	266	133	100	3.68	183.35
CL0H1D2	200	200	100	3.68	171.95
CL3H1D2	400	200	100	3.68	155.30
CL3H4D2	400	400	100	3.68	87.01
CL5H4D2	800	400	100	3.68	69.62
CL6H5D2	1000	1000	100	3.68	19.08
CL7H5D2	2000	1000	100	3.68	14.01

Table 4-5: Results for [40/60] % fabric Group V solid laminate (AS4/8552)

FEM model	L	H	D	t	1 st (q _{BK})
ID. Name:	mm	mm	mm	mm	N/mm
CL00H0D2	133	133	100	2.05	26.55
CL01H0D2	266	133	100	2.05	27.89
CL0H1D2	200	200	100	2.05	24.94
CL3H1D2	400	200	100	2.05	23.10
CL3H4D2	400	400	100	2.05	12.72
CL5H4D2	800	400	100	2.05	9.91
CL6H5D2	1000	1000	100	2.05	2.81
CL7H5D2	2000	1000	100	2.05	1.97

Buckling shear flows obtained for [40/60] fabric are largely smaller than those obtained for the previous tapes. This great difference is mainly due to the decrease in thickness between tape and fabric laminates.

4.1.2 Panels with flanged circular hole

Following tables include results obtained from flanged hole FEM models.

Table 4-6: Results for [0/40/60] % tape Group V solid laminate (AS4/8552)

FEM model	L	H	D	d	t	1 st (q _{BK})
ID. Name:	mm	mm	mm	mm	mm	N/mm
FL0H1D2S1	200	200	100	5.0	3.68	225.94
FL3H1D2S1	400	200	100	5.0	3.68	231.84
FL3H4D2S1	400	400	100	5.0	3.68	93.80
FL5H4D2S1	800	400	100	5.0	3.68	83.06
FL6H5D2S1	1000	1000	100	5.0	3.68	16.64
FL7H5D2S1	2000	1000	100	5.0	3.68	13.94
FL0H1D2S2	200	200	100	10.0	3.68	358.16
FL3H1D2S2	400	200	100	10.0	3.68	346.89
FL3H4D2S2	400	400	100	10.0	3.68	110.65
FL5H4D2S2	800	400	100	10.0	3.68	91.48
FL6H5D2S2	1000	1000	100	10.0	3.68	17.25
FL7H5D2S2	2000	1000	100	10.0	3.68	13.99
FL0H1D2S3	200	200	100	20.0	3.68	464.76
FL3H1D2S3	400	200	100	20.0	3.68	423.58
FL3H4D2S3	400	400	100	20.0	3.68	124.80
FL5H4D2S3	800	400	100	20.0	3.68	95.38
FL6H5D2S3	1000	1000	100	20.0	3.68	17.75
FL7H5D2S3	2000	1000	100	20.0	3.68	14.05

Table 4-7: Results for [0/80/20] % tape Group V solid laminate (AS4/8552)

FEM model	L	H	D	d	t	1 st (q _{BK})
ID. Name:	mm	mm	mm	mm	mm	N/mm
FL0H1D2S1	200	200	100	5.0	3.68	295.60
FL3H1D2S1	400	200	100	5.0	3.68	269.09
FL3H4D2S1	400	400	100	5.0	3.68	114.90
FL5H4D2S1	800	400	100	5.0	3.68	87.08
FL6H5D2S1	1000	1000	100	5.0	3.68	20.49
FL7H5D2S1	2000	1000	100	5.0	3.68	14.66
FL0H1D2S2	200	200	100	10.0	3.68	447.79
FL3H1D2S2	400	200	100	10.0	3.68	382.65
FL3H4D2S2	400	400	100	10.0	3.68	132.54
FL5H4D2S2	800	400	100	10.0	3.68	96.84
FL6H5D2S2	1000	1000	100	10.0	3.68	21.18
FL7H5D2S2	2000	1000	100	10.0	3.68	14.96
FL0H1D2S3	200	200	100	20.0	3.68	568.16
FL3H1D2S3	400	200	100	20.0	3.68	471.74
FL3H4D2S3	400	400	100	20.0	3.68	147.60
FL5H4D2S3	800	400	100	20.0	3.68	103.12
FL6H5D2S3	1000	1000	100	20.0	3.68	21.74
FL7H5D2S3	2000	1000	100	20.0	3.68	15.21

Table 4-8: Results for [40/60] % fabric Group V solid laminate (AGP 193 – PW / 8552)

FEM model	L	H	D	d	t	1 st (q _{BK})
ID. Name:	mm	mm	mm	mm	mm	N/mm
FL0H1D2S1	200	200	100	5.0	2.05	62.33
FL3H1D2S1	400	200	100	5.0	2.05	51.15
FL3H4D2S1	400	400	100	5.0	2.05	19.00
FL5H4D2S1	800	400	100	5.0	2.05	13.38
FL6H5D2S1	1000	1000	100	5.0	2.05	3.10
FL7H5D2S1	2000	1000	100	5.0	2.05	2.09
FL0H1D2S2	200	200	100	10.0	2.05	81.59
FL3H1D2S2	400	200	100	10.0	2.05	63.93
FL3H4D2S2	400	400	100	10.0	2.05	20.65
FL5H4D2S2	800	400	100	10.0	2.05	14.22
FL6H5D2S2	1000	1000	100	10.0	2.05	3.15
FL7H5D2S2	2000	1000	100	10.0	2.05	2.12
FL0H1D2S3	200	200	100	20.0	2.05	100.35
FL3H1D2S3	400	200	100	20.0	2.05	78.22
FL3H4D2S3	400	400	100	20.0	2.05	23.01
FL5H4D2S3	800	400	100	20.0	2.05	15.03
FL6H5D2S3	1000	1000	100	20.0	2.05	3.24
FL7H5D2S3	2000	1000	100	20.0	2.05	2.15

As in the case of shear webs with no flanged circular hole, buckling load for fabric webs is rather smaller than for tape webs due to thickness decrease.

4.1.3 Reinforced webs with circular hole

Geometry of these FEM models was shown in Figure 1-3.

Results for these shear webs are included below.

Table 4-9: Results for [30/60/10] % tape Group V solid laminate (AS4/8552)

FEM model	L	H	D	L _r	t	1 st (q _{BK})
ID. Name:	mm	mm	mm	mm	mm	N/mm
RL0H1D2R1	200	200	100	150.	3.68	757.39
RL3H1D2R1	400	200	100	150.	3.68	472.90
RL3H4D2R1	400	400	100	150.	3.68	164.62
RL5H4D2R1	800	400	100	150.	3.68	94.15
RL6H5D2R1	1000	1000	100	150.	3.68	21.23
RL7H5D2R1	2000	1000	100	150.	3.68	13.13
RL3H1D2R2	400	200	100	200.	3.68	719.77
RL3H4D2R2	400	400	100	200.	3.68	225.44
RL5H4D2R2	800	400	100	200.	3.68	110.09
RL6H5D2R2	1000	1000	100	200.	3.68	23.48
RL7H5D2R2	2000	1000	100	200.	3.68	13.23
RL5H4D2R3	800	400	100	400.	3.68	220.58
RL6H5D2R3	1000	1000	100	400.	3.68	32.54
RL7H5D2R3	2000	1000	100	400.	3.68	15.04

Table 4-10: Results for [10/80/10] % tape Group V solid laminate (AS4/8552)

FEM model	L	H	D	L_r	t	$1^{st} (q_{BK})$
ID. Name:	mm	mm	mm	mm	mm	N/mm
RL0H1D2R1	200	200	100	150.	3.68	778.64
RL3H1D2R1	400	200	100	150.	3.68	444.67
RL3H4D2R1	400	400	100	150.	3.68	169.17
RL5H4D2R1	800	400	100	150.	3.68	94.75
RL6H5D2R1	1000	1000	100	150.	3.68	22.35
RL7H5D2R1	2000	1000	100	150.	3.68	12.99
RL3H1D2R2	400	200	100	200.	3.68	684.17
RL3H4D2R2	400	400	100	200.	3.68	230.30
RL5H4D2R2	800	400	100	200.	3.68	110.98
RL6H5D2R2	1000	1000	100	200.	3.68	24.62
RL7H5D2R2	2000	1000	100	200.	3.68	13.39
RL5H4D2R3	800	400	100	400.	3.68	212.53
RL6H5D2R3	1000	1000	100	400.	3.68	33.10
RL7H5D2R3	2000	1000	100	400.	3.68	15.22

4.1.4 Panels with elongated hole

Two elongation cases have been analysed herein: $L_h/D = 1.5$ and $L_h/D = 2.0$. Geometry of these FEM models was shown in Figure 1-4.

Table 4-11: Results for [30/60/10] % tape Group V solid laminate (AS4/8552)

FEM model	L	H	D	L_h	t	$1^{st} (q_{BK})$
ID. Name:	mm	mm	mm	mm	mm	N/mm
EL0H1D2P2	200	200	100	150.	3.68	100.20
EL3H1D2P2	400	200	100	150.	3.68	100.85
EL3H4D2P2	400	400	100	150.	3.68	65.82
EL5H4D2P2	800	400	100	150.	3.68	51.85
EL6H5D2P2	1000	1000	100	150.	3.68	16.73
EL7H5D2P2	2000	1000	100	150.	3.68	11.45
EL3H1D2P3	400	200	100	200.	3.68	74.08
EL3H4D2P3	400	400	100	200.	3.68	53.96
EL5H4D2P3	800	400	100	200.	3.68	44.66
EL6H5D2P3	1000	1000	100	200.	3.68	15.85
EL7H5D2P3	2000	1000	100	200.	3.68	11.03

Table 4-12: Results for [10/80/10] % tape Group V solid laminate (AS4/8552)

FEM model	L	H	D	L _h	t	1 st (q _{BK})
ID. Name:	mm	mm	mm	mm	mm	N/mm
EL0H1D2P2	200	200	100	150.	3.68	109.48
EL3H1D2P2	400	200	100	150.	3.68	100.04
EL3H4D2P2	400	400	100	150.	3.68	70.37
EL5H4D2P2	800	400	100	150.	3.68	51.4
EL6H5D2P2	1000	1000	100	150.	3.68	17.75
EL7H5D2P2	2000	1000	100	150.	3.68	11.32
EL3H1D2P3	400	200	100	200.	3.68	74.72
EL3H4D2P3	400	400	100	200.	3.68	57.97
EL5H4D2P3	800	400	100	200.	3.68	44.36
EL6H5D2P3	1000	1000	100	200.	3.68	16.83
EL7H5D2P3	2000	1000	100	200.	3.68	10.91

4.1.5 Beaded webs with hole

The web geometry was shown in Figure 1-5 and the analysed models were summarised in Table 3-1.

Results obtained from Linear buckling NASTRAN runs are shown below.

Table 4-13: Results for [30/60/10] % tape Group V solid laminate (AS4/8552)

FEM model	L	H	D	L _h	A	t	1 st (q _{BK})
ID. Name:	mm	mm	mm	mm	mm	mm	N/mm
BL3H1D2P2A1	400	200	100	150.	200	3.68	157.73
BL5H4D2P2A4	800	400	100	150.	400	3.68	80.33
BL7H5D2P2A5	2000	1000	100	150.	1000	3.68	19.47
BL5H4D2P3A4	800	400	100	200.	400	3.68	69.20
BL7H5D2P3A5	2000	1000	100	200.	1000	3.68	18.52

Table 4-14: Results for [10/80/10] % tape Group V solid laminate (AS4/8552)

FEM model	L	H	D	L _h	A	t	1 st (q _{BK})
ID. Name:	mm	mm	mm	mm	mm	mm	N/mm
BL3H1D2P2A1	400	200	100	150.	200	3.68	165.15
BL5H4D2P2A4	800	400	100	150.	400	3.68	84.30
BL7H5D2P2A5	2000	1000	100	150.	1000	3.68	20.48
BL5H4D2P3A4	800	400	100	200.	400	3.68	72.52
BL7H5D2P3A5	2000	1000	100	200.	1000	3.68	19.47

4.2 Linear static FEM model results

Damage tolerance is analysed through these results. Particularly, except for flanged holes, strain concentration at the hole edge is studied. This maximum strain is obtained from the stress NASTRAN output at the ROD elements located along the hole perimeter:

- $\epsilon = \sigma / E$,

where the Young modulus for a standard aluminium alloy is $E = 71020 \text{ N/mm}^2$. All the analyses have been made with average ply properties.

In the case of panels with flanged circular hole, strain at the flange base, which is where the maximum appears, is presented.

4.2.1 Flat webs with circular lightening hole

As it was made for buckling runs, results are presented in tabular format.

Table 4-15: Results for [30/60/10] % tape Group V solid laminate (AS4/8552)

FEM model	L	H	D	t	ϵ
ID. Name:	mm	mm	mm	mm	$\mu\epsilon$
CL00H0D2	133	133	100	3.68	14032.
CL01H0D2	266	133	100	3.68	10178.
CL0H1D2	200	200	100	3.68	4578.
CL3H1D2	400	200	100	3.68	3837.
CL3H4D2	400	400	100	3.68	2579.
CL5H4D2	800	400	100	3.68	2449.
CL6H5D2	1000	1000	100	3.68	2181.
CL7H5D2	2000	1000	100	3.68	2162.

Table 4-16: Results for [10/80/10] % tape Group V solid laminate (AS4/8552)

FEM model	L	H	D	t	ϵ
ID. Name:	mm	mm	mm	mm	$\mu\epsilon$
CL00H0D2	133	133	100	3.68	13021.
CL01H0D2	266	133	100	3.68	13158.
CL0H1D2	200	200	100	3.68	4439.
CL3H1D2	400	200	100	3.68	4155.
CL3H4D2	400	400	100	3.68	2481.
CL5H4D2	800	400	100	3.68	2427.
CL6H5D2	1000	1000	100	3.68	2097.
CL7H5D2	2000	1000	100	3.68	2090.

Table 4-17: Results for [0/40/60] % tape Group V solid laminate (AS4/8552)

FEM model	L	H	D	t	ϵ
ID. Name:	mm	mm	mm	mm	$\mu\epsilon$
CL0H1D2	200	200	100	3.68	6744.
CL3H1D2	400	200	100	3.68	6732.
CL3H4D2	400	400	100	3.68	3436.
CL5H4D2	800	400	100	3.68	3429.
CL6H5D2	1000	1000	100	3.68	2808.
CL7H5D2	2000	1000	100	3.68	2807.

Table 4-18: Results for [0/80/20] % tape Group V solid laminate (AS4/8552)

FEM model	L	H	D	t	ϵ
ID. Name:	mm	mm	mm	mm	$\mu\epsilon$
CL0H1D2	200	200	100	3.68	4879.
CL3H1D2	400	200	100	3.68	4815.
CL3H4D2	400	400	100	3.68	2605.
CL5H4D2	800	400	100	3.68	2590.
CL6H5D2	1000	1000	100	3.68	2179.
CL7H5D2	2000	1000	100	3.68	2177.

Table 4-19: Results for [40/60] % fabric Group V solid laminate (AGP 193 – PW / 8552)

FEM model	L	H	D	t	ϵ
ID. Name:	mm	mm	mm	mm	$\mu\epsilon$
CL0H1D2	200	200	100	2.05	9444.
CL3H1D2	400	200	100	2.05	8749.
CL3H4D2	400	400	100	2.05	5452.
CL5H4D2	800	400	100	2.05	5308.
CL6H5D2	1000	1000	100	2.05	4634.
CL7H5D2	2000	1000	100	2.05	4614.

4.2.2 Panels with flanged circular hole

Tables 4-20 to 4-22 show the strain concentration for panels with flanged circular holes. In this case, the presence of the flange makes the panel behave in a different way. Strain concentration does not appear at the hole edge, but at the flange base. This is mainly due to the bending introduced by the flange.

Table 4-20: Results for [0/40/60] % tape Group V solid laminate (AS4/8552)

FEM model	L	H	D	d	t	ϵ
ID. Name:	mm	mm	mm	mm	mm	$\mu\epsilon$
FL0H1D2S1	200	200	100	5.0	3.68	4810.
FL3H1D2S1	400	200	100	5.0	3.68	4830.
FL3H4D2S1	400	400	100	5.0	3.68	2560.
FL5H4D2S1	800	400	100	5.0	3.68	2560.
FL6H5D2S1	1000	1000	100	5.0	3.68	2030.
FL7H5D2S1	2000	1000	100	5.0	3.68	2020.
FL0H1D2S2	200	200	100	10.0	3.68	6120.
FL3H1D2S2	400	200	100	10.0	3.68	6150.
FL3H4D2S2	400	400	100	10.0	3.68	2850.
FL5H4D2S2	800	400	100	10.0	3.68	2840.
FL6H5D2S2	1000	1000	100	10.0	3.68	2150.
FL7H5D2S2	2000	1000	100	10.0	3.68	2150.
FL0H1D2S3	200	200	100	20.0	3.68	9370.
FL3H1D2S3	400	200	100	20.0	3.68	9460.
FL3H4D2S3	400	400	100	20.0	3.68	3130.
FL5H4D2S3	800	400	100	20.0	3.68	3120.
FL6H5D2S3	1000	1000	100	20.0	3.68	2180.
FL7H5D2S3	2000	1000	100	20.0	3.68	2180.

Table 4-21: Results for [0/80/20] % tape Group V solid laminate (AS4/8552)

FEM model	L	H	D	d	t	ϵ
ID. Name:	mm	mm	mm	mm	mm	$\mu\epsilon$
FL0H1D2S1	200	200	100	5.0	3.68	3540.
FL3H1D2S1	400	200	100	5.0	3.68	3670.
FL3H4D2S1	400	400	100	5.0	3.68	1660.
FL5H4D2S1	800	400	100	5.0	3.68	1650.
FL6H5D2S1	1000	1000	100	5.0	3.68	1340.
FL7H5D2S1	2000	1000	100	5.0	3.68	1340.
FL0H1D2S2	200	200	100	10.0	3.68	4900.
FL3H1D2S2	400	200	100	10.0	3.68	5020.
FL3H4D2S2	400	400	100	10.0	3.68	1870.
FL5H4D2S2	800	400	100	10.0	3.68	1860.
FL6H5D2S2	1000	1000	100	10.0	3.68	1440.
FL7H5D2S2	2000	1000	100	10.0	3.68	1440.
FL0H1D2S3	200	200	100	20.0	3.68	7810.
FL3H1D2S3	400	200	100	20.0	3.68	7990.
FL3H4D2S3	400	400	100	20.0	3.68	2070.
FL5H4D2S3	800	400	100	20.0	3.68	2060.
FL6H5D2S3	1000	1000	100	20.0	3.68	1470.
FL7H5D2S3	2000	1000	100	20.0	3.68	1470.

Table 4-22: Results for [40/60] % fabric Group V solid laminate (AGP 193 – PW / 8552)

FEM model	L	H	D	d	t	ϵ
ID. Name:	mm	mm	mm	mm	mm	$\mu\epsilon$
FL0H1D2S1	200	200	100	5.0	2.05	7320.
FL3H1D2S1	400	200	100	5.0	2.05	6940.
FL3H4D2S1	400	400	100	5.0	2.05	3960.
FL5H4D2S1	800	400	100	5.0	2.05	3820.
FL6H5D2S1	1000	1000	100	5.0	2.05	3210.
FL7H5D2S1	2000	1000	100	5.0	2.05	3190.
FL0H1D2S2	200	200	100	10.0	2.05	8350.
FL3H1D2S2	400	200	100	10.0	2.05	8060.
FL3H4D2S2	400	400	100	10.0	2.05	4110.
FL5H4D2S2	800	400	100	10.0	2.05	3940.
FL6H5D2S2	1000	1000	100	10.0	2.05	3220.
FL7H5D2S2	2000	1000	100	10.0	2.05	3200.
FL0H1D2S3	200	200	100	20.0	2.05	11300.
FL3H1D2S3	400	200	100	20.0	2.05	11500.
FL3H4D2S3	400	400	100	20.0	2.05	4350.
FL5H4D2S3	800	400	100	20.0	2.05	4140.
FL6H5D2S3	1000	1000	100	20.0	2.05	3190.
FL7H5D2S3	2000	1000	100	20.0	2.05	3160.

4.2.3 Reinforced webs with circular hole

Results for these shear webs are included below.

Table 4-23: Results for [30/60/10] % tape Group V solid laminate (AS4/8552)

FEM model	L	H	D	L _r	t	ε
ID. Name:	mm	mm	mm	mm	mm	με
RL0H1D2R1	200	200	100	150.	3.68	2612.
RL3H1D2R1	400	200	100	150.	3.68	2240.
RL3H4D2R1	400	400	100	150.	3.68	1767.
RL5H4D2R1	800	400	100	150.	3.68	1699.
RL6H5D2R1	1000	1000	100	150.	3.68	1537.
RL7H5D2R1	2000	1000	100	150.	3.68	1525.
RL3H1D2R2	400	200	100	200.	3.68	1875.
RL3H4D2R2	400	400	100	200.	3.68	1576.
RL5H4D2R2	800	400	100	200.	3.68	1539.
RL6H5D2R2	1000	1000	100	200.	3.68	1441.
RL7H5D2R2	2000	1000	100	200.	3.68	1432.
RL5H4D2R3	800	400	100	400.	3.68	1166.
RL6H5D2R3	1000	1000	100	400.	3.68	1210.
RL7H5D2R3	2000	1000	100	400.	3.68	1212.

Table 4-24: Results for [10/80/10] % tape Group V solid laminate (AS4/8552)

FEM model	L	H	D	L _r	t	ε
ID. Name:	mm	mm	mm	mm	mm	με
RL0H1D2R1	200	200	100	150.	3.68	2524.
RL3H1D2R1	400	200	100	150.	3.68	2417.
RL3H4D2R1	400	400	100	150.	3.68	1684.
RL5H4D2R1	800	400	100	150.	3.68	1651.
RL6H5D2R1	1000	1000	100	150.	3.68	1453.
RL7H5D2R1	2000	1000	100	150.	3.68	1446.
RL3H1D2R2	400	200	100	200.	3.68	1983.
RL3H4D2R2	400	400	100	200.	3.68	1483.
RL5H4D2R2	800	400	100	200.	3.68	1465.
RL6H5D2R2	1000	1000	100	200.	3.68	1339.
RL7H5D2R2	2000	1000	100	200.	3.68	1333.
RL5H4D2R3	800	400	100	400.	3.68	1108.
RL6H5D2R3	1000	1000	100	400.	3.68	1110.
RL7H5D2R3	2000	1000	100	400.	3.68	1107.

4.2.4 Panels with elongated hole

Results for the two elongation factors are shown in Tables 4-25 and 4-26.

Table 4-25: Results for [30/60/10] % tape Group V solid laminate (AS4/8552)

FEM model	L	H	D	L _h	t	ε
ID. Name:	mm	mm	mm	mm	mm	με
EL0H1D2P2	200	200	100	150.	3.68	11746.
EL3H1D2P2	400	200	100	150.	3.68	5719.
EL3H4D2P2	400	400	100	150.	3.68	3261.
EL5H4D2P2	800	400	100	150.	3.68	2936.
EL6H5D2P2	1000	1000	100	150.	3.68	2426.
EL7H5D2P2	2000	1000	100	150.	3.68	2391.
EL3H1D2P3	400	200	100	200.	3.68	8031.
EL3H4D2P3	400	400	100	200.	3.68	4298.
EL5H4D2P3	800	400	100	200.	3.68	3505.
EL6H5D2P3	1000	1000	100	200.	3.68	2701.
EL7H5D2P3	2000	1000	100	200.	3.68	2638.

Table 4-26: Results for [10/80/10] % tape Group V solid laminate (AS4/8552)

FEM model	L	H	D	L _h	t	ε
ID. Name:	mm	mm	mm	mm	mm	με
EL0H1D2P2	200	200	100	150.	3.68	10842.
EL3H1D2P2	400	200	100	150.	3.68	7016.
EL3H4D2P2	400	400	100	150.	3.68	3238.
EL5H4D2P2	800	400	100	150.	3.68	3076.
EL6H5D2P2	1000	1000	100	150.	3.68	2408.
EL7H5D2P2	2000	1000	100	150.	3.68	2393.
EL3H1D2P3	400	200	100	200.	3.68	10704.
EL3H4D2P3	400	400	100	200.	3.68	4307.
EL5H4D2P3	800	400	100	200.	3.68	3857.
EL6H5D2P3	1000	1000	100	200.	3.68	2731.
EL7H5D2P3	2000	1000	100	200.	3.68	2703.

4.2.5 Beaded webs with hole

Linear static results are summarised in Tables 4-27 and 4-28.

Table 4-27: Results for [30/60/10] % tape Group V solid laminate (AS4/8552)

FEM model	L	H	D	L _h	A	t	ε
ID. Name:	mm	mm	mm	mm	mm	mm	με
BL3H1D2P2A1	400	200	100	150.	200	3.68	5720.
BL5H4D2P2A4	800	400	100	150.	400	3.68	2936.
BL7H5D2P2A5	2000	1000	100	150.	1000	3.68	2391.
BL5H4D2P3A4	800	400	100	200.	400	3.68	3505.
BL7H5D2P3A5	2000	1000	100	200.	1000	3.68	2638.

Table 4-28: Results for [10/80/10] % tape Group V solid laminate (AS4/8552)

FEM model	L	H	D	L _h	A	t	ε
ID. Name:	mm	mm	mm	mm	mm	mm	με
BL3H1D2P2A1	400	200	100	150.	200	3.68	7029.
BL5H4D2P2A4	800	400	100	150.	400	3.68	3076.
BL7H5D2P2A5	2000	1000	100	150.	1000	3.68	2393.
BL5H4D2P3A4	800	400	100	200.	400	3.68	3857.
BL7H5D2P3A5	2000	1000	100	200.	1000	3.68	2703.

5 Webs without holes. Buckling results

5.1 Buckling of flat shear webs

This section summarises the results obtained for shear webs without considering the lightening holes. Shear webs are flat, and they match in geometry –except from the hole– and laminates with those shown for lightening shear web FEM models.

Buckling analysis was made with geometrical data shown in Table 5-1 for flat shear webs. All of them were under an applied shear flow of ± 1.0 N/mm. An example of a typical input file could be seen in Figure 5-1.

Analysis Files:	L	H	t	L/H	Laminate (%-plies)
	(mm)			–	
pl7h5t361	2000.	1000.	3.68	2.00	[30/60/10]
pl7h5t181			3.68	2.00	[10/80/10]
pl7h5t046			3.68	2.00	[0/40/60]
pl7h5t082			3.68	2.00	[0/80/20]
pl7h5f46			2.05	2.00	[40/60]
pl6h5t361	1000.	1000.	3.68	1.00	[30/60/10]
pl6h5t181			3.68	1.00	[10/80/10]
pl6h5t046			3.68	2.00	[0/40/60]
pl6h5t082			3.68	2.00	[0/80/20]
pl6h5f46			2.05	1.00	[40/60]
pl5h4t361	800.	400.	3.68	2.00	[30/60/10]
pl5h4t181			3.68	2.00	[10/80/10]
pl5h4t046			3.68	2.00	[0/40/60]
pl5h4t082			3.68	2.00	[0/80/20]
pl5h4f46			2.05	2.00	[40/60]
pl3h4t361	400.	400.	3.68	1.00	[30/60/10]
pl3h4t181			3.68	1.00	[10/80/10]
pl3h4t046			3.68	2.00	[0/40/60]
pl3h4t082			3.68	2.00	[0/80/20]
pl3h4f46			2.05	1.00	[40/60]
pl3h1t361	400.	200.	3.68	2.00	[30/60/10]
pl3h1t181			3.68	2.00	[10/80/10]
pl3h1t046			3.68	2.00	[0/40/60]
pl3h1t082			3.68	2.00	[0/80/20]
pl3h1f46			2.05	2.00	[40/60]
pl0h1t361	200.	200.	3.68	1.00	[30/60/10]
pl0h1t181			3.68	1.00	[10/80/10]
			3.68	2.00	[0/40/60]
			3.68	2.00	[0/80/20]
pl0h1f46			2.05	1.00	[40/60]
pl01h01t361	266.	133.	3.68	2.00	[30/60/10]
pl01h01t181			3.68	2.00	[10/80/10]
pl00h01t361	133.	133.	3.68	1.00	[30/60/10]
pl00h01t181			3.68	1.00	[10/80/10]

Table 5-1: Geometries of Flat Shear Webs. (Neither hole nor reinforcement beads).

File: pl5h4t361.dat						
4						
1,	114500.0,	7150.0,	3300.0,	0.300,	0.184,	0.
2,	114500.0,	7150.0,	3300.0,	0.300,	0.184,	45.
3,	114500.0,	7150.0,	3300.0,	0.300,	0.184,	-45.
4,	114500.0,	7150.0,	3300.0,	0.300,	0.184,	90.
1						
2						
206966.512,	65800.608,	0.0,	127510.528,	0.0,	70006.480,	
0.000,	0.000,	0.0,	0.00,	0.0,	0.000,	
199485.330,	81441.213,	4035.057,	163617.507,	4035.057,	86187.680,	
3.68						
800.,	400.,					
1.,	0.,	0.,				
0						
0						
PL5H4T361						
2						
0.0,	0.0,	0.0	0.0,	1.		
0.0,	0.0,	0.0	0.0,	-1.		

Figure 5-1: Typical Input File for Shear Web Buckling Analysis. (Ref. 18).

Buckling of shear webs without hole has been obtained using Ref. 18, (PANPA Program), considering "B" values for ply properties. (See Table 2-5). Results obtained for all geometries and laminates are summarised in Table 5-2 for non-beaded flat shear webs.

Size		Tape (30/60/10)	Tape (10/80/10)	Tape (0/40/60)	Tape (0/80/20)	Fabric (40/60)
		t = 3.68mm	t = 3.68mm	t = 3.68mm	t = 3.68mm	t = 2.05mm
L	H	q ₀	q ₀	q ₀	q ₀	q ₀
mm	mm	N/mm	N/mm	N/mm	N/mm	N/mm
133.	133.	1089.62	1157.12	-	-	-
266.	133.	704.13	694.58	-	-	-
200.	200.	481.86	511.71	422.58	521.88	76.83
400.	200.	311.39	307.16	345.65	369.47	51.85
400.	400.	120.46	127.93	105.64	130.47	19.21
800.	400.	77.85	76.79	86.41	92.37	12.96
1000.	1000.	19.27	20.47	16.90	20.88	3.07
2000.	1000.	12.46	12.29	13.83	14.78	2.07

Table 5-2. Buckling shear flow for CFC webs without hole.

As it can be seen from data shown in previous table, in terms of buckling shear flow, differences between the four laminates made from CFC tape are small. Again, difference in fabric thickness causes a huge decrease in buckling shear flow for this laminate.

5.2 Buckling of beaded shear webs

By using program PANPA, (Ref. 18), it was made the buckling analysis of beaded webs under shear flow.

In this analysis, two reinforcing beads of the same material property values than the basic web laminate were considered. The area of these beads is not relevant, since the web is loaded in shear. Regarding the beads inertia, values high enough as to produce local buckling between beads have been considered.

Table 5-4 summarises laminates and the geometrical layout considered.

.Analysis Files:	L mm	H mm	A mm	t mm	Laminate (%-plies)
Sl7h5t361	2000.	1000.	1000.	3.68	[30/60/10]
Sl7h5t181				3.68	[10/80/10]
Sl5h4t361	800.	400.	400.	3.68	[30/60/10]
Sl5h4t181				3.68	[10/80/10]
Sl3h1t361	400.	200.	200.	3.68	[30/60/10]
Sl3h1t181				3.68	[10/80/10]

Table 5-4. Analysed beaded webs without hole.

Size			Tape (30/60/10)	Tape (10/80/10)
			t = 3.68mm	t = 3.68mm
L	H	A	q _o	q _o
mm	mm	mm	N/mm	N/mm
400.	200.	200.	563.57	593.41
800.	400.	400.	140.89	148.35
2000.	1000.	1000.	22.54	23.73

Table 5-5. Shear buckling flows for beaded flat panels without hole.

Table 5-5 above summarises all data obtained for beaded webs under shear flow.

6 Buckling design curves

One of the objectives of all the previous results is the development of a simple design guideline for the described shear webs. For this purpose, buckling shear flows for shear webs with lightening hole and without it have been compared. Curves have been built showing the ratio between both buckling shear flows. This procedure simplifies the panel design since only buckling load of the no-hole panel needs to be calculated. This calculation is easily carried out with in-house numerical tools as Ref. 18 (PANPA program). Non-dimensional parameters are used in order to extend the results to other geometric configurations:

- q_{bk}/q_o : ratio between buckling shear flow of panel with hole and panel without hole.
- q_{bf}/q_{bk} : ratio between buckling shear flow of panel with flanged hole and panel with no flanged hole of the same inner diameter
- D/H : hole diameter – panel width ratio
- L/H : panel length – panel width ratio
- L_v/D : reinforcement side length – hole diameter ratio
- L_h/D : for elongated holes, ratio between the hole long length and the hole short length
- D/d : hole diameter – flange height ratio

Notice that panel thickness is not considered in the previous parameters. Thickness influence is eliminated as buckling loads between panels of the same laminate are compared.

6.1 Flat webs with flat lightening hole

Two lay-ups have been considered for the study: $(30/60/10)_s$ and $(10/80/10)_s$, both of them in group V CASA tape (AS4/8552). Following figures show the correction factor for buckling shear flow due to the presence of a hole. Figure 6-1 applies for square panels whereas Figure 6-2 applies for rectangular ones. Hole elongation factor goes from 1.0 (circular hole) to 2.0.

As it can be observed, the presence of a hole produces a decrease on the buckling shear flow, which becomes null when the hole edge reaches any of the panel borders. The buckling load decreases also as the hole elongation's factor increases, since a greater lost of material is produced.

Behaviour differences between both materials are completely negligible, which shows that the material and stacking influences are considered when the buckling allowable flows for the panel without hole are calculated.

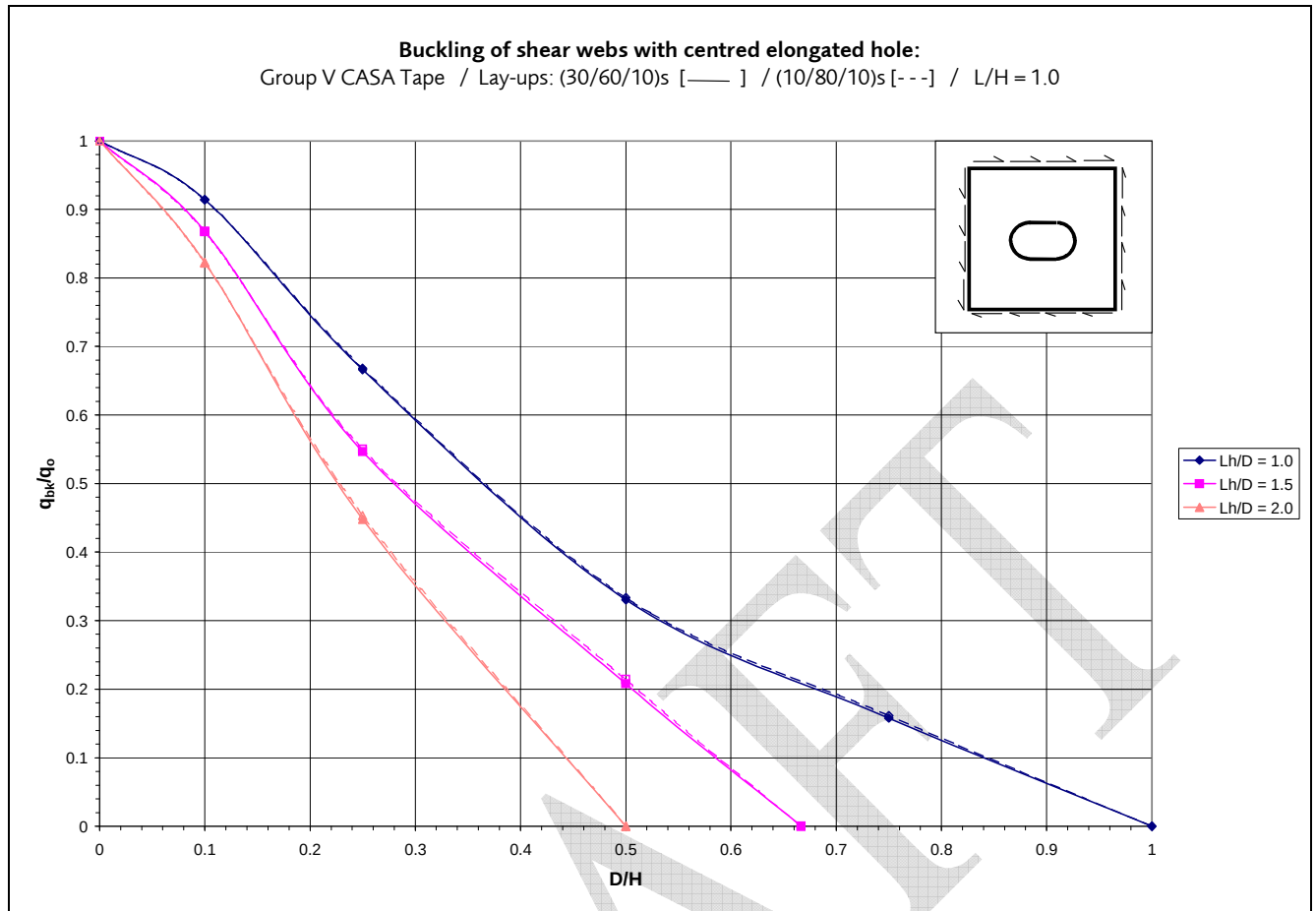


Figure 6-1

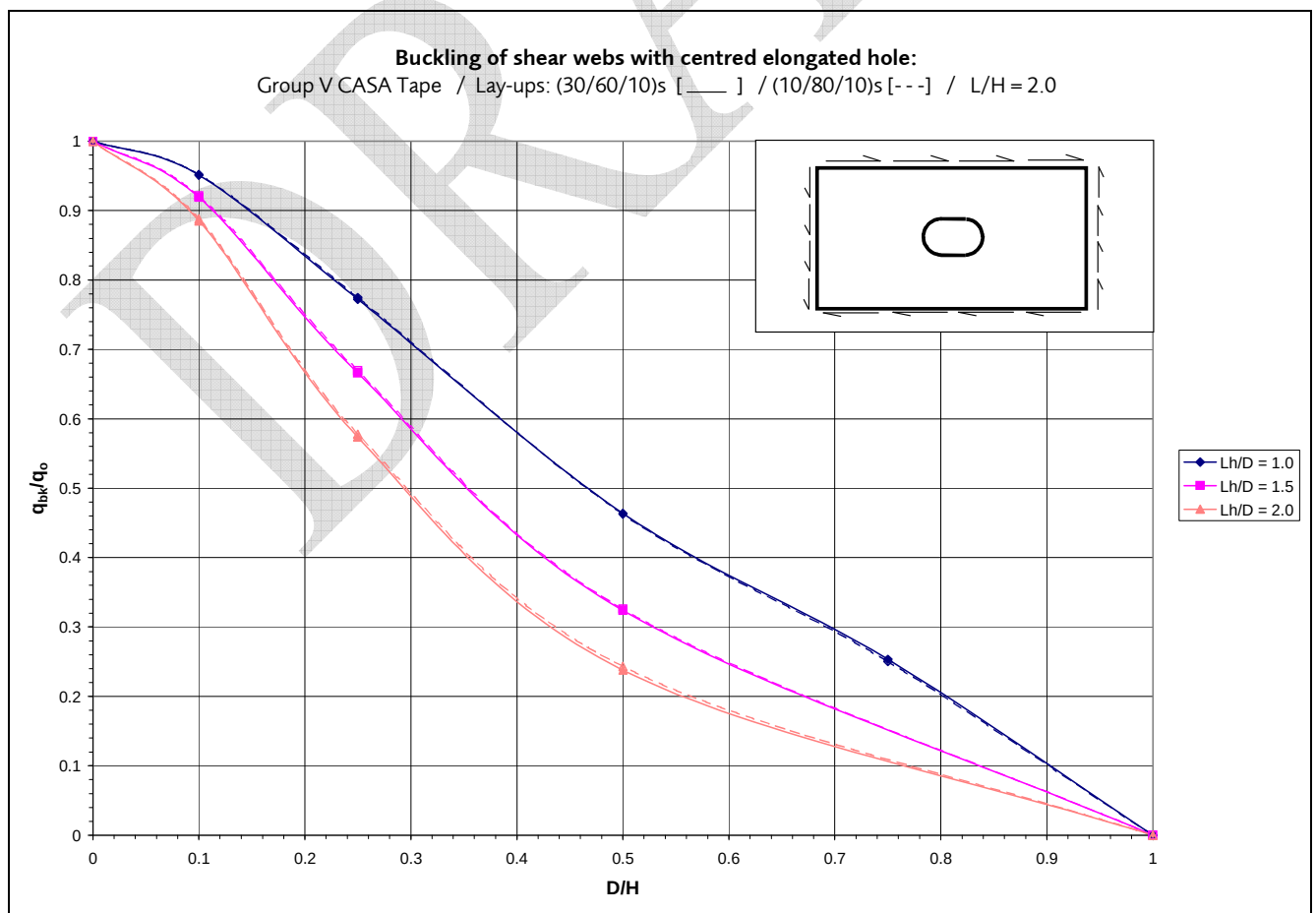


Figure 6-2

Finally, as complementary information, curves are issued for circular holes for $(0/40/60)_s$ and $(0/80/20)_s$ Group V tape lay-ups (AS4/8552) and $(40/60)_s$ Group V fabric lay-up (AGP 193 – PW / 8552) (Figure 6-3).

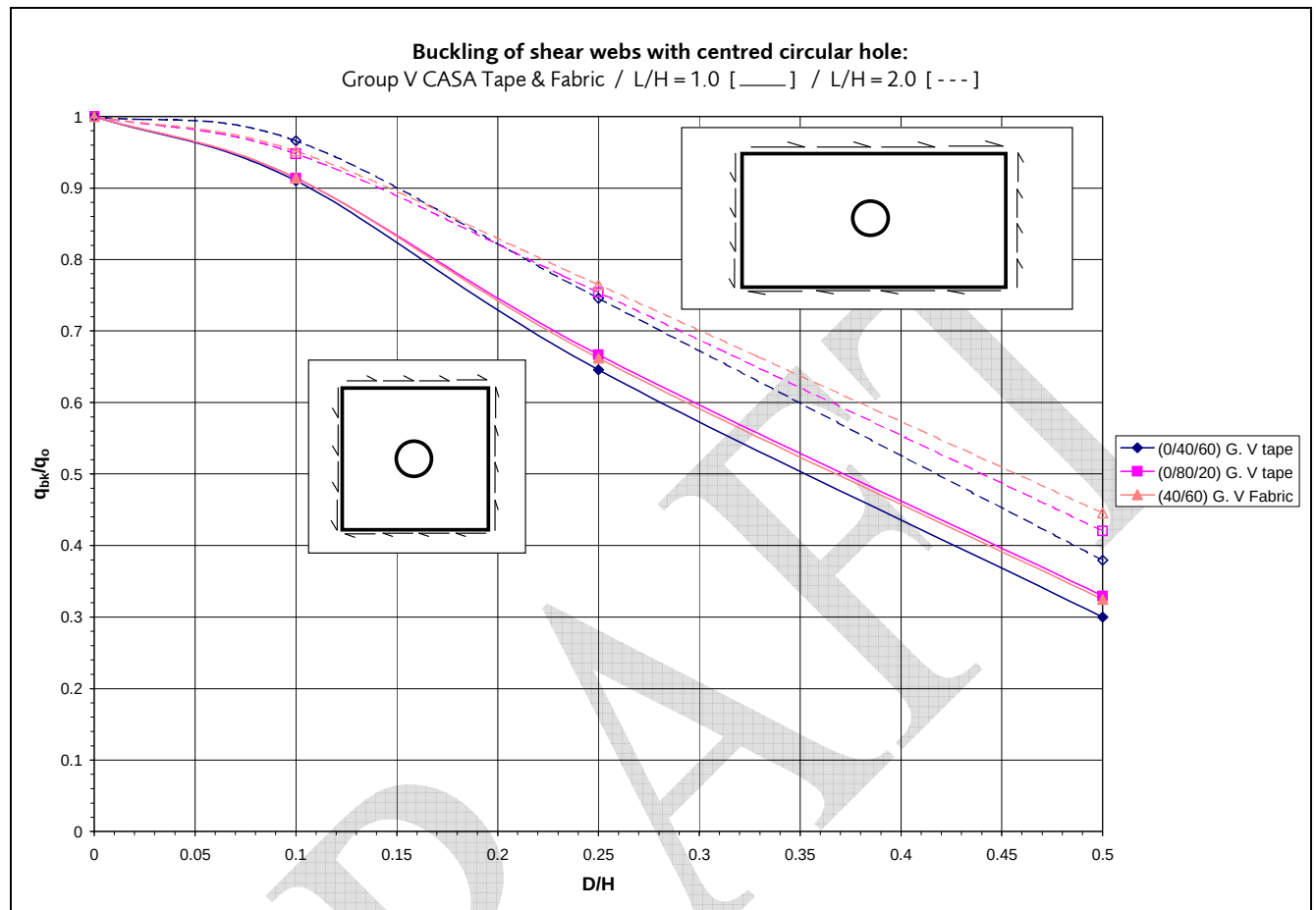


Figure 6-3

6.2 Beaded flat webs with elongated lightening hole

The shape of these curves is very similar to the ones corresponding to the square panel delimited by the beads (see Figure 6-1). Moreover, for both elongation factors, buckling load increases slightly compared with the respective square configuration. Differences grow as D/H is incremented. Anyway, the buckling load ratio obtained is lower than the buckling load ratio for a rectangular panel of the same dimensions. This means that beads make the panel behave as if only the web between beads were effective. In that way, hole size related to the effective web becomes bigger than related to the real web, reducing the buckling load ratio between panel with hole and without hole.

Results obtained are shown in Figure 6-4. In this figure, q_0 stands for the buckling load of a rectangular panel of the same dimensions and identical beads configuration. Once again, there are not significant differences between both materials.

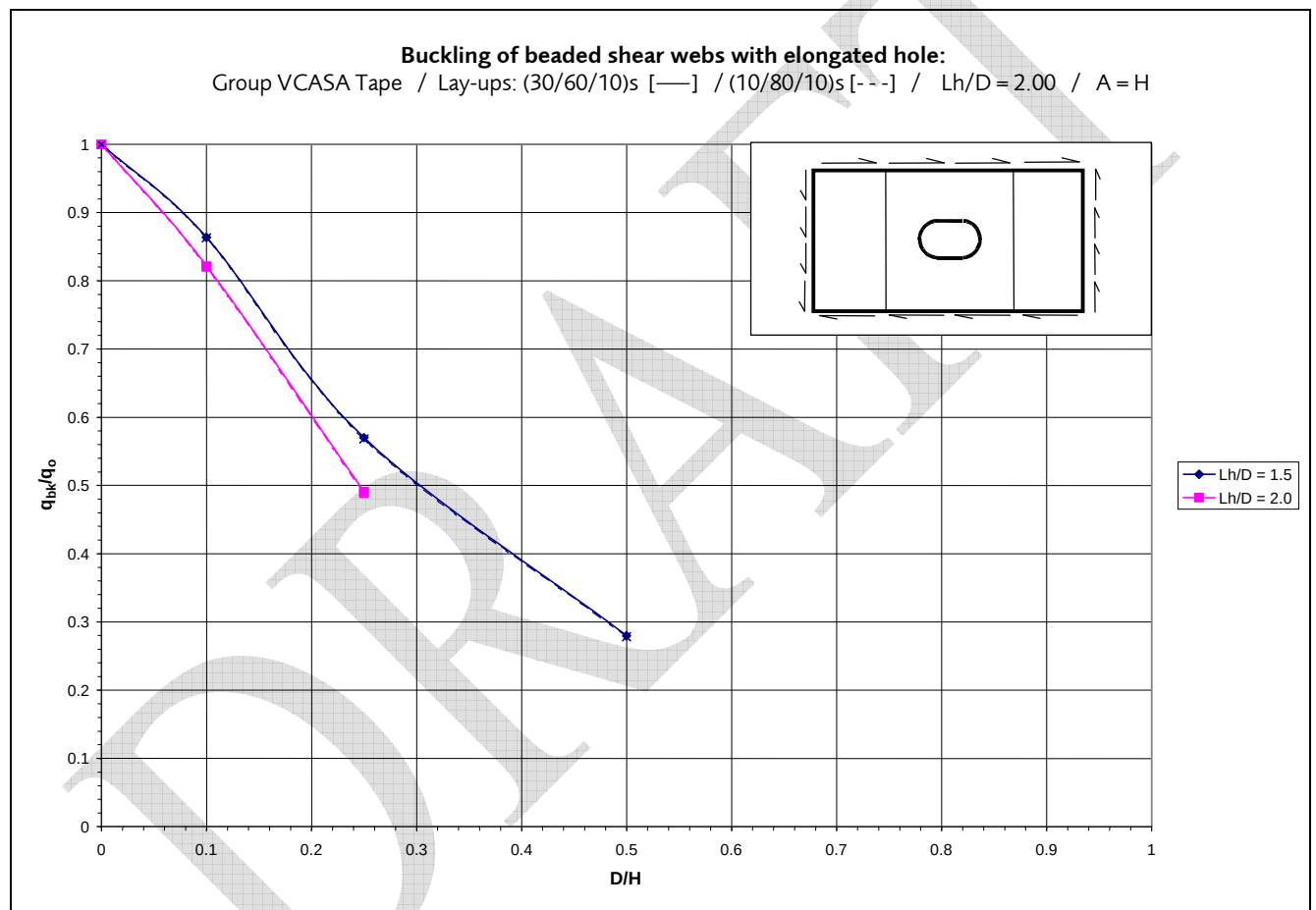


Figure 6-4

6.3 Reinforced webs with circular lightening hole

The presence of the reinforcement changes drastically the buckling load evolution with the D/H parameter, as it can be seen in Figures 6-5 and 6-6. This effect was to be expected since the reinforcement area increases with the hole size. For a square panel and a reinforcement parameter of $L_r/D = 1.5$, (Figure 6-5), the reinforcement effect makes the buckling load increase quickly until the size of the hole is big enough to weaken this influence and decrease the curve slope. The same behaviour is expected for the other configurations, but only the initial phase (steep increase) is detected in the analysed range of the parameters. Notice also that reinforcement influence is bigger in square panels than in rectangular ones.

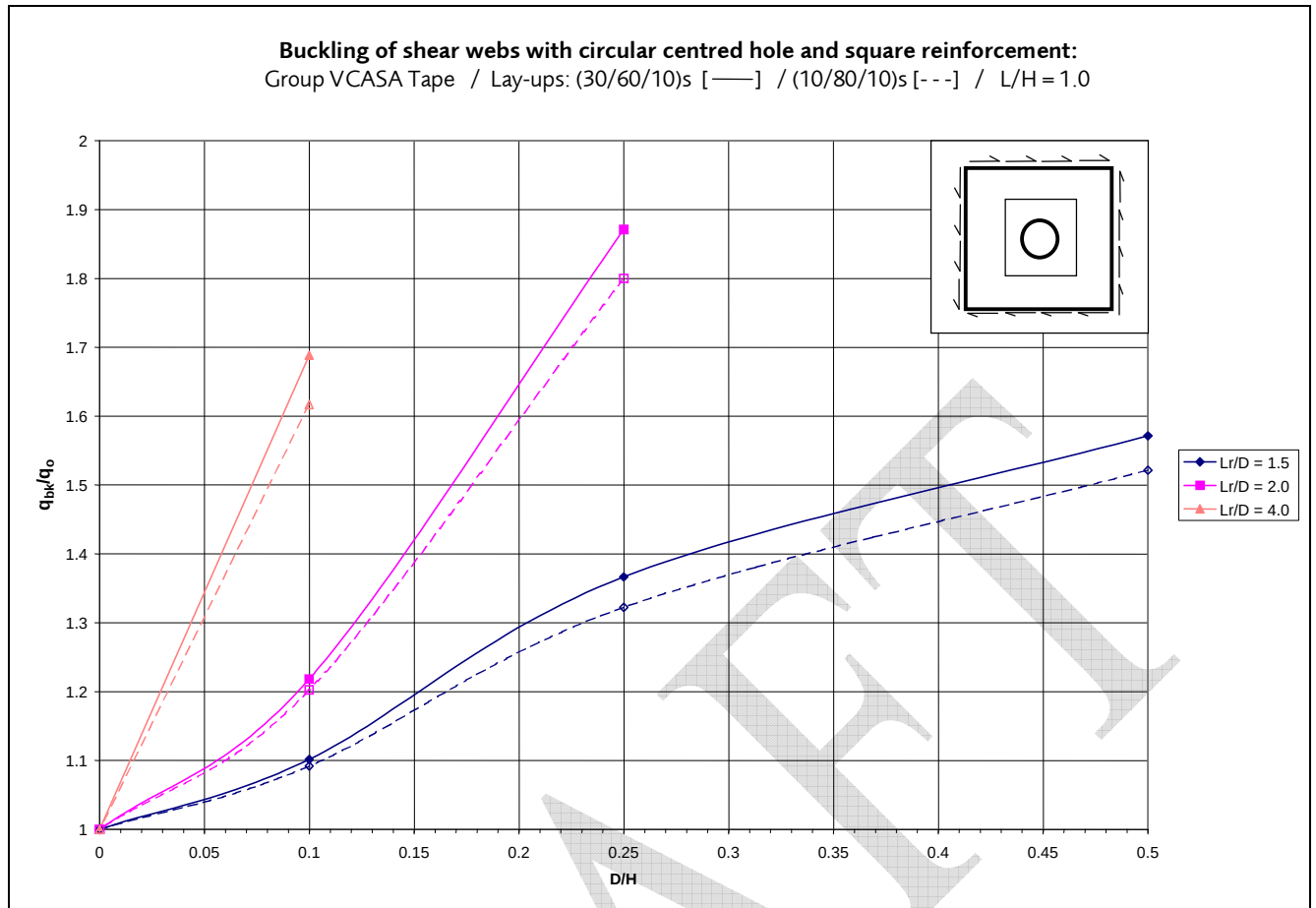


Figure 6-5

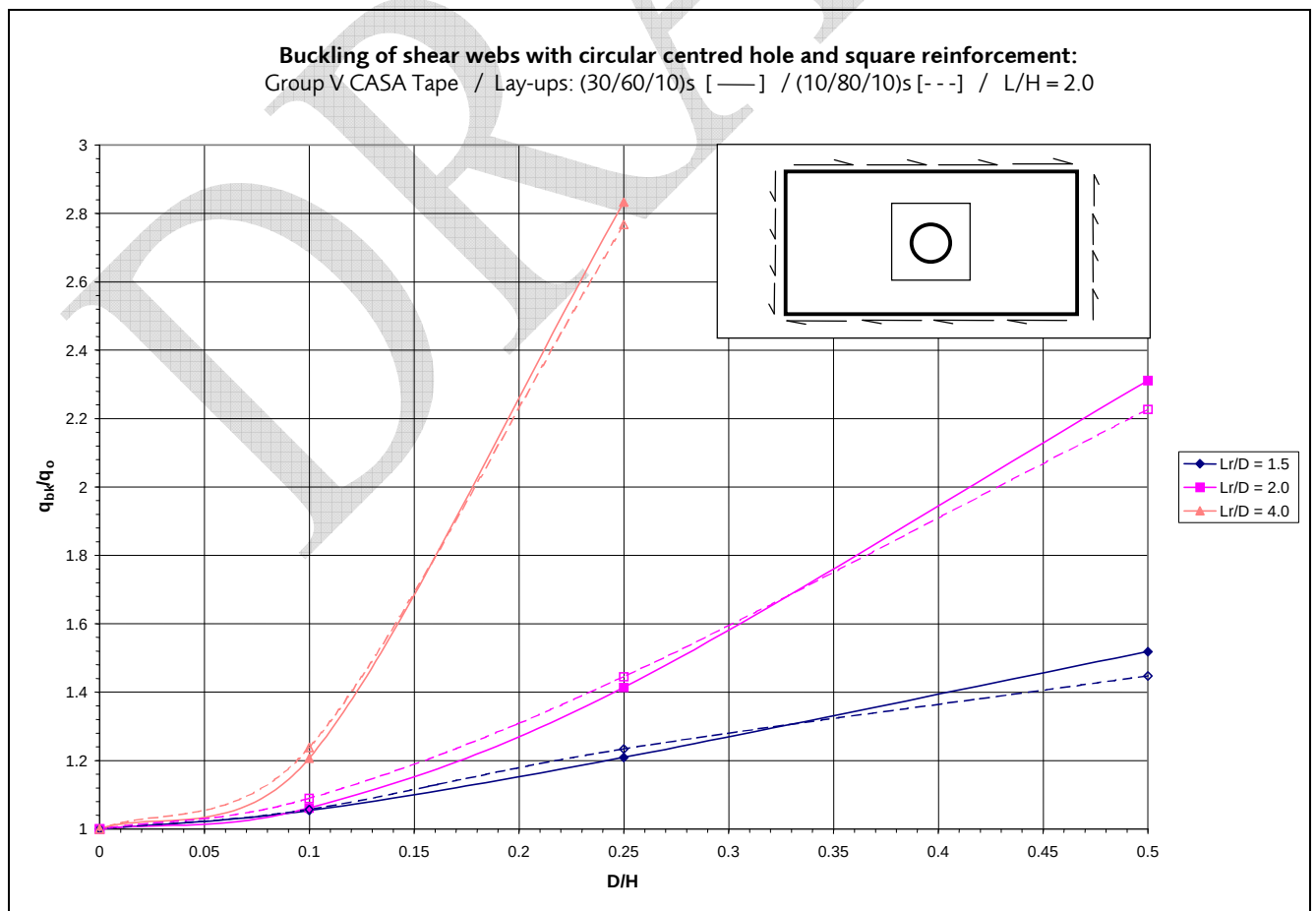


Figure 6-6

6.4 Flat webs with flanged circular holes

In this case, three different materials have been studied. By the one hand, two laminates of group V CASA tape (AS4/8552) are considered: $(0/40/60)_s$ and $(0/80/20)_s$. In the other hand, a laminate of group V CASA fabric (AGP193-PW/8552) is considered: $(40/60)_s$. This is a commonly used lay-up for ribs, where flange reinforced holes are a usual solution.

The buckling load evolution is similar than the one observed for reinforced panels. Nevertheless, flanges are not as effective as thickness reinforcements.

A much better behaviour is observed for the fabric laminate than for the tape ones. This is mainly due to the presence of 0° plies, which makes the element much more stable. For this reason, separated charts are issued for tape and for fabric laminates.

Figure 6-7 shows results obtained for square panels of tape laminate. As it can be observed, for low values of D/H , the effect of the increased stiffness due to the flange is greater than the effect of the hole, so buckling shear flow grows initially. When D/H reaches a critical value, the hole effect becomes more important and the buckling load slows down its growing or, even, begins to decrease. Similar behaviour is observed for other configurations and materials (see Figures 6-8 to 6-10).

It is also interesting to analyse the isolated effect of the flange. For that purpose, a comparison has been made between panels with flanged hole (q_{bk}) and panels with no flanged hole of the same inner diameter (q'_{bk}). As it can be seen from Figure 6-11 to 6-14, buckling shear flow increases with D/H for the same flange height.

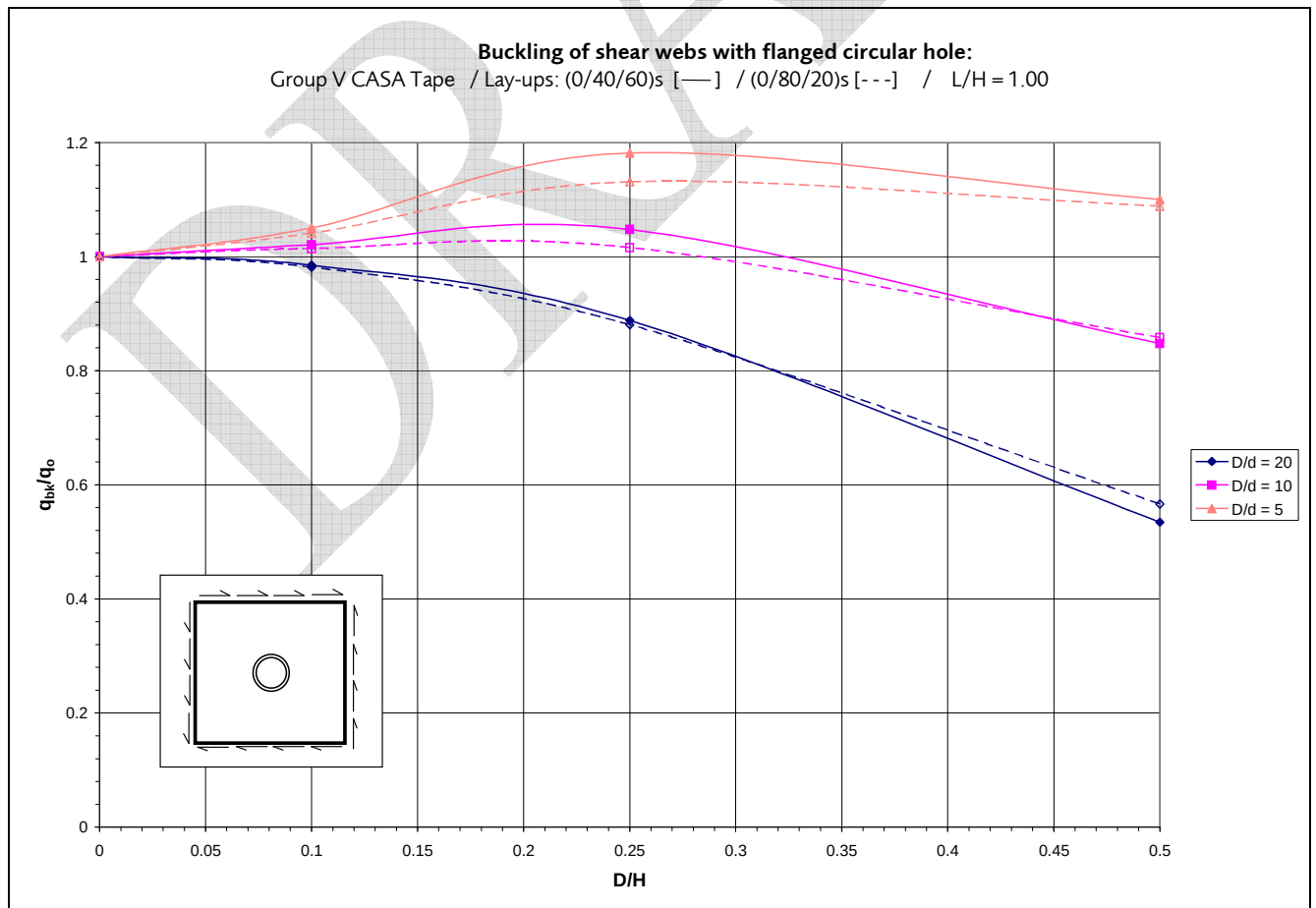


Figure 6-7

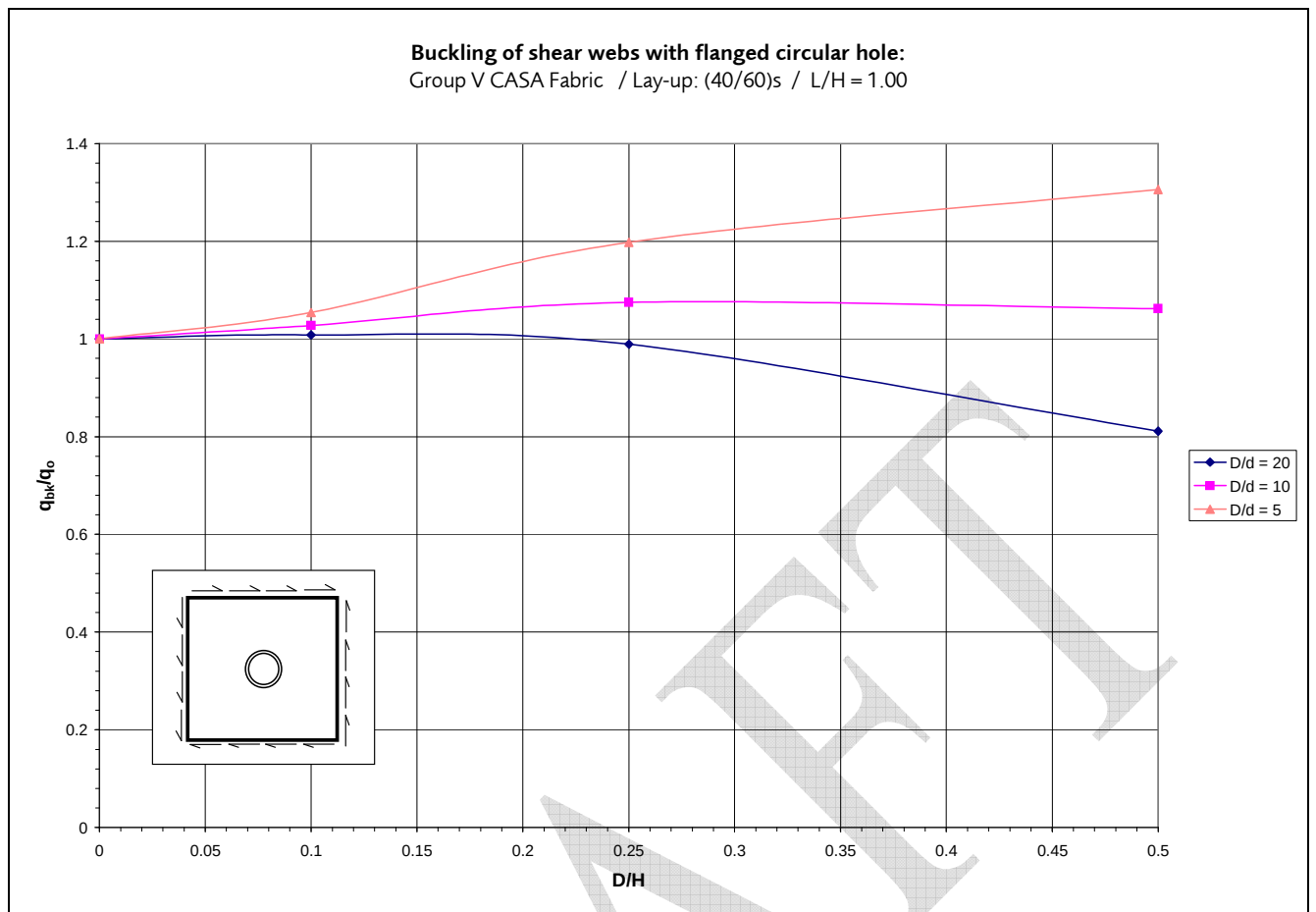


Figure 6-8

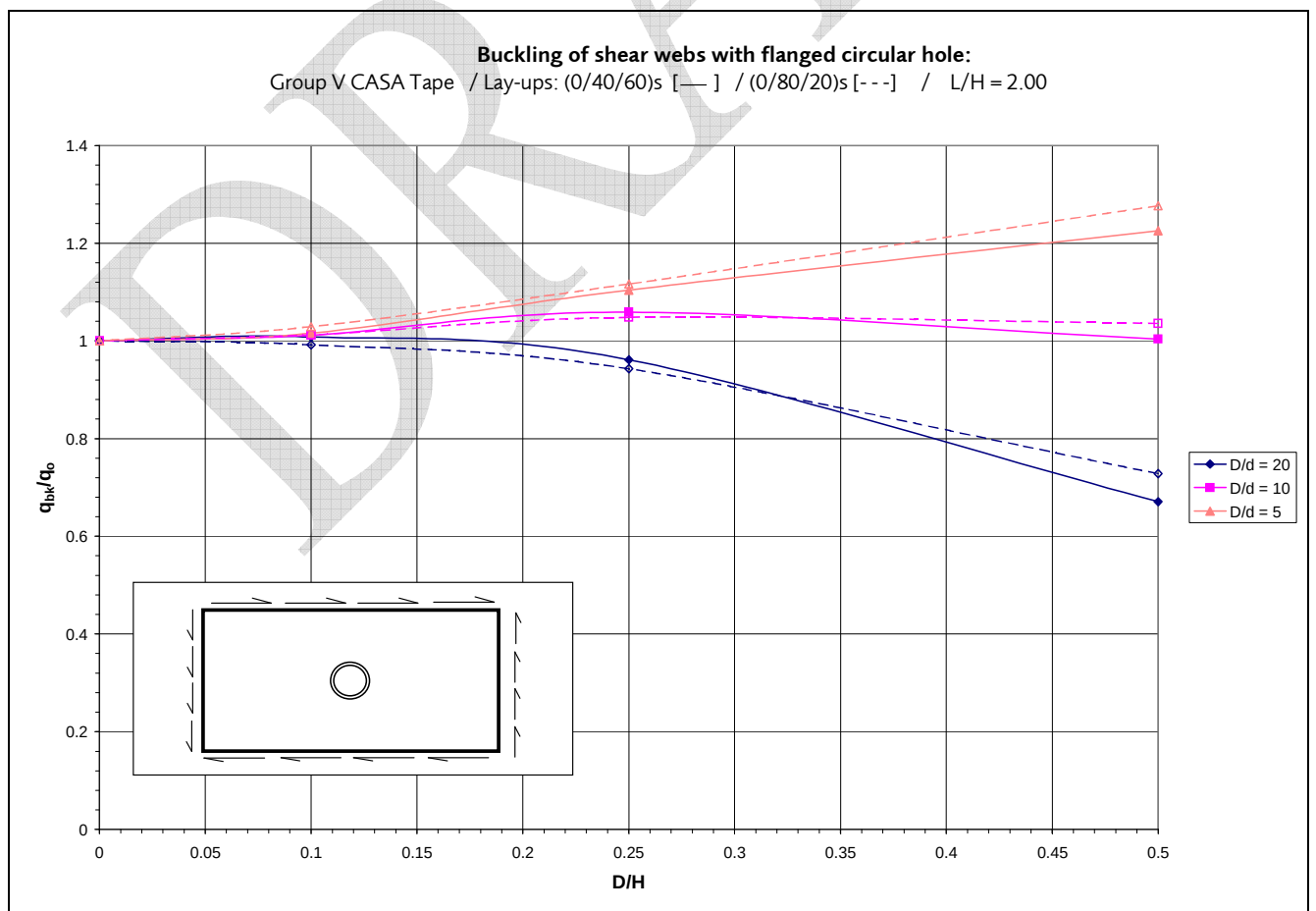


Figure 6-9

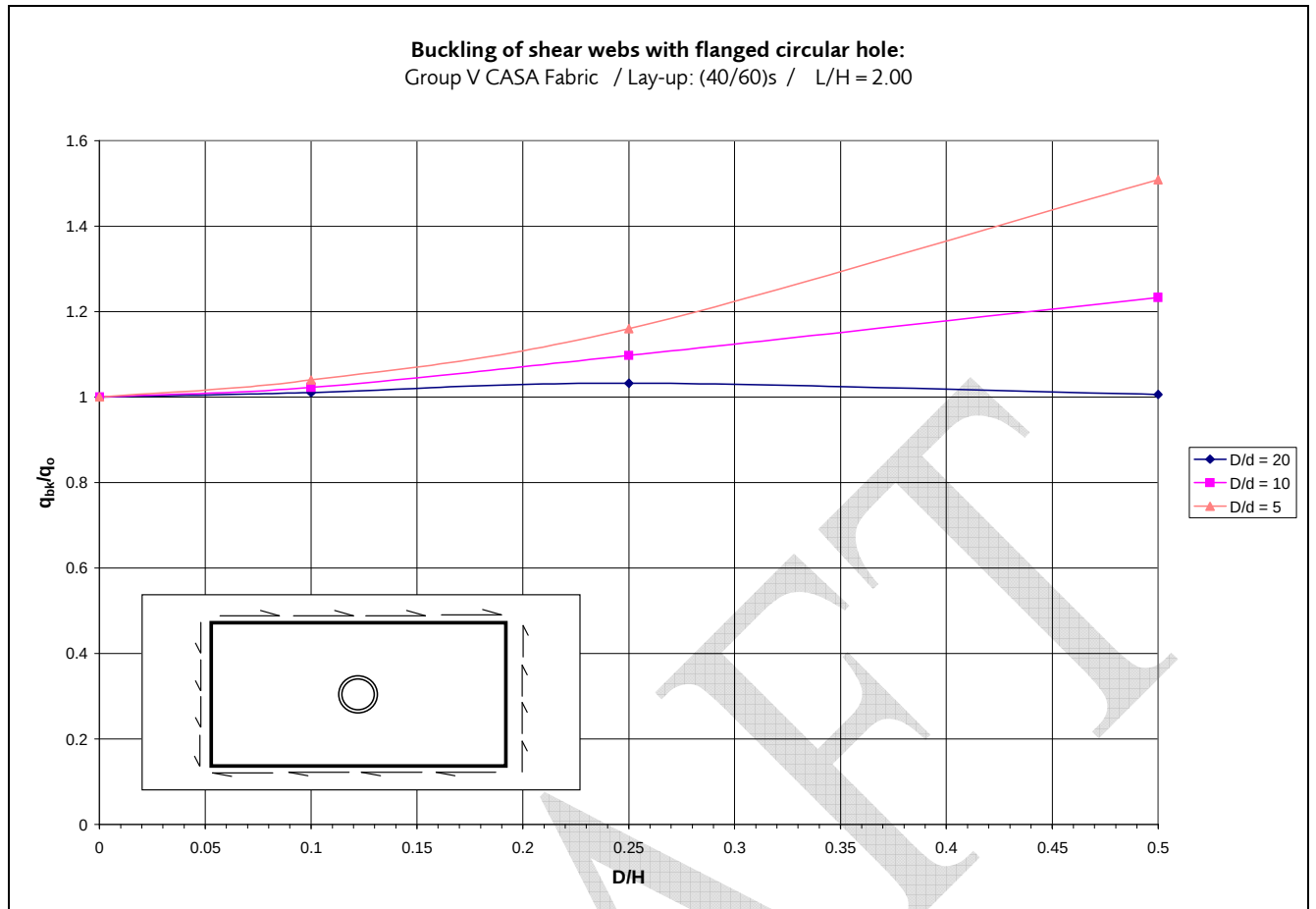


Figure 6-10

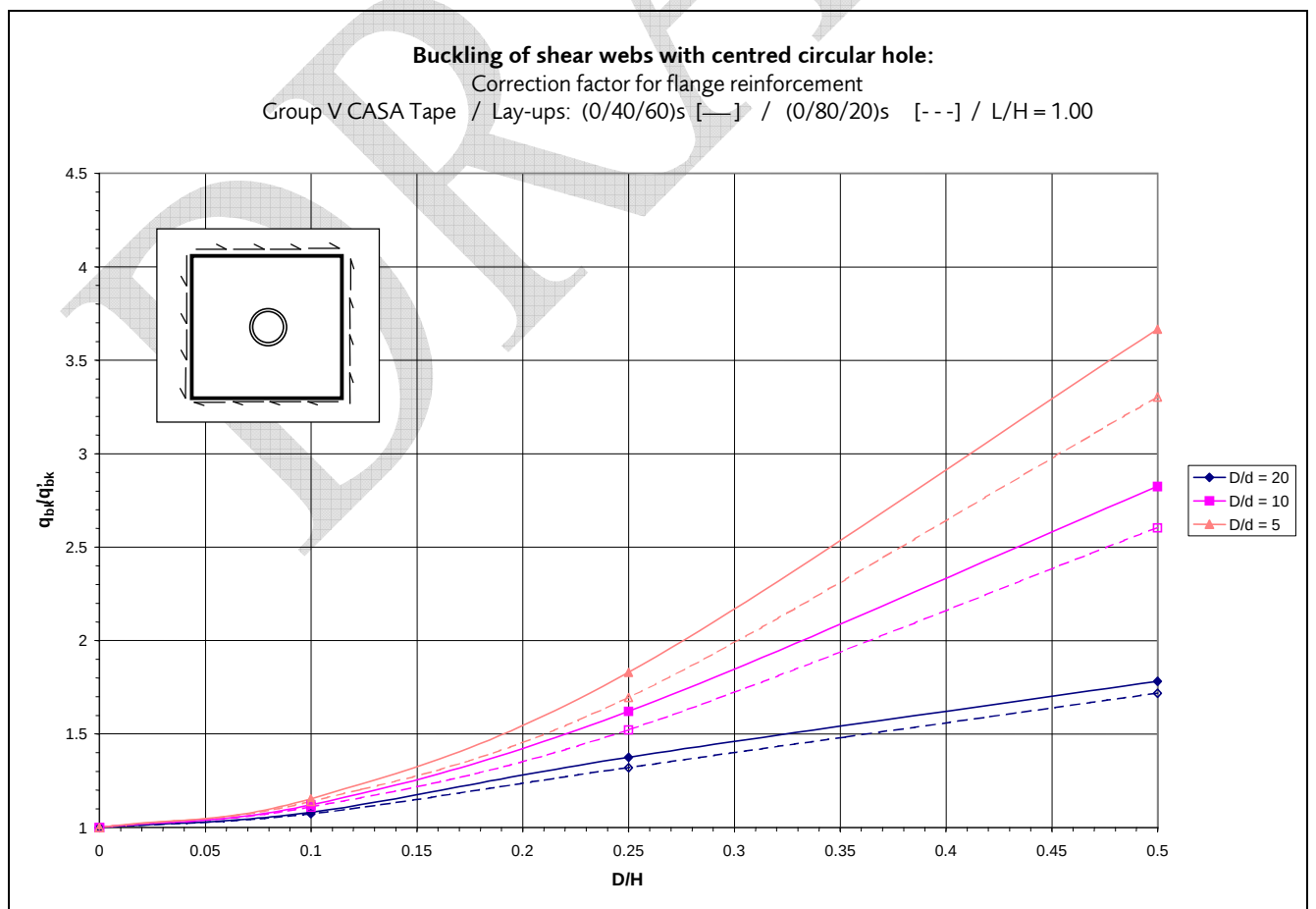


Figure 6-11

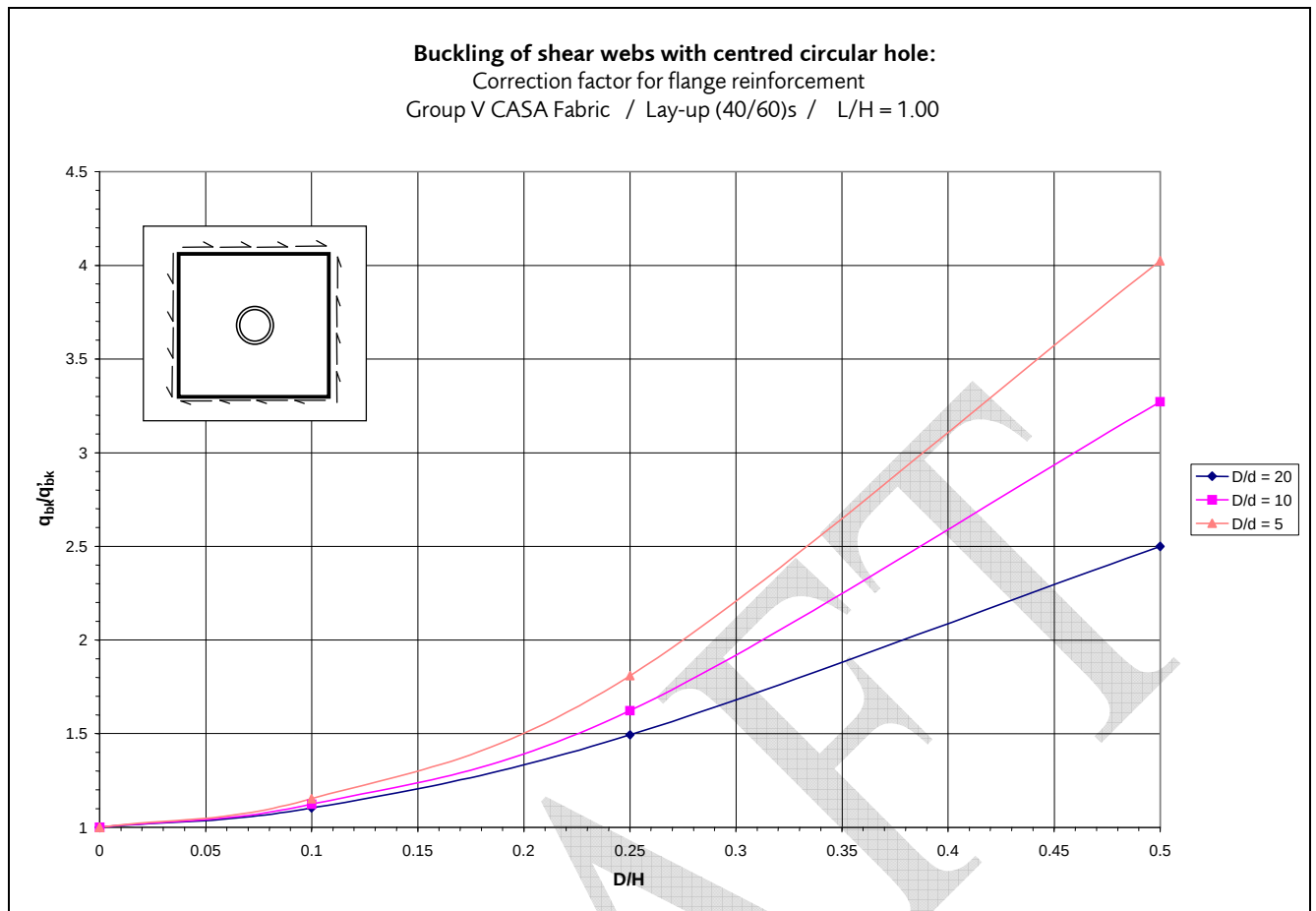


Figure 6-12

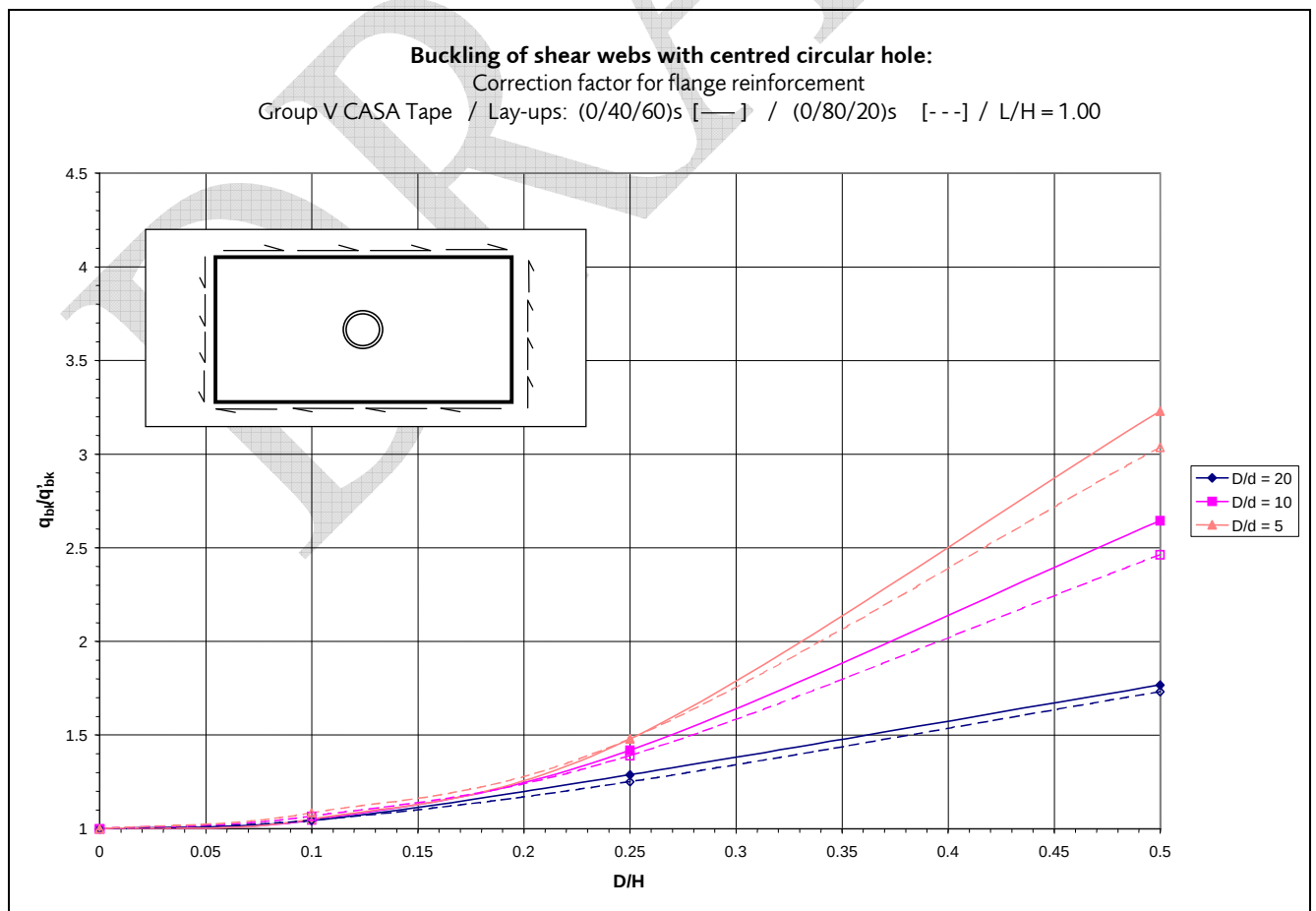


Figure 6-13

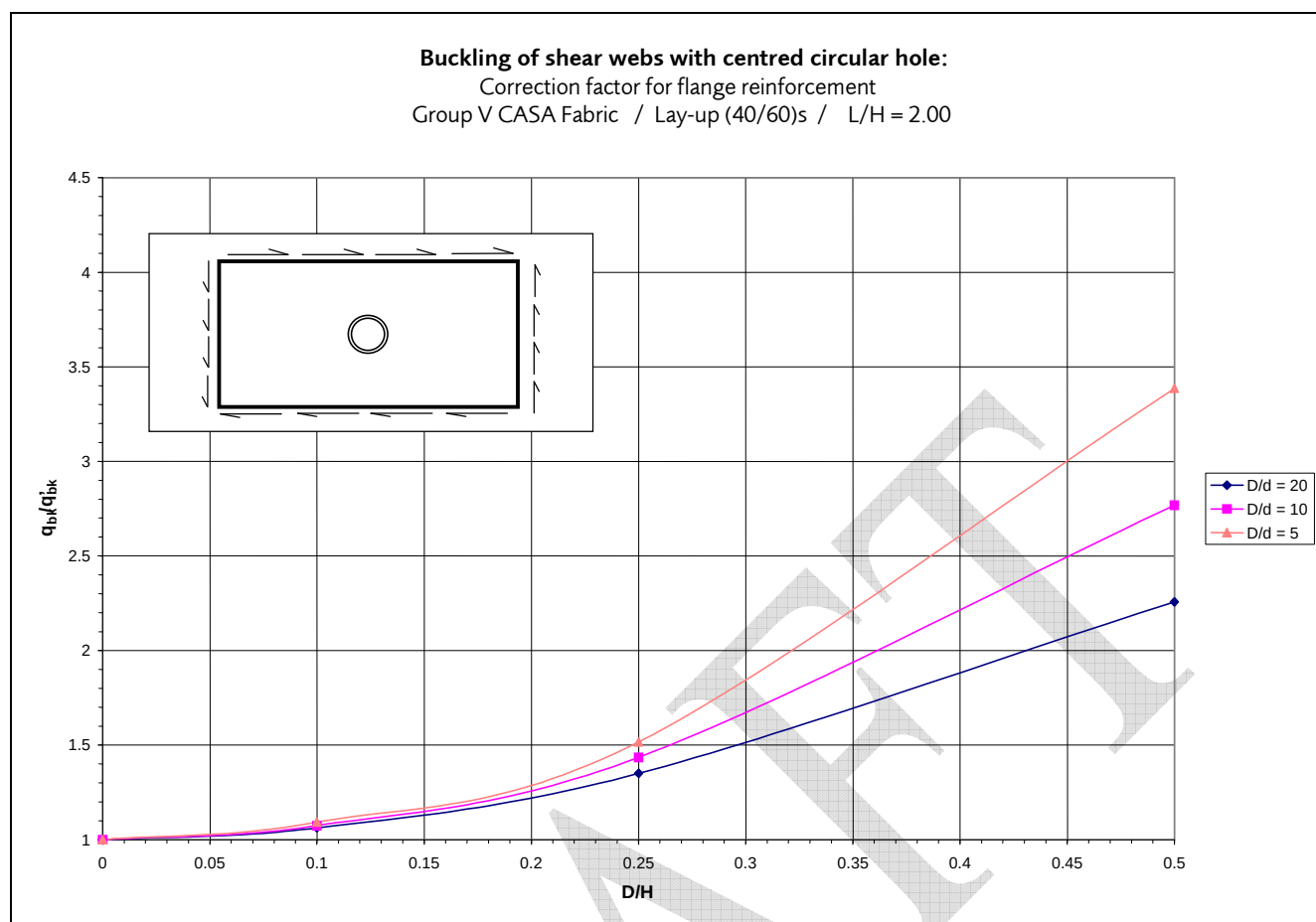


Figure 6-14

7 Strength checking design curves

In order to obtain design curves to carry out the strength checking, maximum strain at hole edge has been compared with a reference shear strain γ_{ref} . This reference shear strain corresponds to the applied shear flow (100 N/mm), the laminate shear modulus (obtained from ANCOLAB) and the panel thickness. As for buckling design curves, non-dimensional parameters have been used:

- ε/γ_{ref} : ratio between maximum strain at hole edge and reference shear strain, where

$$\gamma_{ref} = \frac{100}{G t}$$
- $\varepsilon_{fl}/\varepsilon_{n_fl}$: ratio between maximum strain at panel with flanged hole and panel with no flanged hole of the same inner diameter
- D/H: hole diameter – panel width ratio
- L/H: panel length – panel width ratio
- L_v/D : reinforcement side length – hole diameter ratio
- L_h/D : for elongated holes, ratio between the hole long length and the hole short length
- D/d: hole diameter – flange height ratio

From these curves, maximum strain at the panel is easily obtained so it can be compared with the allowable strain of the laminate.

The same laminates as for buckling analysis have been considered for each configuration. Reference shear strain for each material is detailed below:

- $(30/60/10)_s \Rightarrow (G = 21884 \text{ N/mm}^2; t = 3.68 \text{ mm}) \Rightarrow \gamma_{ref} = 1242 \mu\epsilon$
- $(10/80/10)_s \Rightarrow (G = 27863 \text{ N/mm}^2; t = 3.68 \text{ mm}) \Rightarrow \gamma_{ref} = 975 \mu\epsilon$
- $(0/40/60)_s \Rightarrow (G = 15906 \text{ N/mm}^2; t = 3.68 \text{ mm}) \Rightarrow \gamma_{ref} = 1708 \mu\epsilon$
- $(0/80/20)_s \Rightarrow (G = 27863 \text{ N/mm}^2; t = 3.68 \text{ mm}) \Rightarrow \gamma_{ref} = 975 \mu\epsilon$
- $(40/60)_s \Rightarrow (G = 17931 \text{ N/mm}^2; t = 2.05 \text{ mm}) \Rightarrow \gamma_{ref} = 2720 \mu\epsilon$

7.1 Flat webs with flat lightening hole

Figures 7-1 and 7-2 show the strain concentration evolution with D/H for square and rectangular shear webs, respectively. Circular holes and two elongated holes are represented. As it was expected, strain level increases with D/H and with elongation factor. Notice that behaviour differences between both materials are much more noticeable than in the buckling case, especially for rectangular panels. Values for D/H = 0.0 have been obtained from EH program, (Ref. 20), which provides the stress and strain distributions along a circular hole at an infinite orthotropic plate (there is no available solution for D/H = 0.0 in the case of elongated holes).

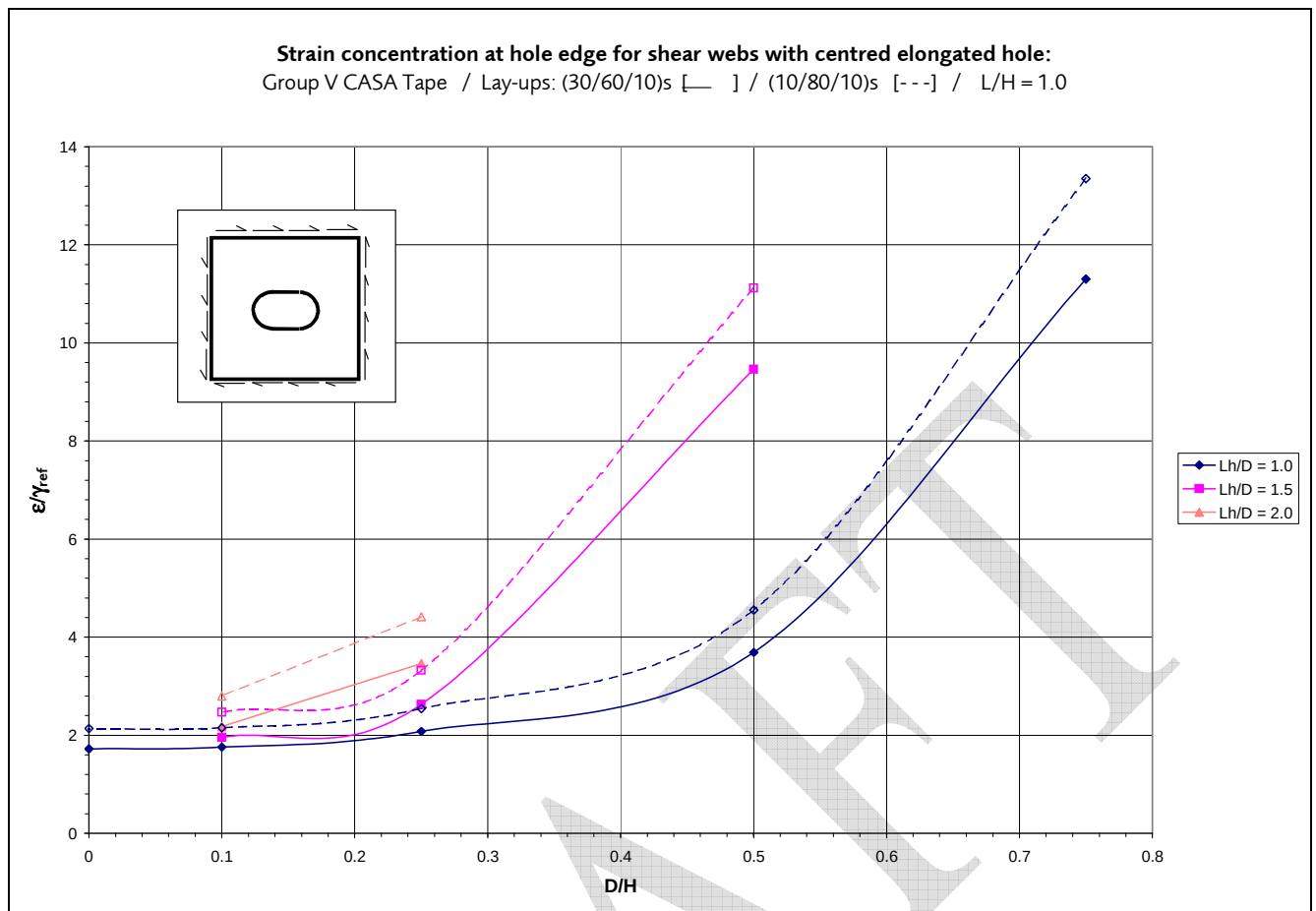


Figure 7-1

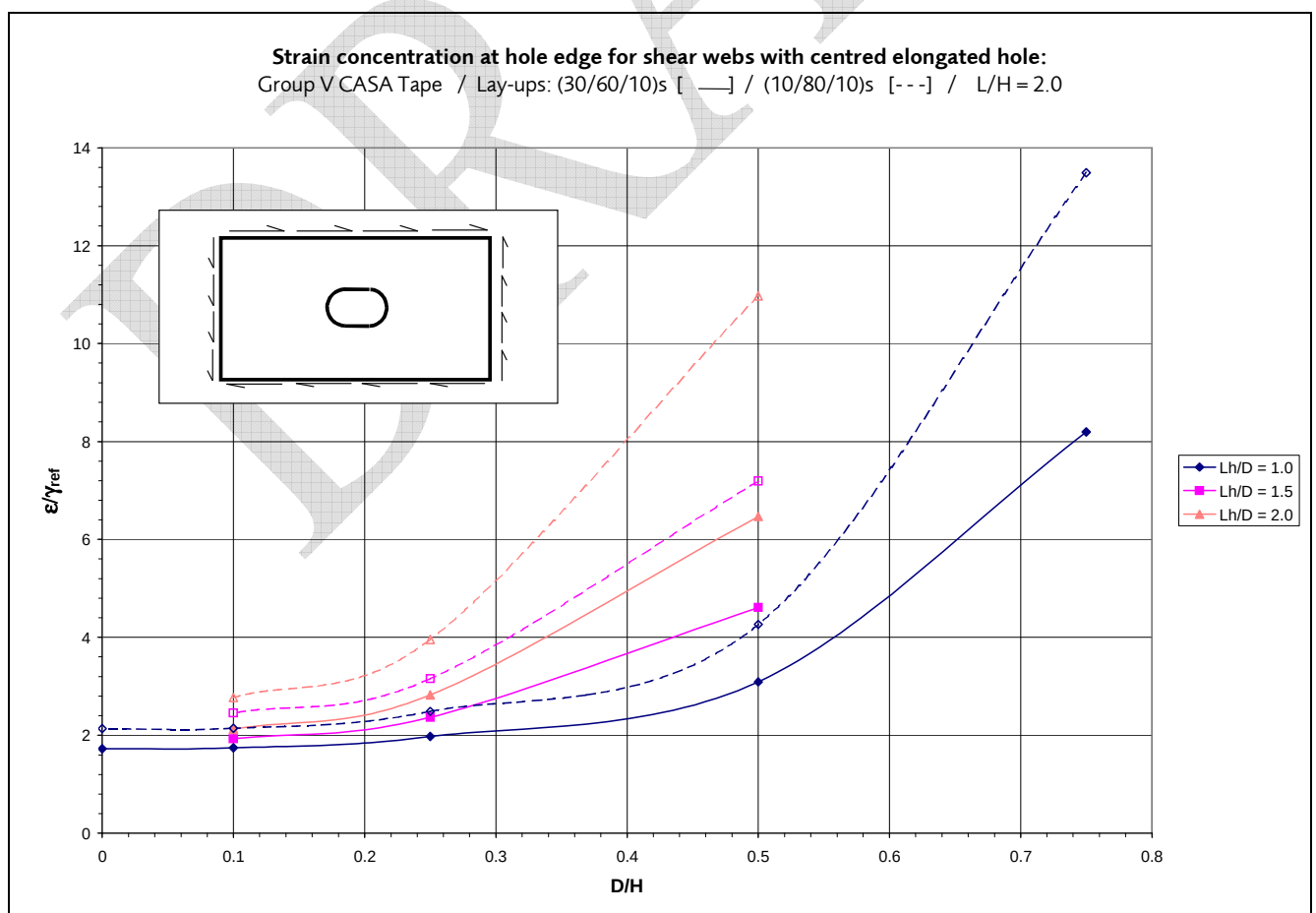


Figure 7-2

7.2 Beaded flat webs with elongated lightening hole

Results for this configuration are exactly the same ones as for rectangular panels without beads. Since the loads are applied in the plane of the plate and there is no flexural-membrane coupling in the laminate, the plate experiences no deformation out of its plane. Thus, the support supplied by the beads has no influence on the behaviour of the structure. (Note that beads are only simulated as supports, with no flexural stiffness at all).

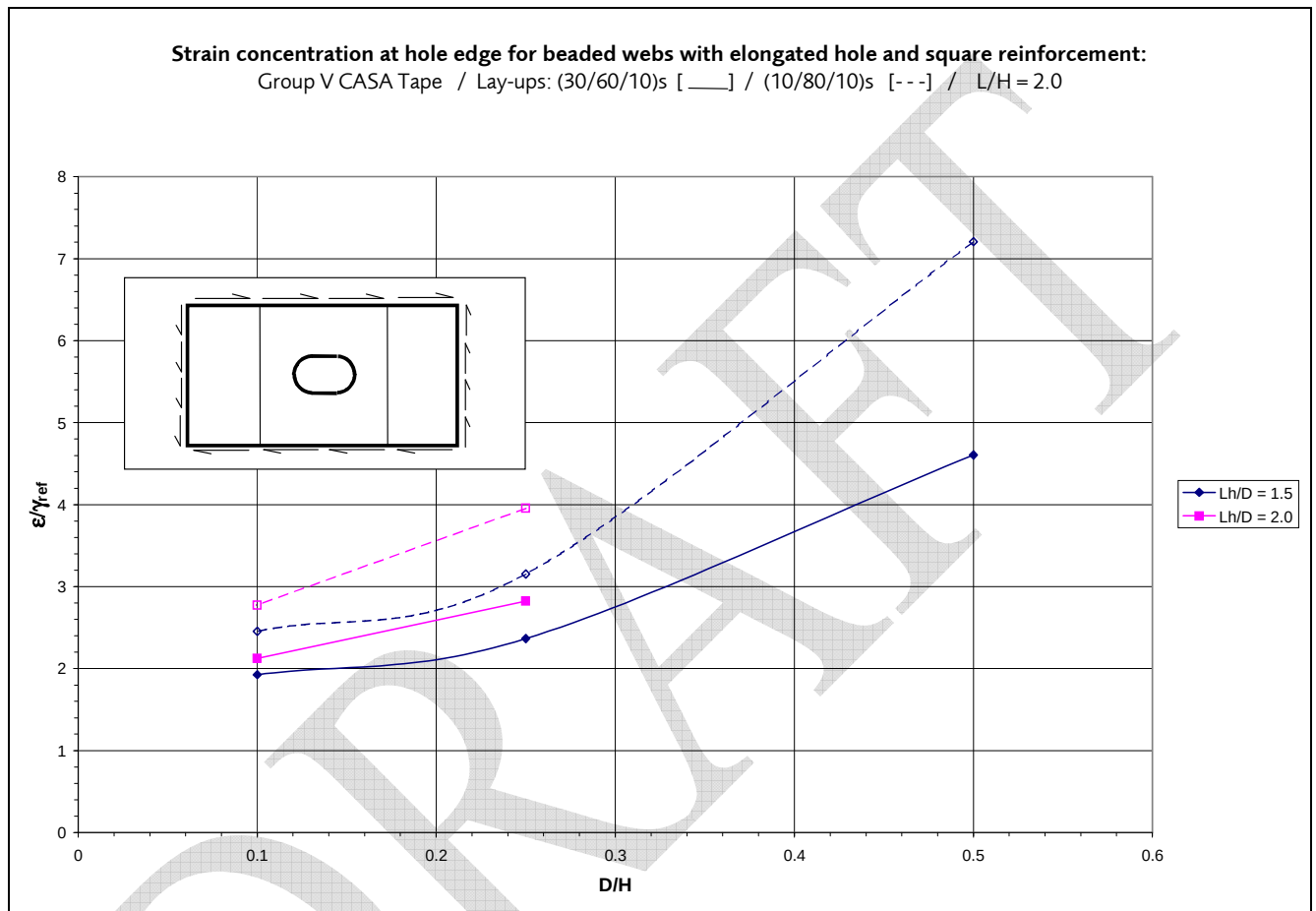


Figure 7-3

7.3 Reinforced webs with circular lightening hole

The presence of the reinforcement is very effective reducing strains concentration, as thickness growth reduces the stress level over the reinforced area. Comparing Figures 7-4 and 7-5 with Figures 7-1 and 7-2 it can be seen that strain level is around 30% of the strain level obtained for the web without reinforcement.

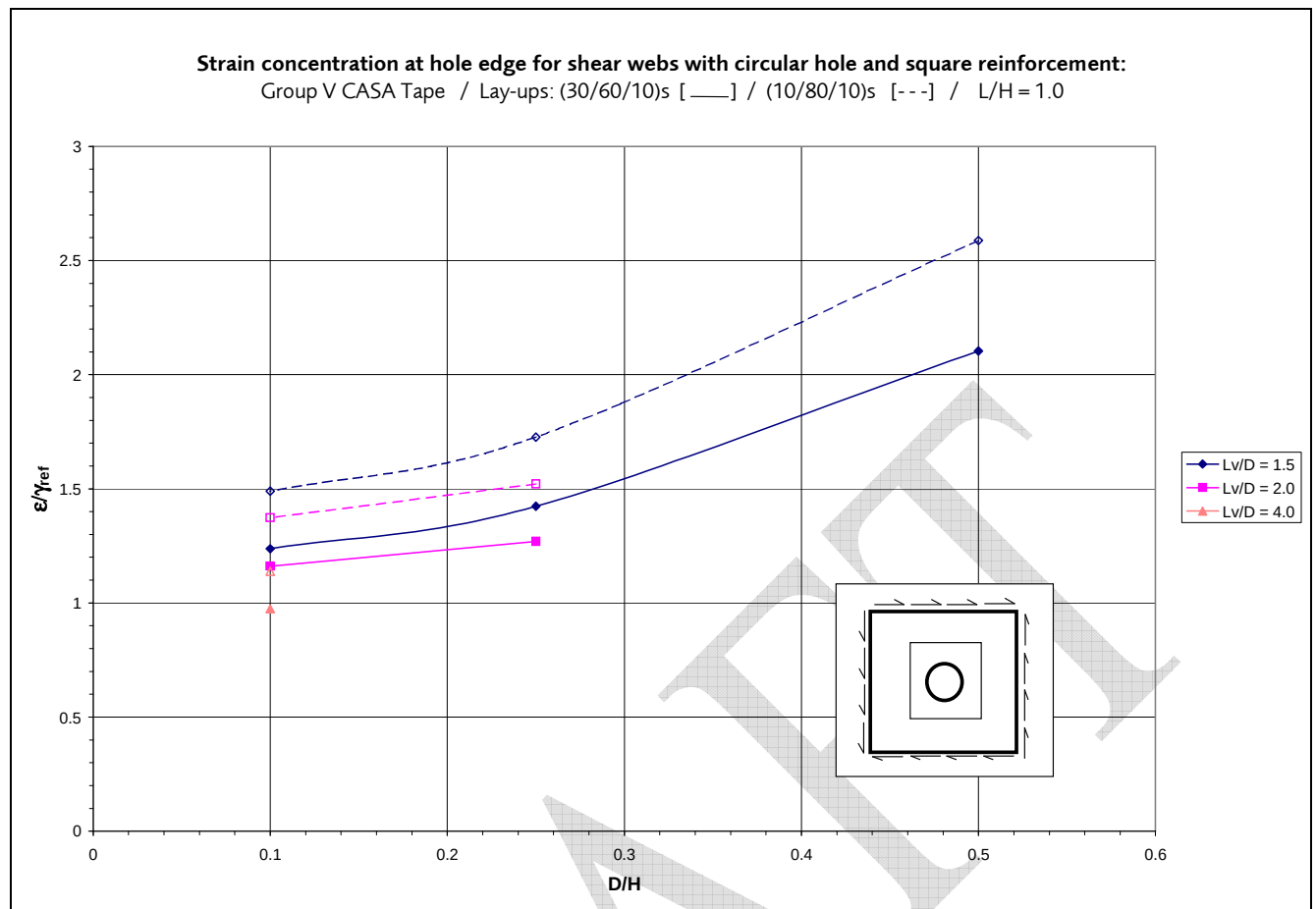


Figure 7-4

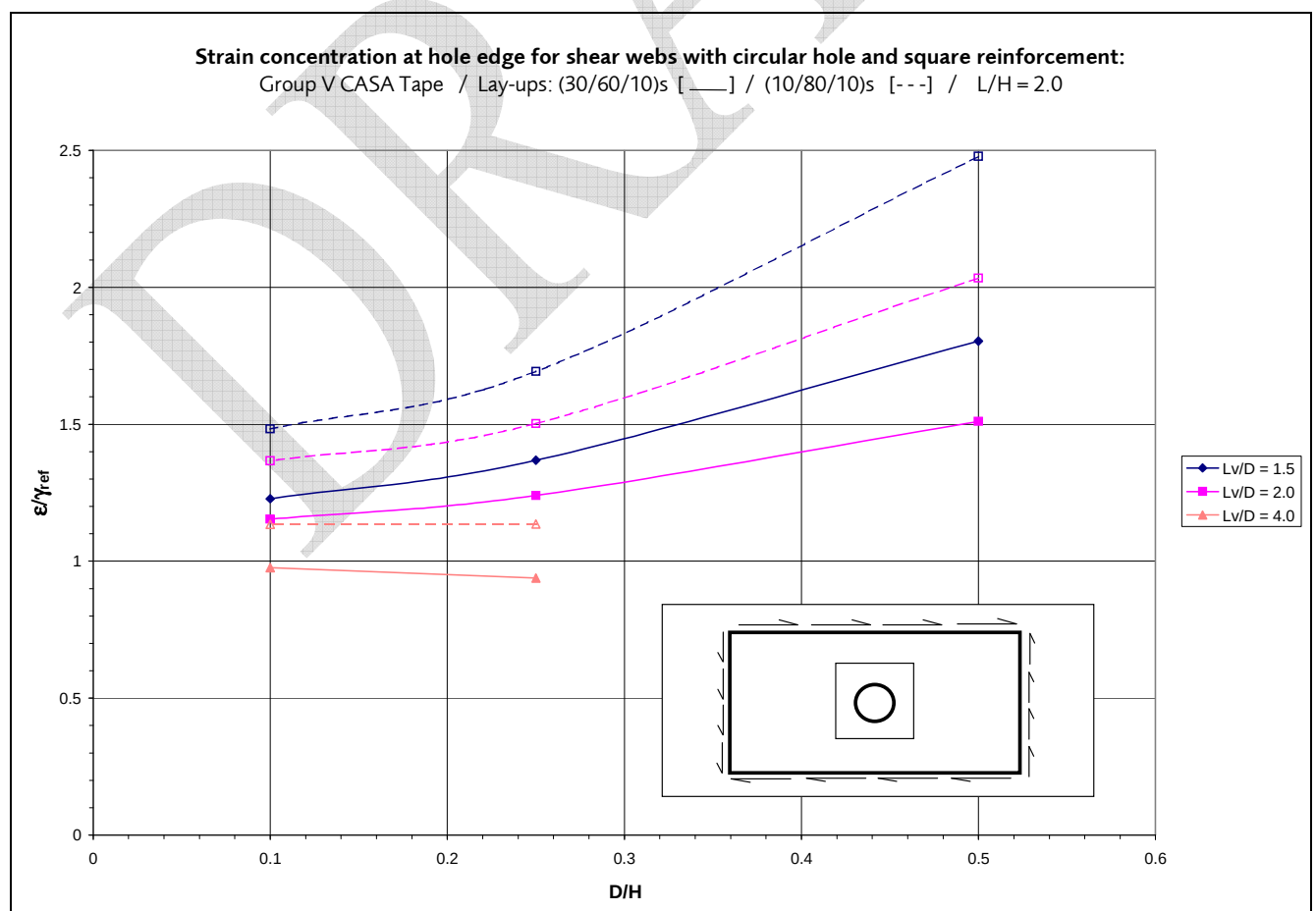


Figure 7-5

7.4 Flat webs with circular flanged holes

As it was already commented, bending introduced by the flange makes maximum strain appear at its base instead of appearing at the hole edge. Figures 7-5 to 7-7 show how the flange reduces the strain level, but it is not as effective as the thickness reinforcement.

Anyway, there is a significant behaviour difference between reinforced panels and flanged panels. Whereas for reinforced panels strains decrease as the reinforced area grows, for flanged panels strains get increased as the flange height becomes greater. Indeed, an increment of the reinforced area reduces the stress level whereas an elevation of the flange height increases bending stresses at the flange base.

As in the buckling analysis, separated charts are issued for each of the tape and fabric laminates. In this case, there are not noticeable differences between square and rectangular panels, so both configurations are represented together.

Once more, it is interesting to analyse the isolated effect of the flange. As before, a comparison has been made between panels with flanged hole and panels with no flanged hole of the same inner diameter. Curves for each material are represented in Figures 7-8, 7-9 and 7-10. It can be seen that there is a critical value of the flange height for each material for which strain concentration keeps always lower than the strain concentration of the non-flanged hole. In some cases, strain level is almost reduced to a 60 % of the original value.

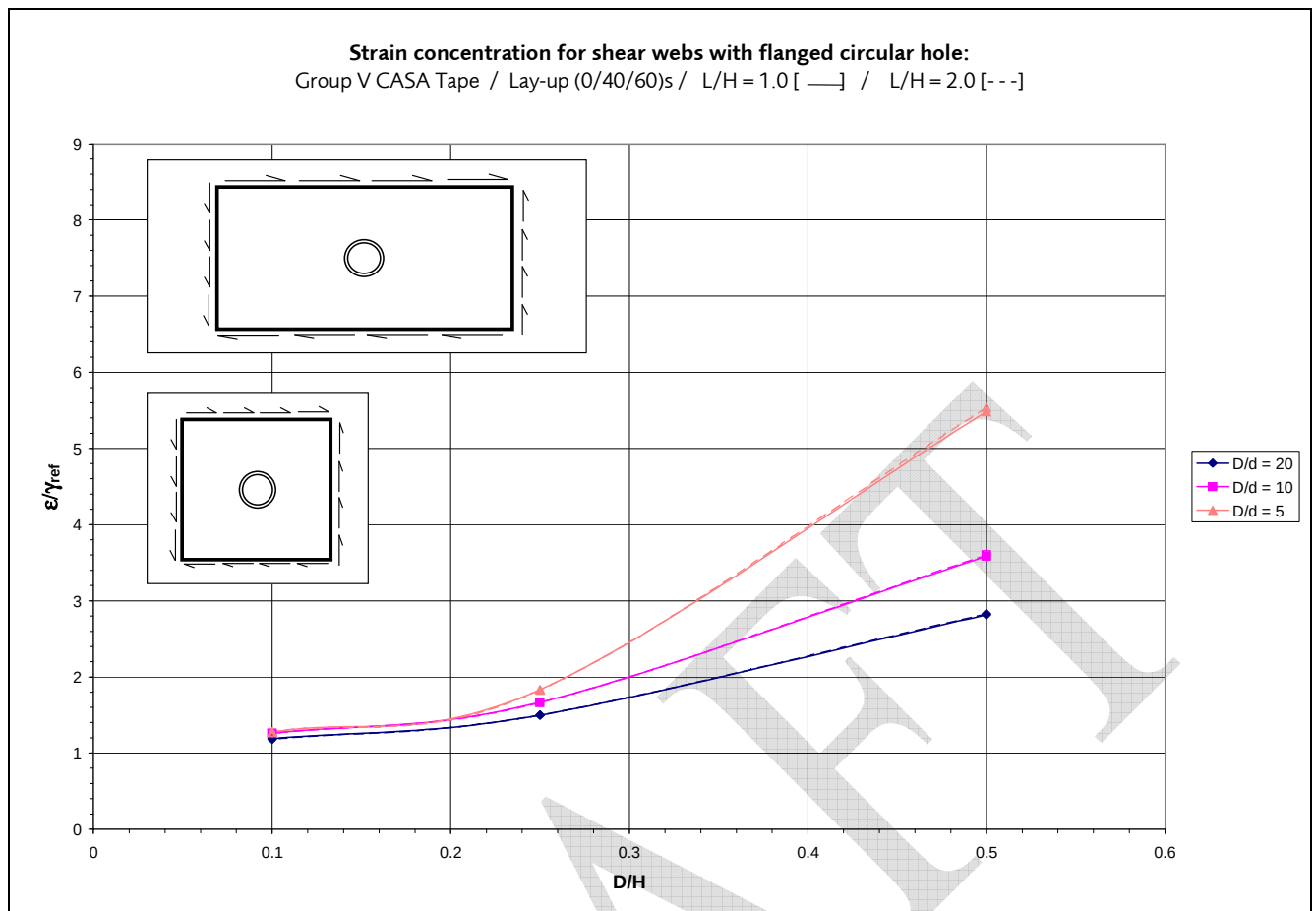


Figure 7-5

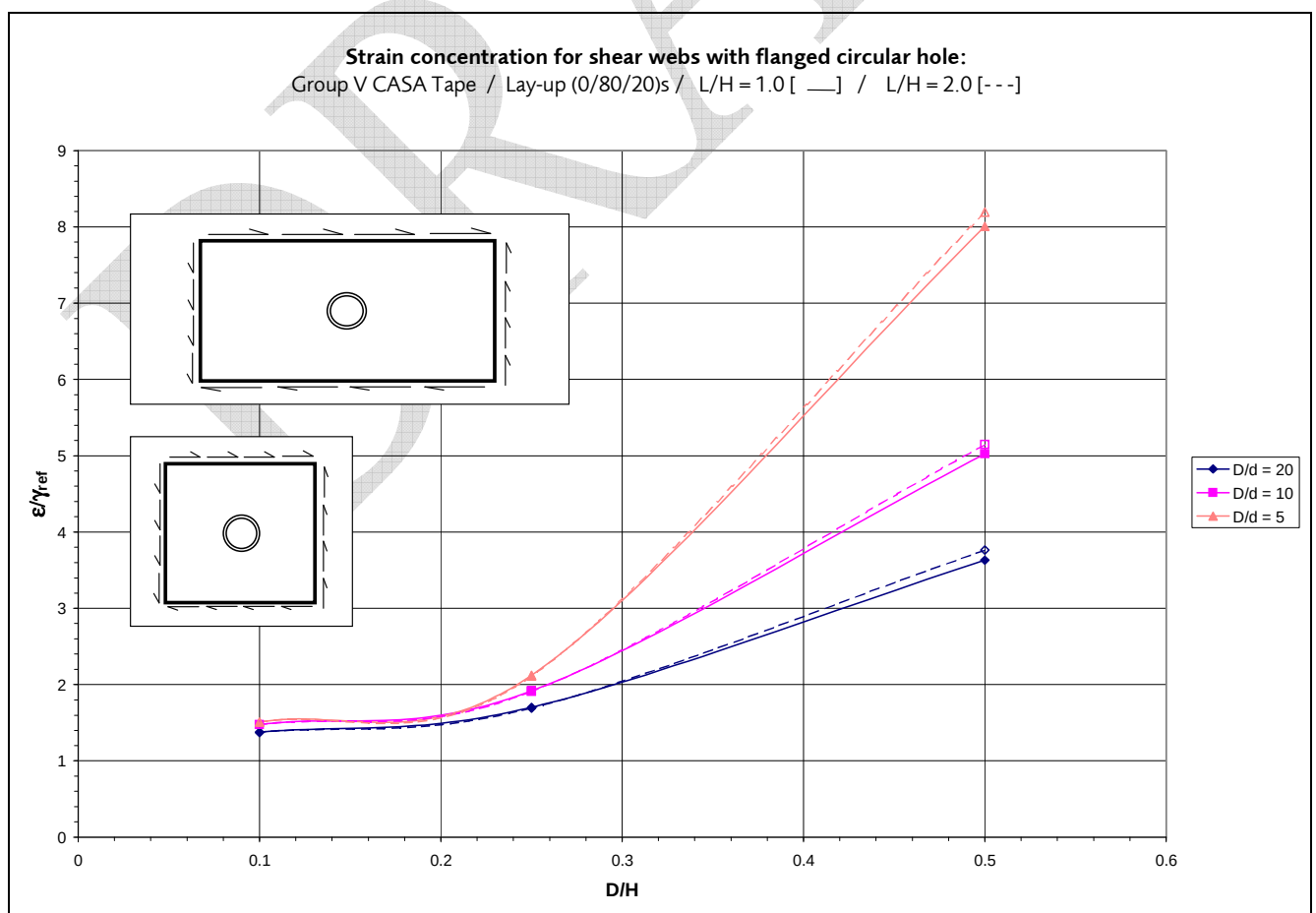


Figure 7-6

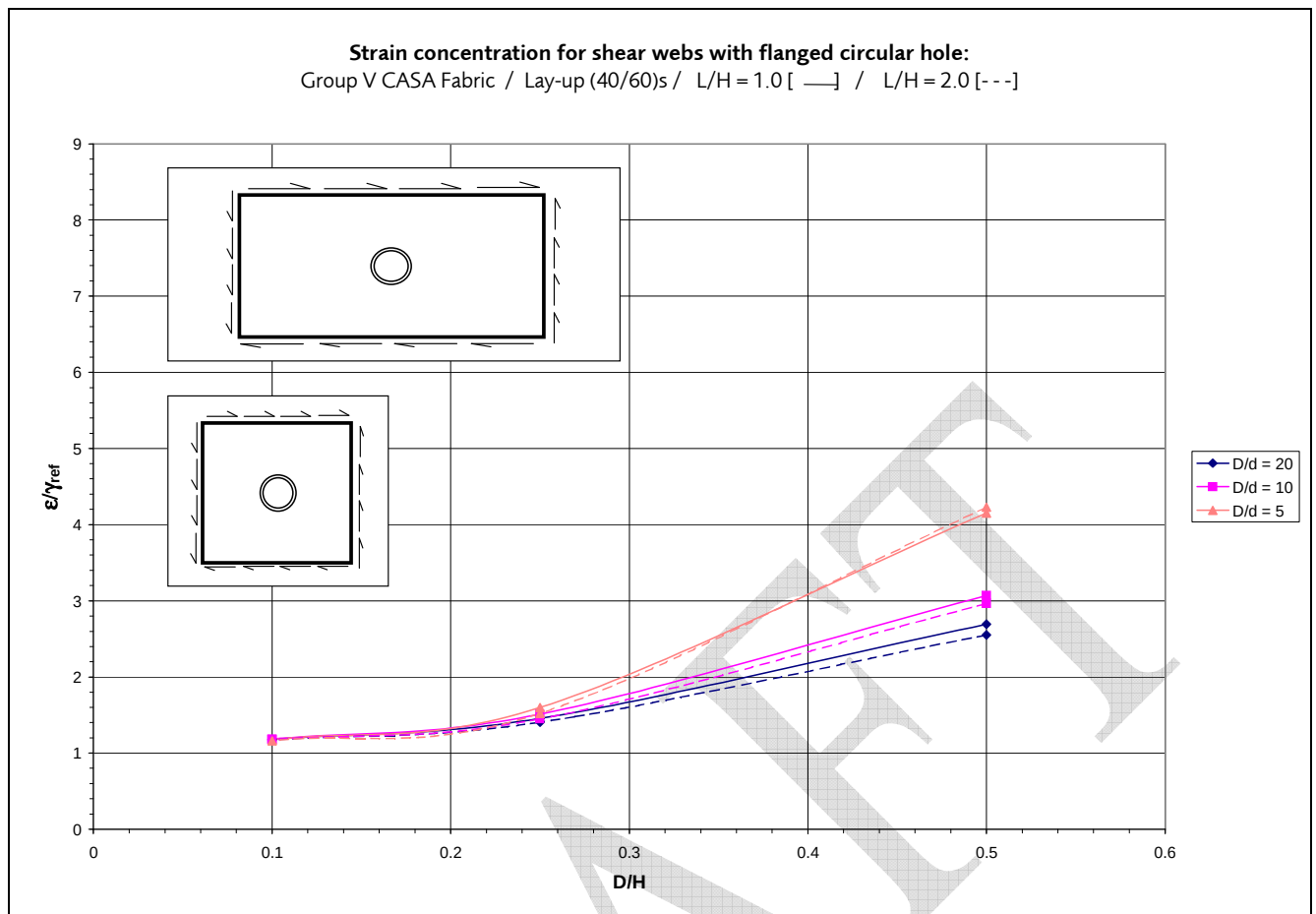


Figure 7-7

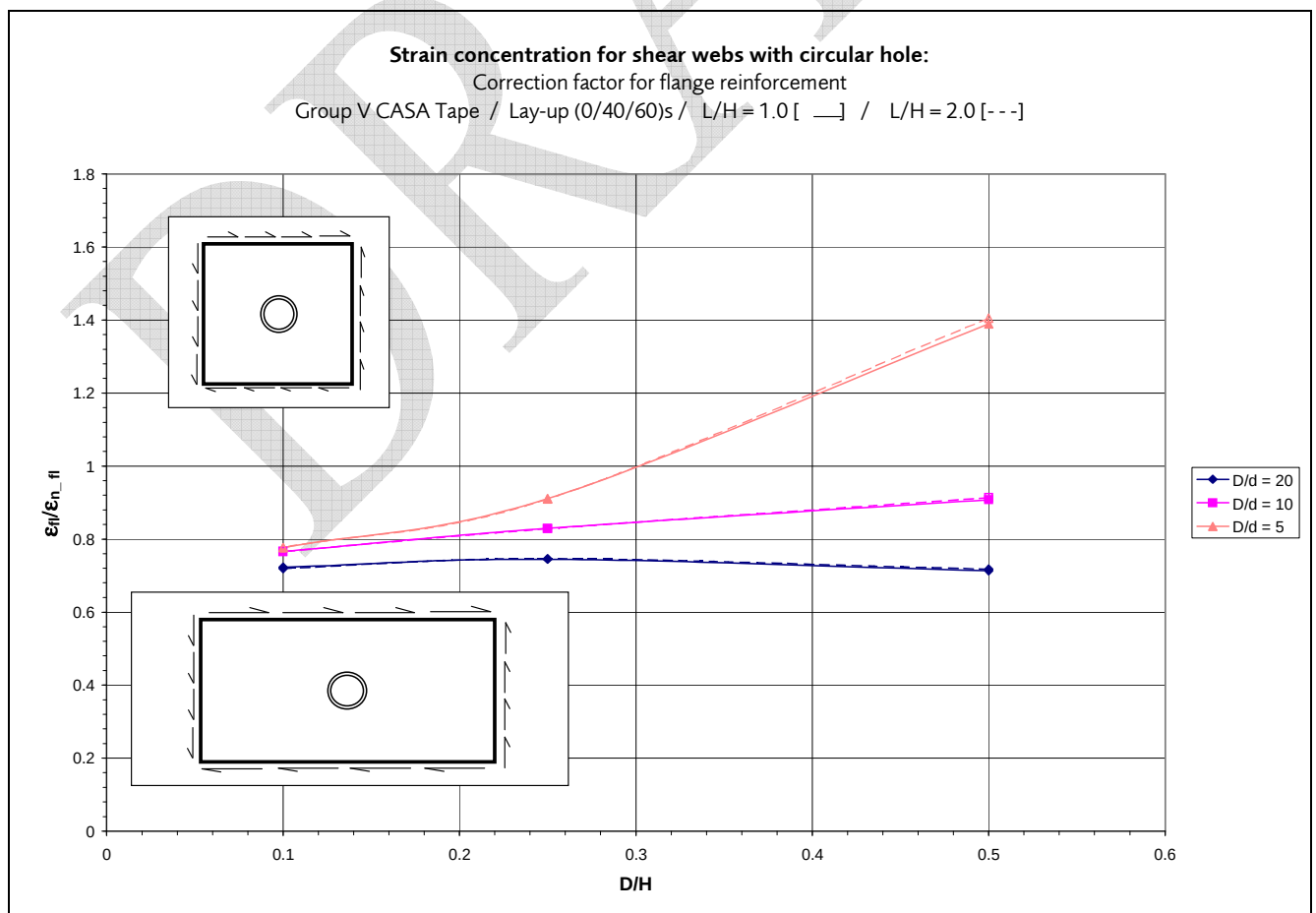


Figure 7-8

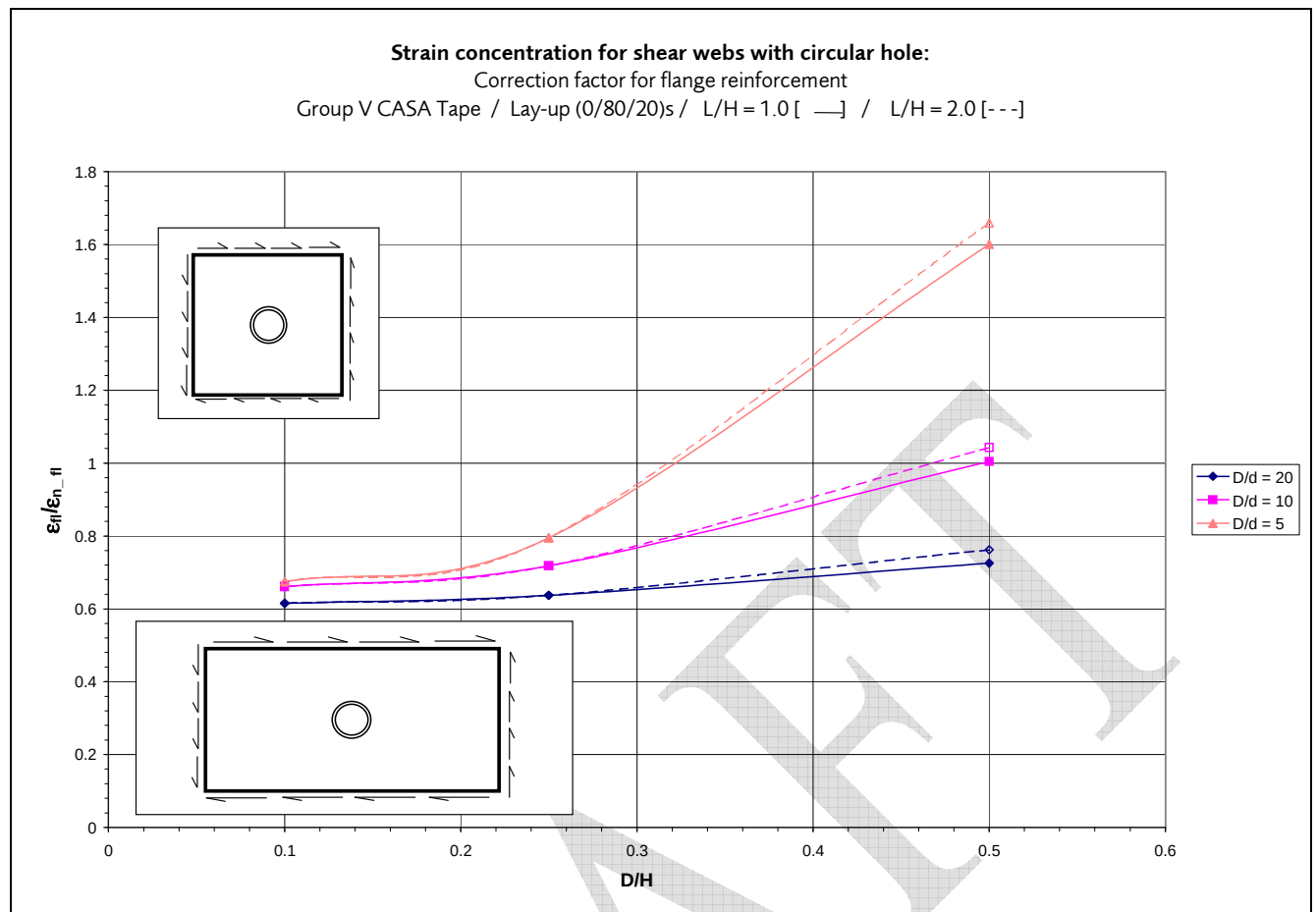


Figure 7-9

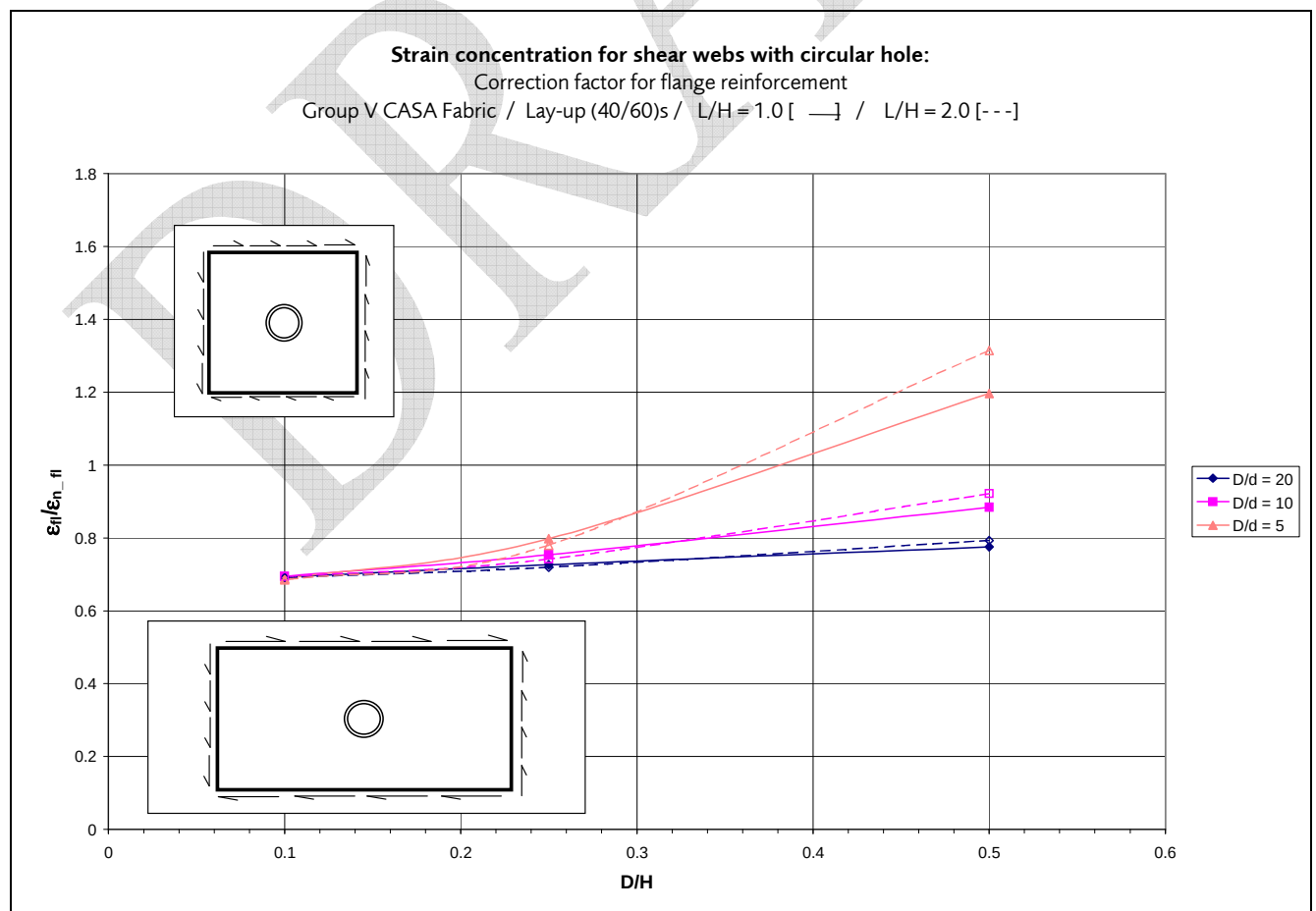


Figure 7-10

8 Validation and comparison with other tools

8.1 Comparison with approximate analytical buckling results

Shear buckling load for plates with centred circular hole has been analytically obtained by using a not official in-house numerical tool. This program is an improved version of Panpa program (Ref. 18), which solves the stability problem applying the Rayleigh-Ritz method. The influence of the hole is taken into account by adding appropriate terms in the strain energy expression. This program was expected to provide quite accurate results only for small hole sizes. FEM results have corroborated this point, as it can be seen in Table 8-1 for a (30/60/10) lay-up.

L/H=1.0				
H (mm)	D/H	q _{bk} (N/mm)		Δ (%)
		FEM	Analytical	
133	0.75	171.83	1791.00	942.3
200	0.5	159.25	476.70	199.3
400	0.25	80.24	98.10	22.3
1000	0.1	17.62	18.55	5.3
L/H=2.0				
H (mm)	D/H	q _{bk} (N/mm)		Δ (%)
		FEM	Analytical	
133	0.75	178.49	755.60	323.3
200	0.5	144.39	262.80	82.0
400	0.25	60.12	68.39	13.8
1000	0.1	11.85	12.22	3.1

Table 8-1. Shear buckling flows for plates with circular hole

Analytical results are non-conservative. Furthermore, when D/H becomes greater than 0.15 or 0.20, the committed errors result inadmissible. This program shows small profitability since acceptable results are only obtained for values of D/H around 0.1.

8.2 Comparison with FEA results

FEA program (see Ref. 19) is an in-house tool, which analytically calculates the stress and strain distribution around a circular or elliptical hole at anisotropic plates submitted to in-plane loads. This program makes use of Letzhniskii theory for anisotropic plates, which gives an exact solution for infinite plates whereas an approximated solution is reached for finite plates.

FEA program has been run for (30/60/10), (10/80/10), (0/40/60) and (0/80/20) tape lay-ups, for (40/60) fabric lay-up and several geometrical configurations. Great accordance has been found between FEA results and FEM results, as it can be seen from Figure 8-1 to Figure 8-6. FEA results for D/H = 0.1 have been obtained assuming an infinite plate, since distance from hole edge to plate border is greater than four diameters. (Notice that no accurate solution has been possible to obtain for D/H = 0.75 in the rectangular plate case, for (10/80/10) lay-up). These results validate the procedure established in this Technical Note.

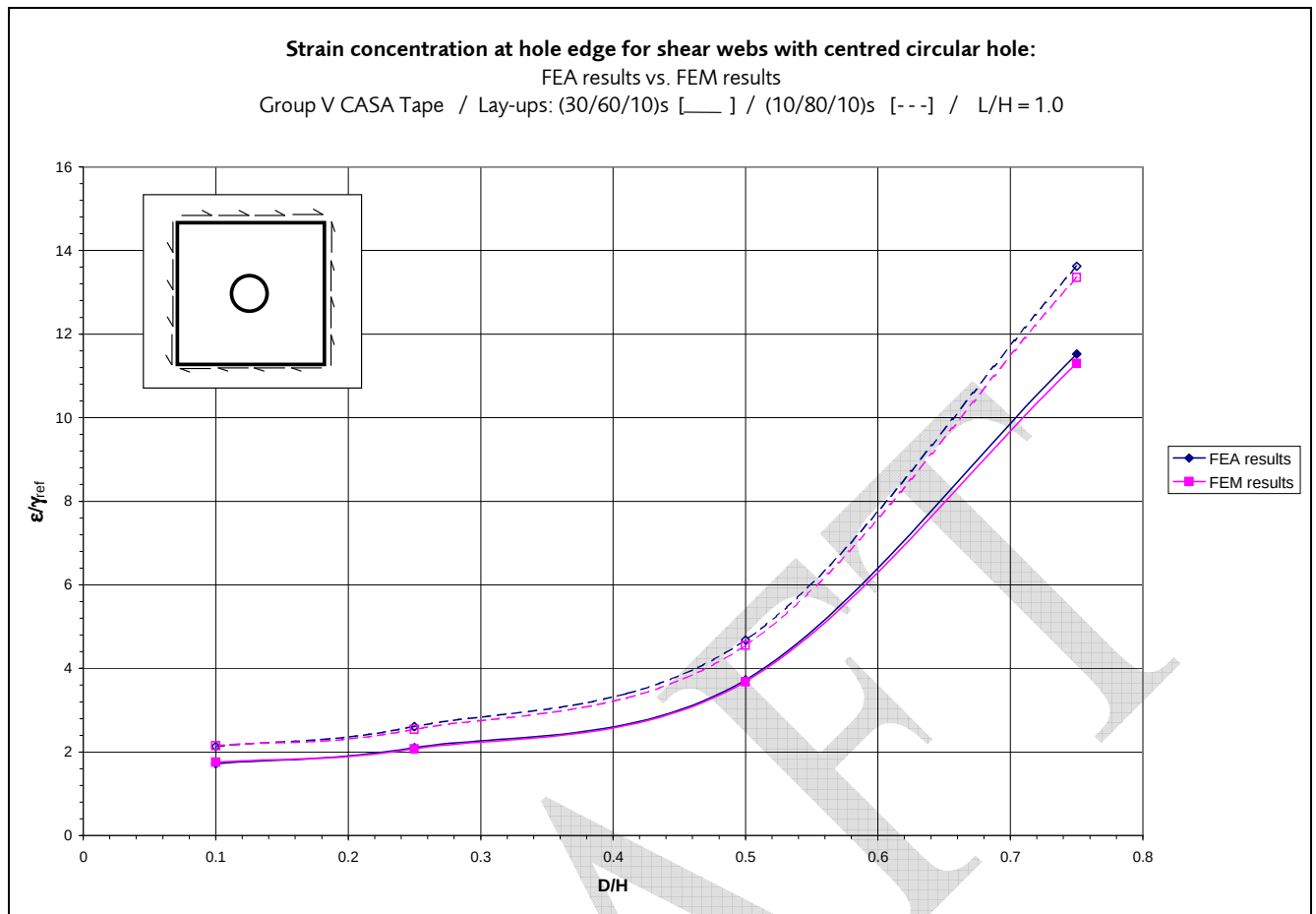


Figure 8-1

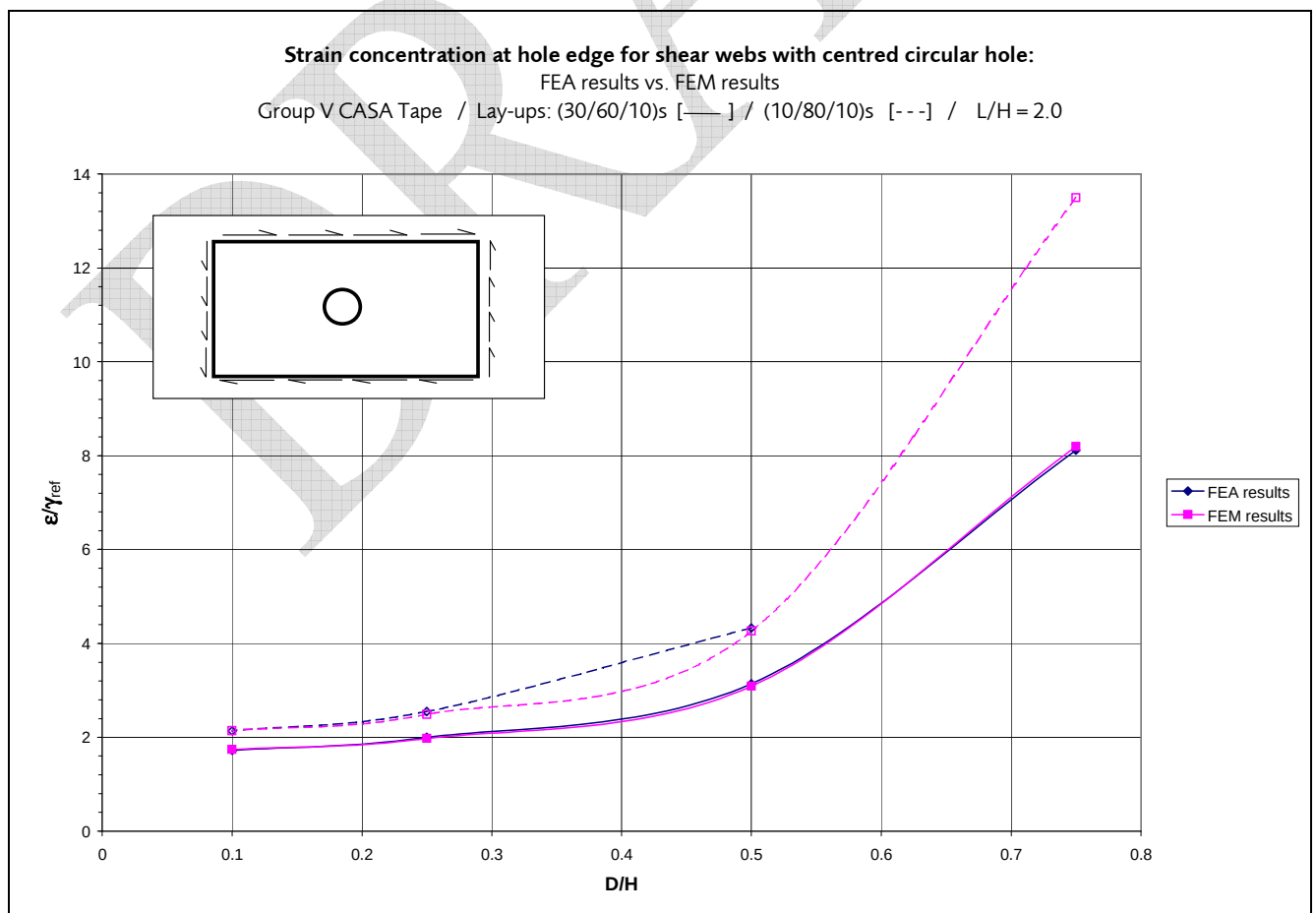


Figure 8-2

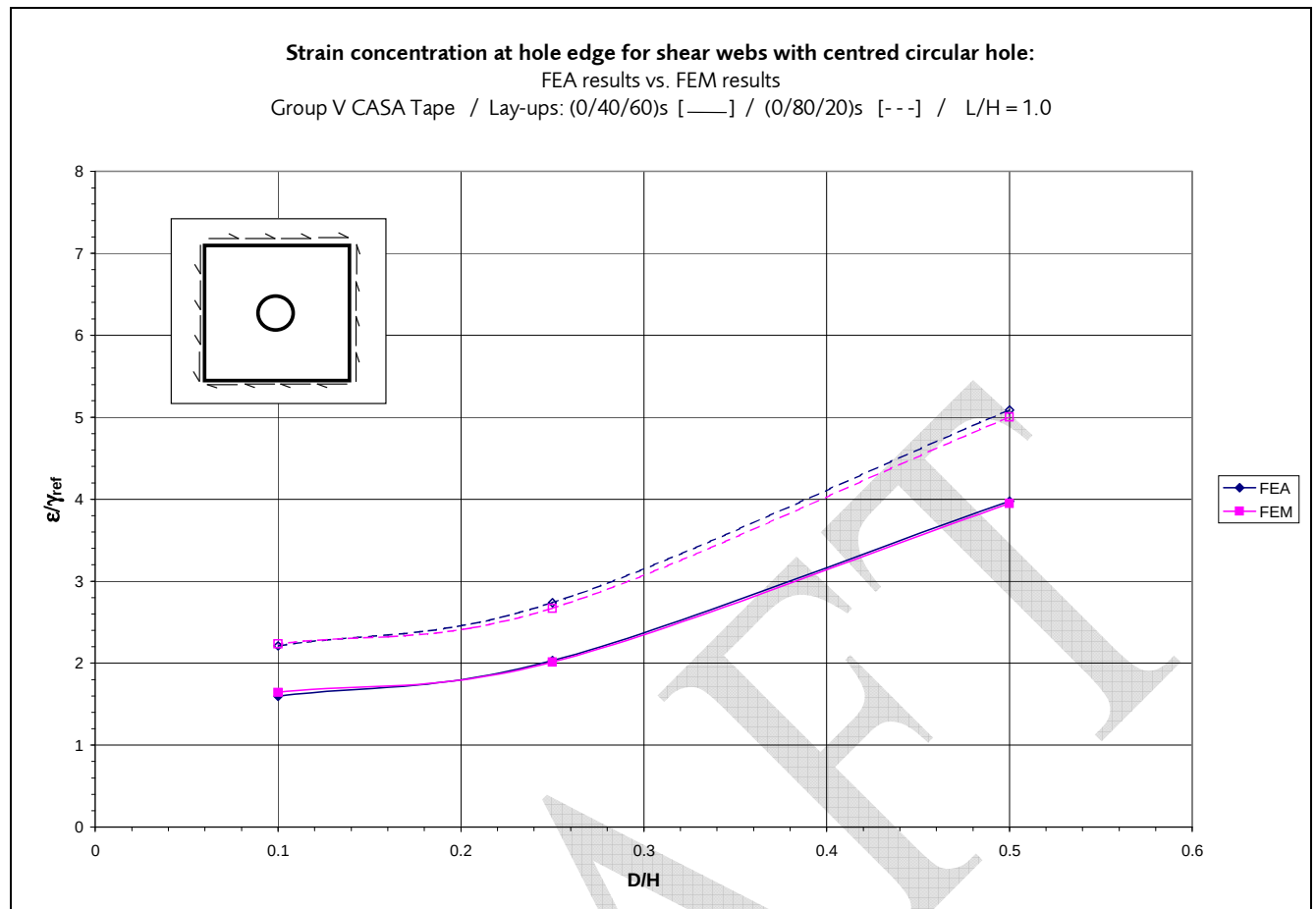


Figure 8-3

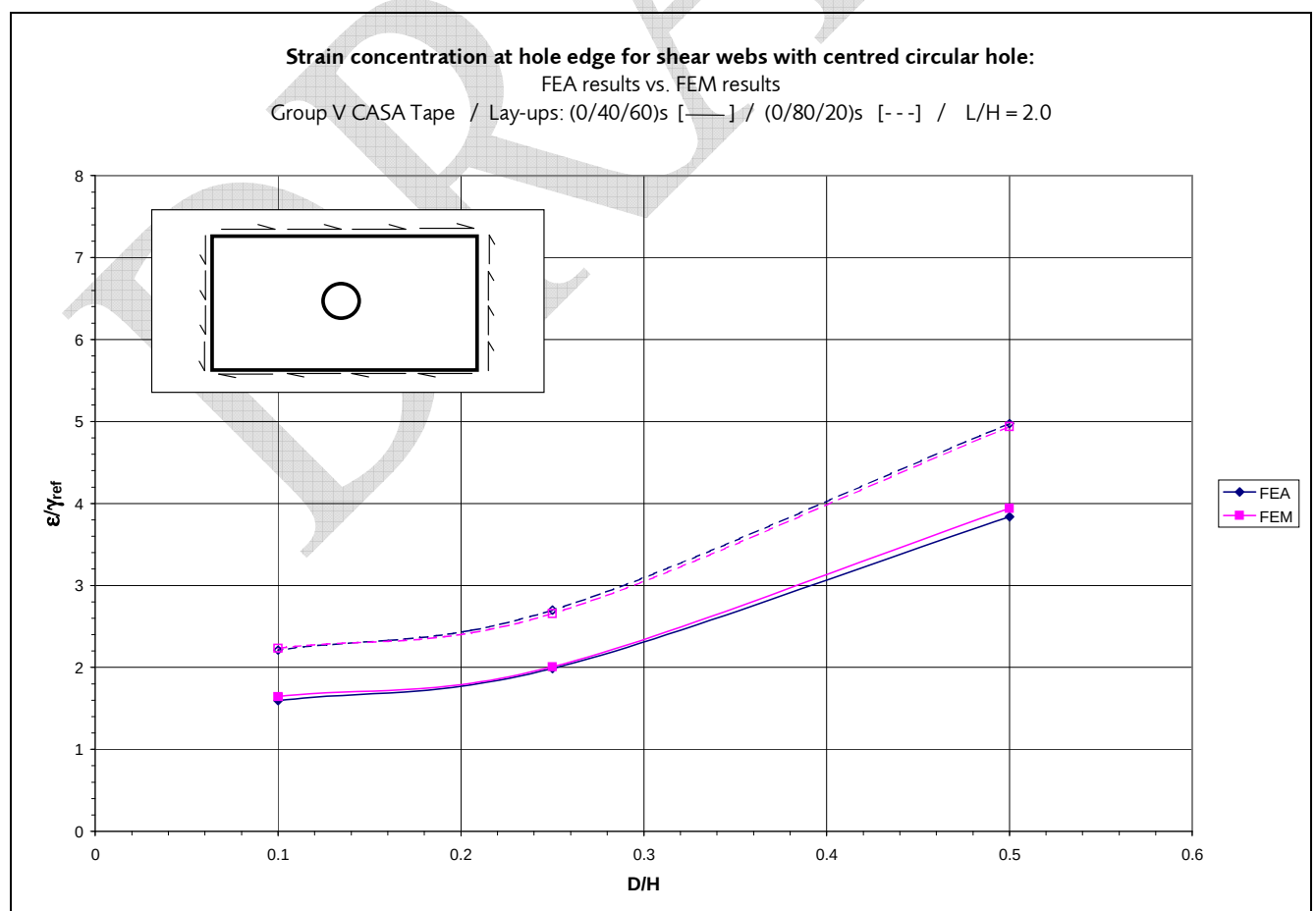


Figure 8-4

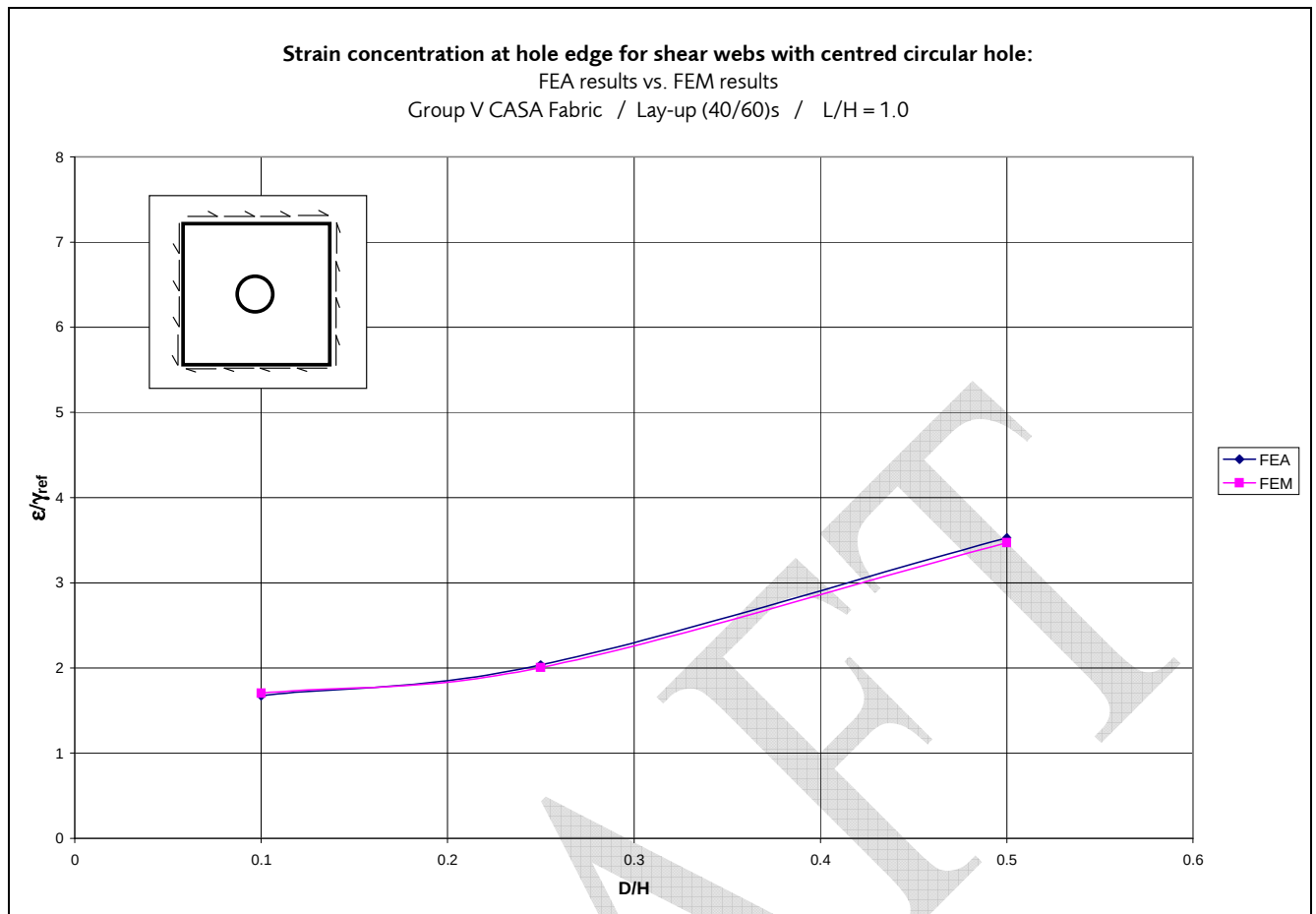


Figure 8-5

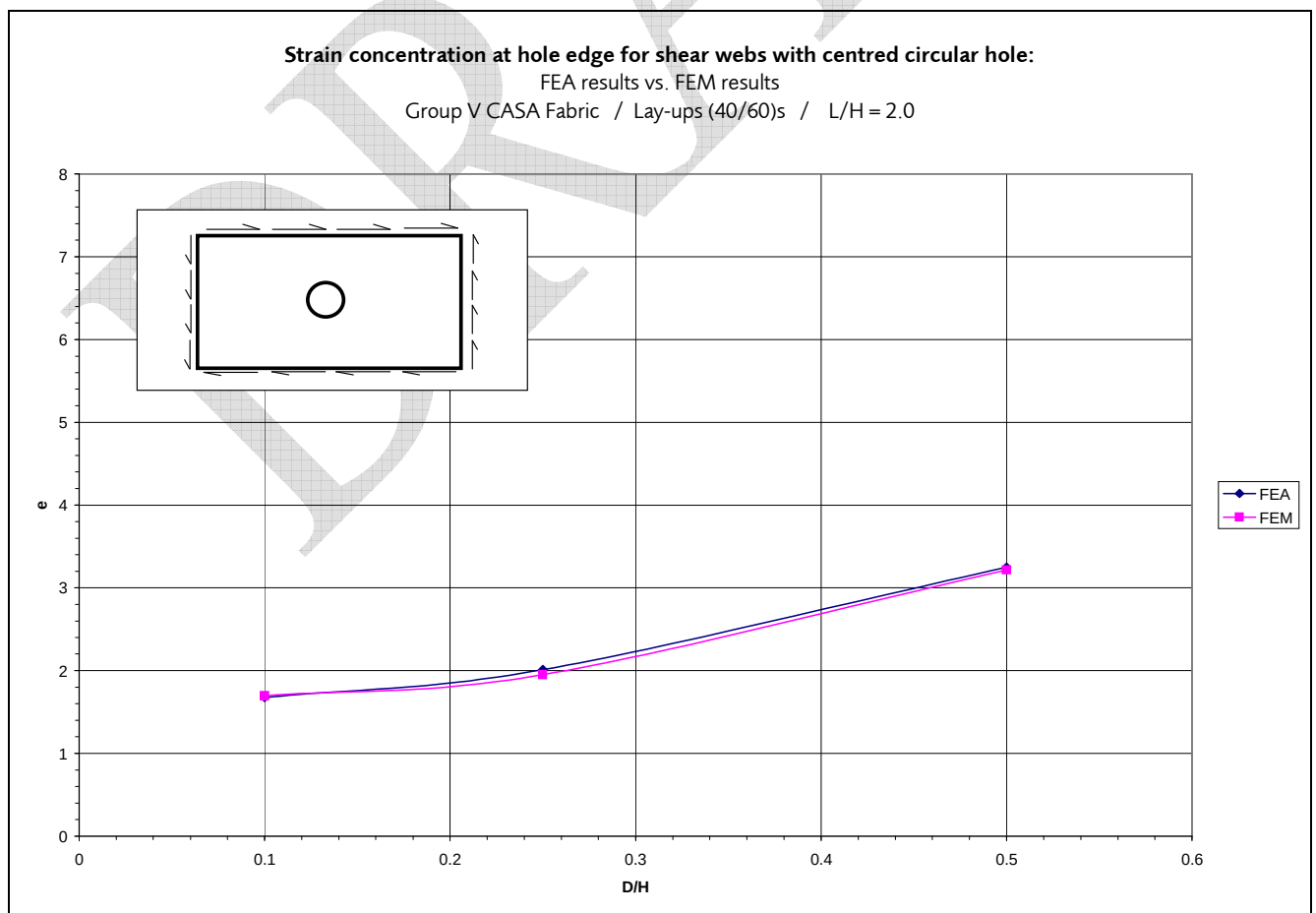


Figure 8-6

9 Comparison with the design method employed in the A340-600 HTP

In order to compare the new design procedure with the existing methods, buckling and strength allowable loads have been calculated for the A340-600 HTP spars panels. Obtained results have been compared with the ones used in the real design (see Ref. 14 and Ref. 15). As the existing method is based on the tests carried out for the A320 panels (see ref. 12), results for these panels have been also compared.

9.1 Existing design procedure

Spars failure criterion for the A340-600 HTP is originally based in the tests that were carried out for the A320 spars (Ref. 12). This method was later improved for the strength checking of the A330/340 HTP spars, making it valid for any material (see Ref. 13). This procedure takes into account not only strength and stability considerations but also interaction between both of them. Furthermore, panel failure is determined by the strain concentration at the hole edge taking into account the interaction with the panel buckling.

9.1.1 Strength analysis

Maximum tangential strain at the hole edge is defined as:

$$- \quad \varepsilon_{\tau \text{ máx}} = K_e \cdot \gamma,$$

where

$$- \quad \gamma = N_{xy} / (G t)$$

$$- \quad K_e = K_{\infty} \cdot f$$

The strain concentration factor for an infinite panel (K_{∞}) depends on its mechanical properties (material, lay-up and ageing degree). This concentration factor may be obtained from EH program (Ref. 20). Correction factor f is introduced to take into account that the panel has finite dimensions:

$$- \quad f = \frac{2 + \left(1 - \frac{D}{H}\right)^3}{3 \left(1 - \frac{D}{H}\right)}$$

Allowable microstrains at the hole edge are 7200 $\mu\epsilon$. The procedure assumes this value, which takes into account hot and wet conditions, for any material (Ref. 13). Once the material and the geometry are known, the strength allowable flow is directly obtained.

This formulation is only valid for circular holes, so comparison between both procedures has been restrained to such panels.

9.1.2 Stability analysis

Allowable buckling shear flow for panels with circular hole is given by:

$$- N_{xy, \text{ buck hole}} = K_h \cdot N_{xy, \text{ buck}}$$

The buckling shear flow for plain panels may be obtained from PANPA program (Ref. 18). The correction factor that takes into account the hole influence is given by the following expression:

$$- K_h = 0.85 \cdot (1 - D/H)$$

9.1.3 Interaction between stability and strength

Interaction curves were obtained from the panels with hole tested for the A320 spars (Ref. 12). Later on, two interaction curves were obtained with no dependence on the material, one corresponding to panel thickness greater than 3.5 mm and the other one to panel thickness lower than 3.5 mm (Ref. 13). These curves represent the relative increment between the maximum microstrains and the medium microstrains as a function of the applied percentage of the buckling load. Final allowable flow is obtained through an iterative process.

Interaction is not considered if the strength flow is lower than a 40% of the buckling flow, so pure strength failure is then considered. By the contrary, if the final allowable load is greater than the buckling load, buckling failure is considered and the buckling shear flow is taken as the allowable flow. All this procedure is thoroughly explained in Ref 14.

9.2 Comparison with A320 tests

As a first step in the new procedure validation, A320 tests results have been compared with the allowable flows obtained from the design procedure employed at the A340-600 and with the ones obtained from the new design method. This comparison shows which of the methods is the most conservative.

The material employed for the A320 spars is CASA Group IV fabric (Hexcel W6T – 282 – 42 – F593 – 14), with (0/100) lay-up. The tests were carried out at RT/AR conditions, which corresponds to the following mechanical properties:

	Average value	B value
E_{1t} (N/mm ²)	52000	45000
E_{2t} (N/mm ²)	52000	45000
E_{1c} (N/mm ²)	46000	39000
E_{2c} (N/mm ²)	46000	39000
G_{12} (N/mm ²)	3100	2600
μ_{12}	0.05	-
ϵ_{1t} (µε)	-	9135
ϵ_{2t} (µε)	-	9135
ϵ_{1c} (µε)	-	10000
ϵ_{2c} (µε)	-	10000
γ_{12} (µε)	-	39548
γ_{13} (µε)	-	20322

Table 9-1. Mechanical properties for CASA Group IV Fabric (RT/AR conditions)

Average properties between tension and compression have been used for the modulus values. Laminate properties are obtained with ANCOLAB program (Ref. 17). Average properties for the (0/100) lay-up, used for strength calculations, are:

- $E_x = E_y = 11069 \text{ N/mm}^2$
- $G_{xy} = 23333 \text{ N/mm}^2$
- $\mu_{xy} = 0.79$

B properties, used for buckling calculations, are:

- $E_x = E_y = 9306 \text{ N/mm}^2$
- $G_{xy} = 20000 \text{ N/mm}^2$
- $\mu_{xy} = 0.79$

A few remarks must be made on the new procedure. Strain concentration at the hole edge is a quite conservative criterion for strength failure, since the crack of a fibre at the hole does not directly imply the panel failure. Moreover, failure would occur when allowable compression strain is reached at a characteristic distance of the hole edge. For that reason, evolution of the strength allowable flow with the hole edge distance has been analysed with FEA program (Ref. 19). Notice that FEA results are reliable only if a good solution convergence is achieved.

Specimens have vertical stiffeners. Distance between stiffeners has been considered as the panel width for buckling calculations, assuming simply supported boundary conditions, which is conservative. Buckling design curves showed in this document show low dependence on the material, so they may be used for this analysis.

Specimens geometry is defined in table 9-2.

Specimen	Location	L (mm)	H (mm)	L_{ef} (mm) ⁽¹⁾	H_{ef} (mm) ⁽¹⁾	D (mm)	t (mm)
R4-R5	Rear spar	602	317	317	200	90	5.76
R4-R5	Front spar	614	292	292	180	90	5.04
R6-R7	Front spar	582	262	262	190	80	5.04
R2-R3	Rear spar	522	342	360	342	110	7.44
R11-R12 ⁽²⁾	Front spar	– ⁽³⁾	– ⁽³⁾	176	128	50	2.4
R11-R12 ⁽²⁾	Front spar	– ⁽³⁾	– ⁽³⁾	176	128	40	2.16
R12-R13	Rear spar	– ⁽³⁾	– ⁽³⁾	140	133	35	2.4

⁽¹⁾ Local panel, dimensions between stiffeners

⁽²⁾ This specimen has two holes

⁽³⁾ These specimens have been tested as a beam in 3-point bending, instead of as picture frame shear panels, due to their high aspect ratio

Table 9-2. A320 test panels geometry

Following figures show the next allowable flows for each specimen:

- Strength allowable flow, derived from FEA results (evolution with hole edge distance is considered), to reach the minimum H/W compression allowable microstrains of the material (10000 $\mu\epsilon$)
- Buckling allowable flow, derived from the design curves previously shown at this document and applying linear interpolation between them
- Allowable flow derived from the existing procedure employed at the A340-600, which takes into account interaction between buckling and strength
- Allowable flow occurred at the specimen test

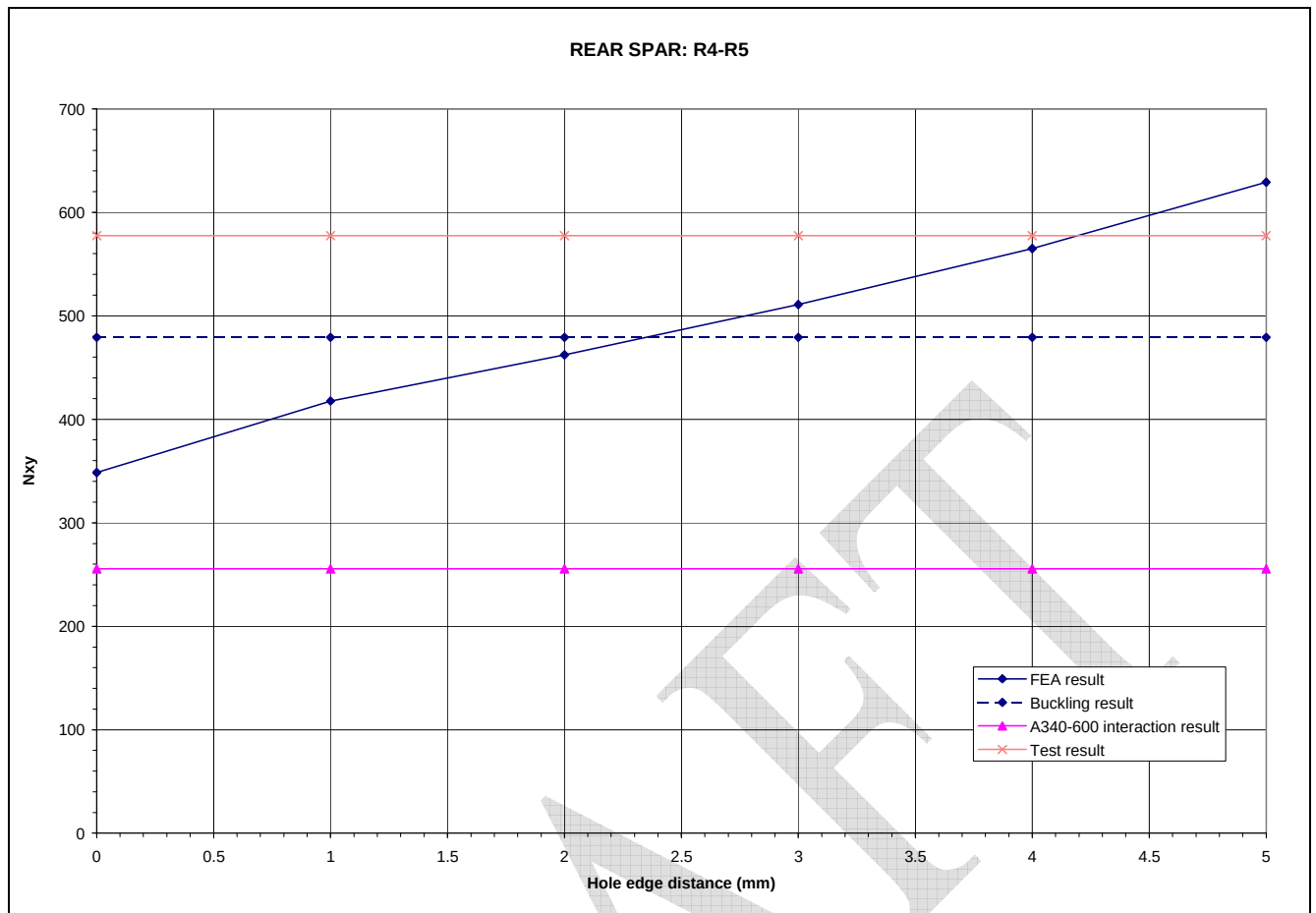


Figure 9-1

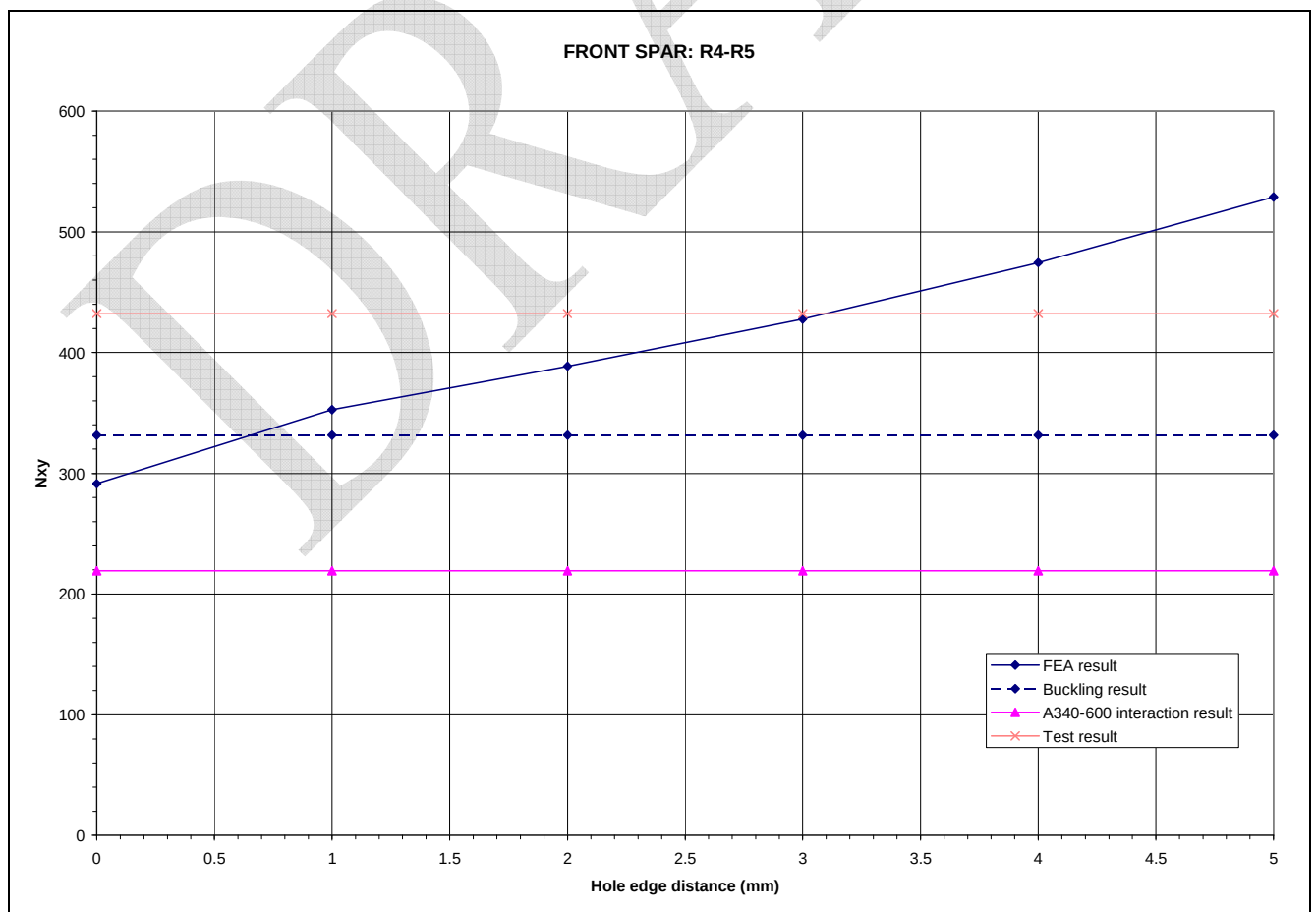


Figure 9-2

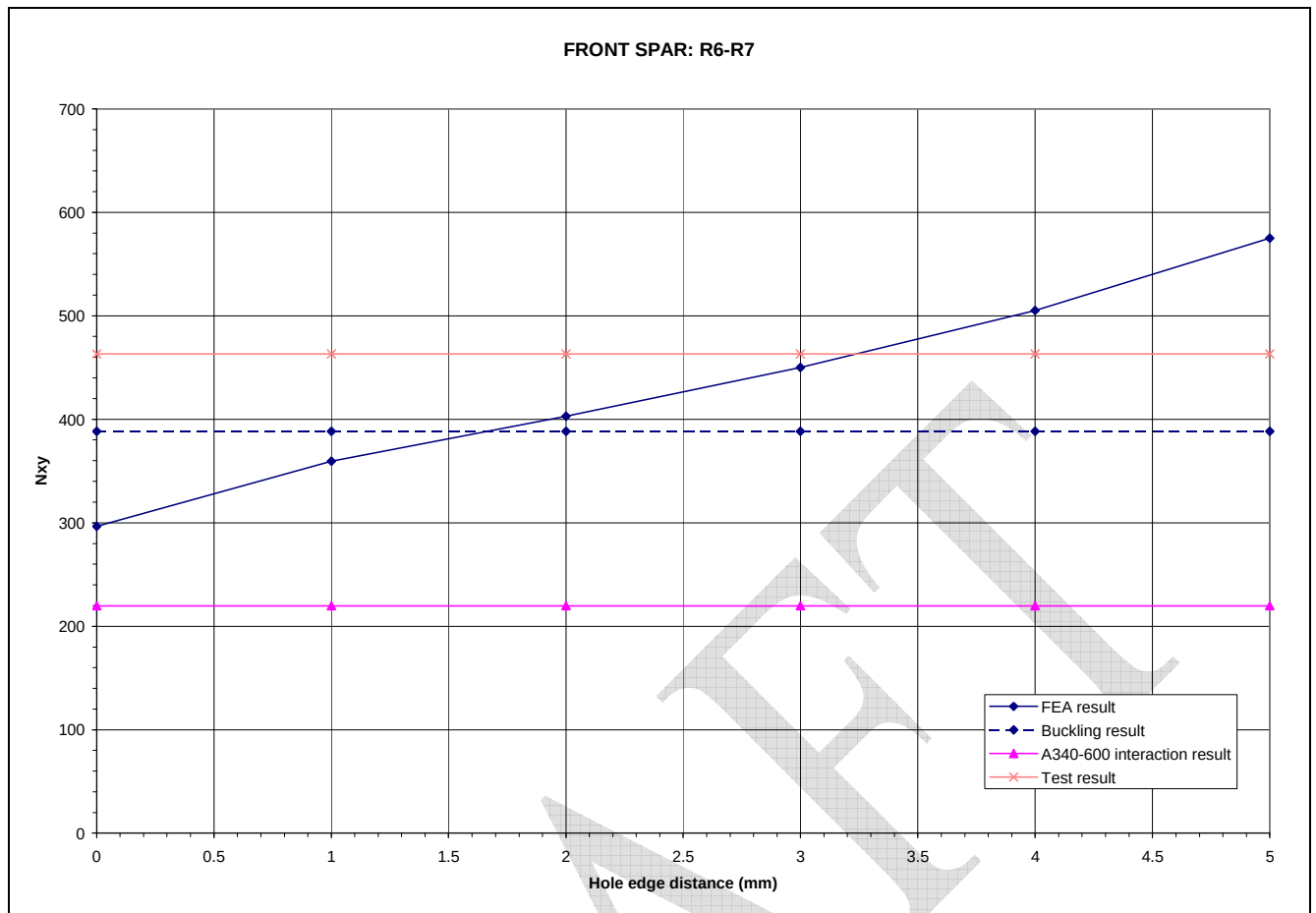


Figure 9-3

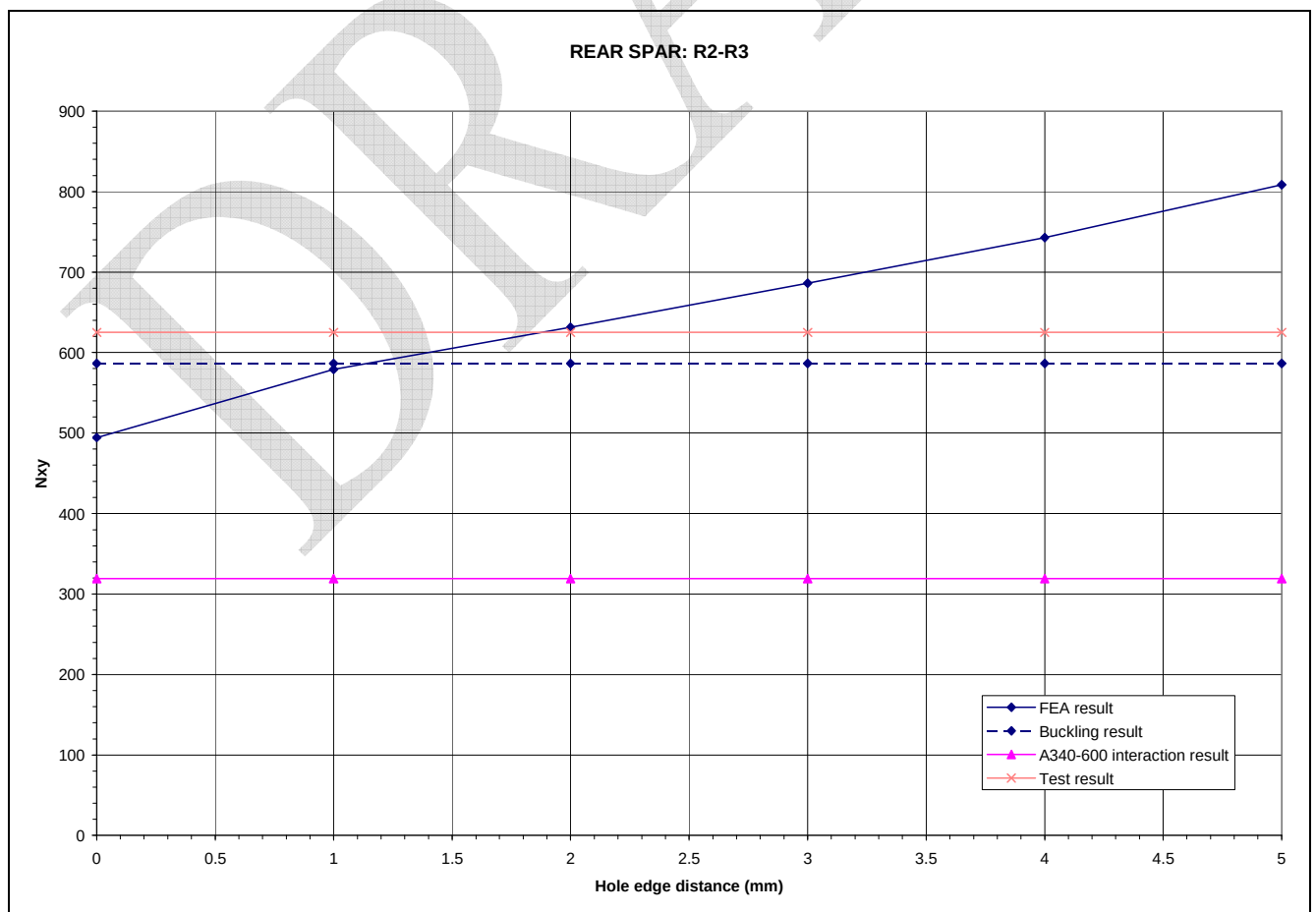


Figure 9-4

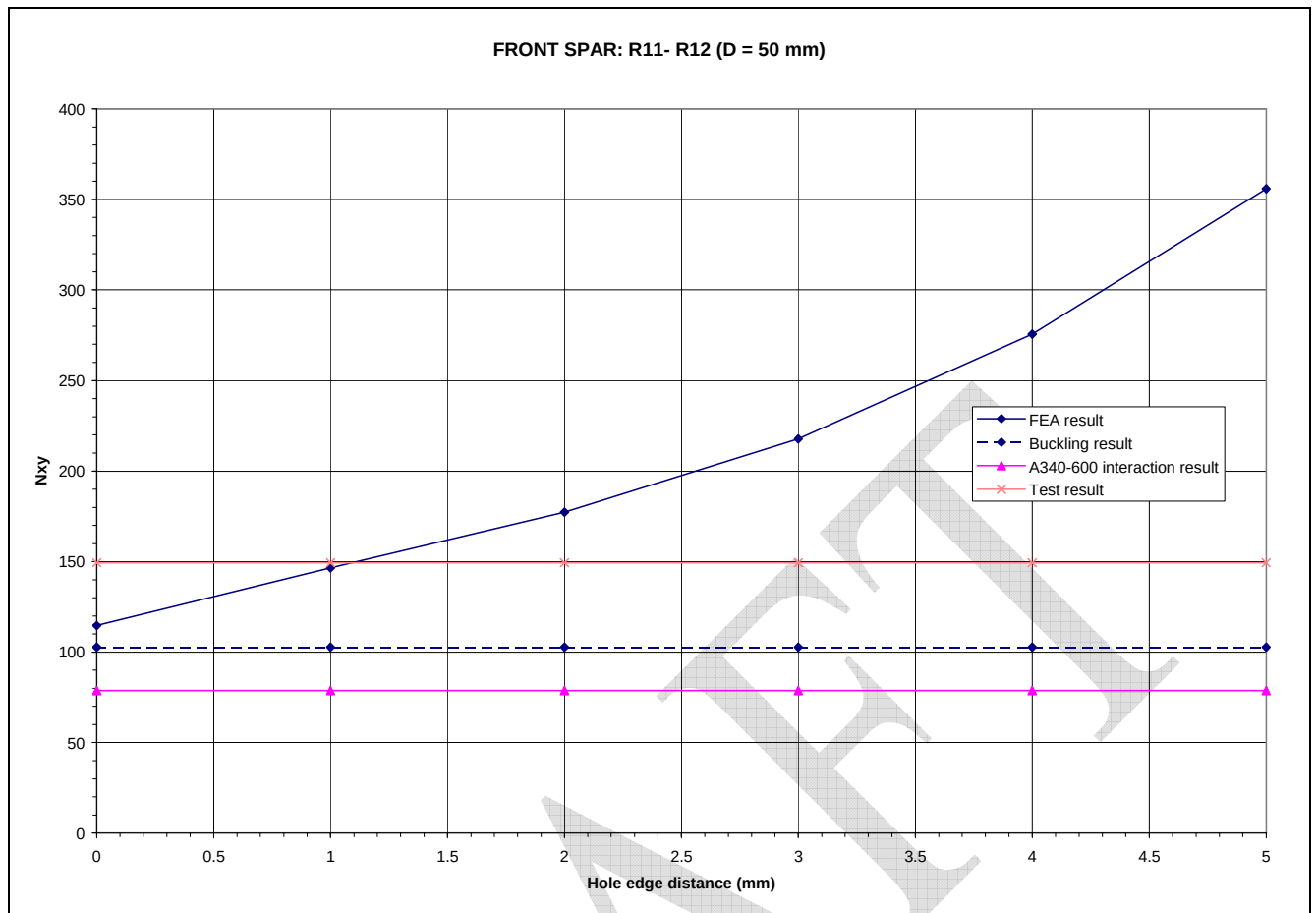


Figure 9-5

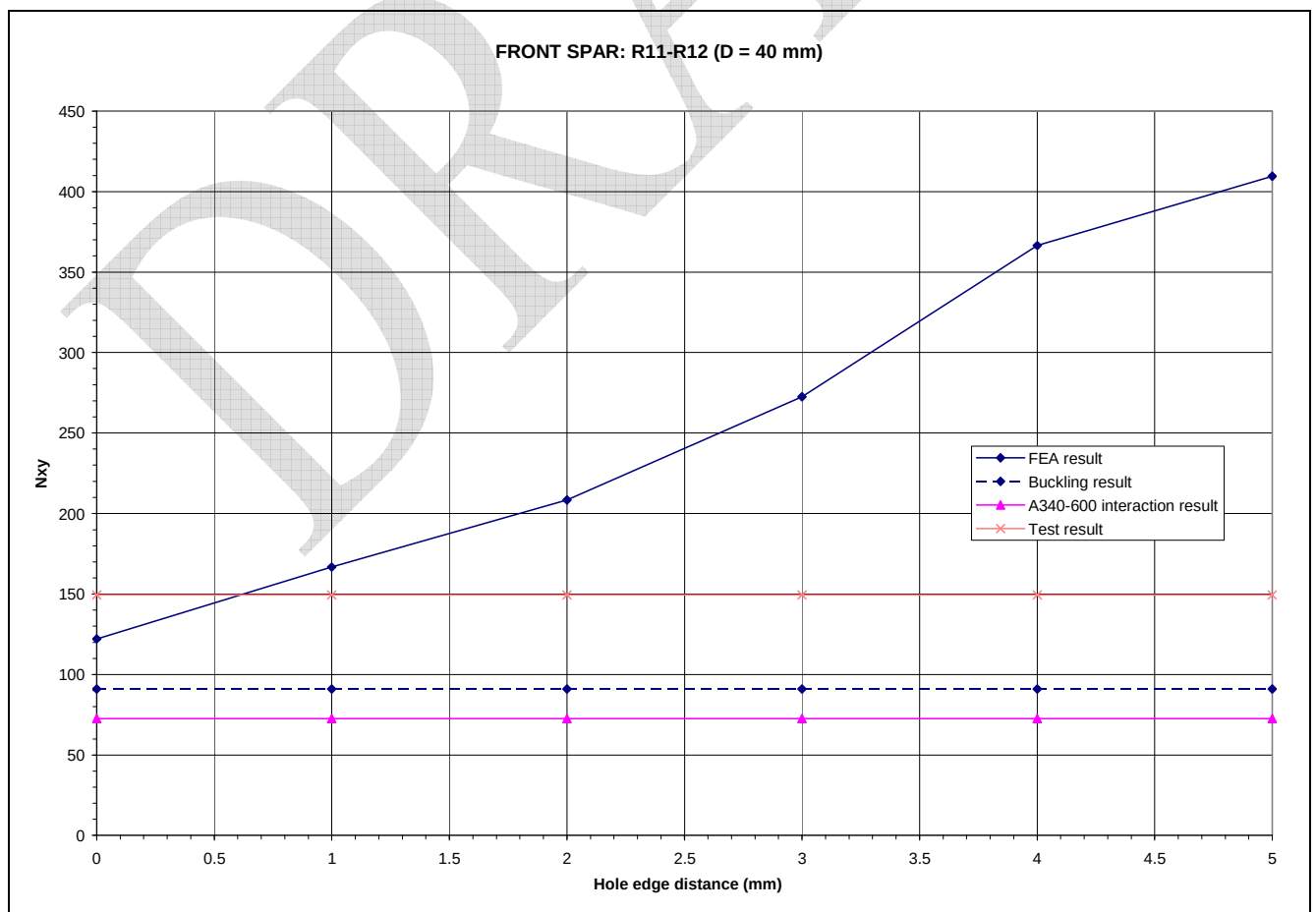


Figure 9-6

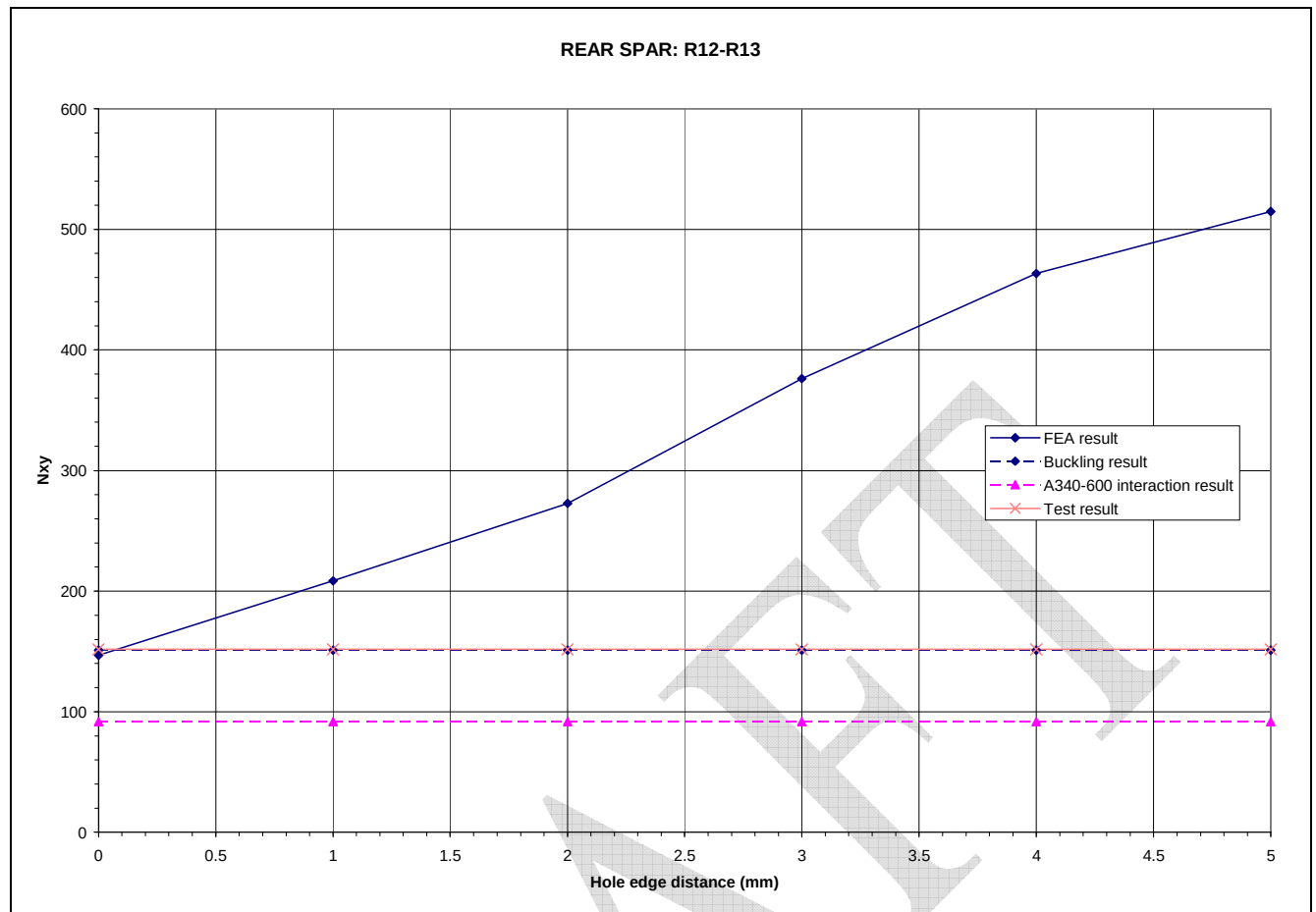


Figure 9-7

As it can be seen in figures 9-1 to 9-7, both procedures are conservative since the test allowable flow is always higher. The interaction method used at the A340-600 is extremely conservative for these panels. The procedure exposed in this document is also conservative but reproduces quite better the test result. For the first four specimens, observing the results at the hole edge, strength is the critical criterion, whereas instability causes the failure in the other three. Premature failure occurred at the last specimen (Figure 9-7) due to flange crippling. This fact explains that the predicted allowable for this panel seems to be almost non-conservative.

The difference between both design methods is mainly due to the allowable strain used for the strength analysis. The new procedure considers a maximum strain of 10000 $\mu\epsilon$ at the hole edge (allowable compression strain of CASA Group IV Fabric) whereas the interaction method does not allow strains bigger than 7200 $\mu\epsilon$. By the contrary, buckling prediction is similar for both methods.

In conclusion, the new procedure provides conservative allowable flows that get nearer to the actual failure flow. Anyway, it would be worthy a deeper study of the characteristic hole edge distance at which maximum strain should be considered. Much more accurate allowable loads would be obtained if characteristic distance dependence on geometry and material were known.

9.3 Comparison with A340-600 results

As a final evaluation of the developed design tool, the procedure has been applied to the A340-600 HTP spars panels. The results have been compared with the ones obtained in the actual design (see Ref. 14 and 15). Spars from the torque box and the elevator are analysed. In this case, characteristic hole edge distance is again considered. Besides, tension and compression allowable strains have been considered, so influence of the maximum allowed strain can be observed.

9.3.1 Torque box spars

The material employed for the front and rear spars is CASA Group V Tape (AS4/8552). In this case, H/W 2 conditions are considered. Mechanical properties are shown in Table 9-3.

	Average value	B value
E_{1t} (N/mm ²)	140000	120000
E_{2t} (N/mm ²)	8700	6800
E_{1c} (N/mm ²)	122000	109000
E_{2c} (N/mm ²)	9000	7500
G_{12} (N/mm ²)	3950	3300
μ_{12}	0.3	-
ϵ_{1t} (μ ϵ)	-	11400
ϵ_{2t} (μ ϵ)	-	3000
ϵ_{1c} (μ ϵ)	-	8100
ϵ_{2c} (μ ϵ)	-	15500
γ_{12} (μ ϵ)	-	19000

Table 9-3. Mechanical properties for CASA Group V Tape (H/W 2 conditions)

The most frequent lay-up is (0/80/20), which has the following average properties:

- $E_x = 22120$ N/mm²
- $E_y = 38040$ N/mm²
- $G_{xy} = 27860$ N/mm²
- $\mu_{xy} = 0.45$

Table 9-4 shows the geometry of the analysed panels. Only panels with centred circular hole have been considered. Besides, buckling prediction is presented. Buckling shear flow is obtained applying a correction factor to the buckling shear flow of the panel without hole (q_0).

It can be observed that both methods predict similar buckling loads. Indeed, the biggest difference is not greater than 9%, (see Δ in Table 9-4).

FRONT SPAR		H (mm)	D (mm)	L/H	D/H	q_o (N/mm)	A340-600	New method	Δ (%)
Panel	L (mm)						q_{bk} (N/mm)	q_{bk} (N/mm)	
41	258	191	80	1.35	0.42	1270	626.9	597.4	4.93
44	247	191	80	1.29	0.42	648	319.9	301.5	6.11
47	233	191	80	1.22	0.42	684	337.6	313.7	7.60
50	221	191	80	1.16	0.42	720	355.4	326.2	8.95
52	230	209	80	1.10	0.38	648	340.2	321.8	5.71
REAR SPAR		H (mm)	D (mm)	L/H	D/H	q_o (N/mm)	A340-600	New method	Δ (%)
Panel	L (mm)						q_{bk} (N/mm)	q_{bk} (N/mm)	
38	267	190	80	1.41	0.42	1180	580.7	557.4	4.19
41	255	190	80	1.34	0.42	1220	600.4	569.3	5.45
44	244	190	80	1.28	0.42	730	359.2	336.9	6.62
47	231	190	80	1.22	0.42	351	172.7	159.8	8.05
50	217	216	80	1.00	0.37	333	178	168.1	5.88

Table 9-4. Buckling load prediction for torque box spar panels

Strength results for the new procedure are presented in graphical way since they depend on the hole edge distance. These figures include two strength allowable curves, one corresponding to the compression allowable strain and the other one to the tension allowable strain. They also include the buckling load shown in table 9-4 and the allowable flow resulting from the interaction method of the A340-600 (see Figures 9-8 to 9-17).

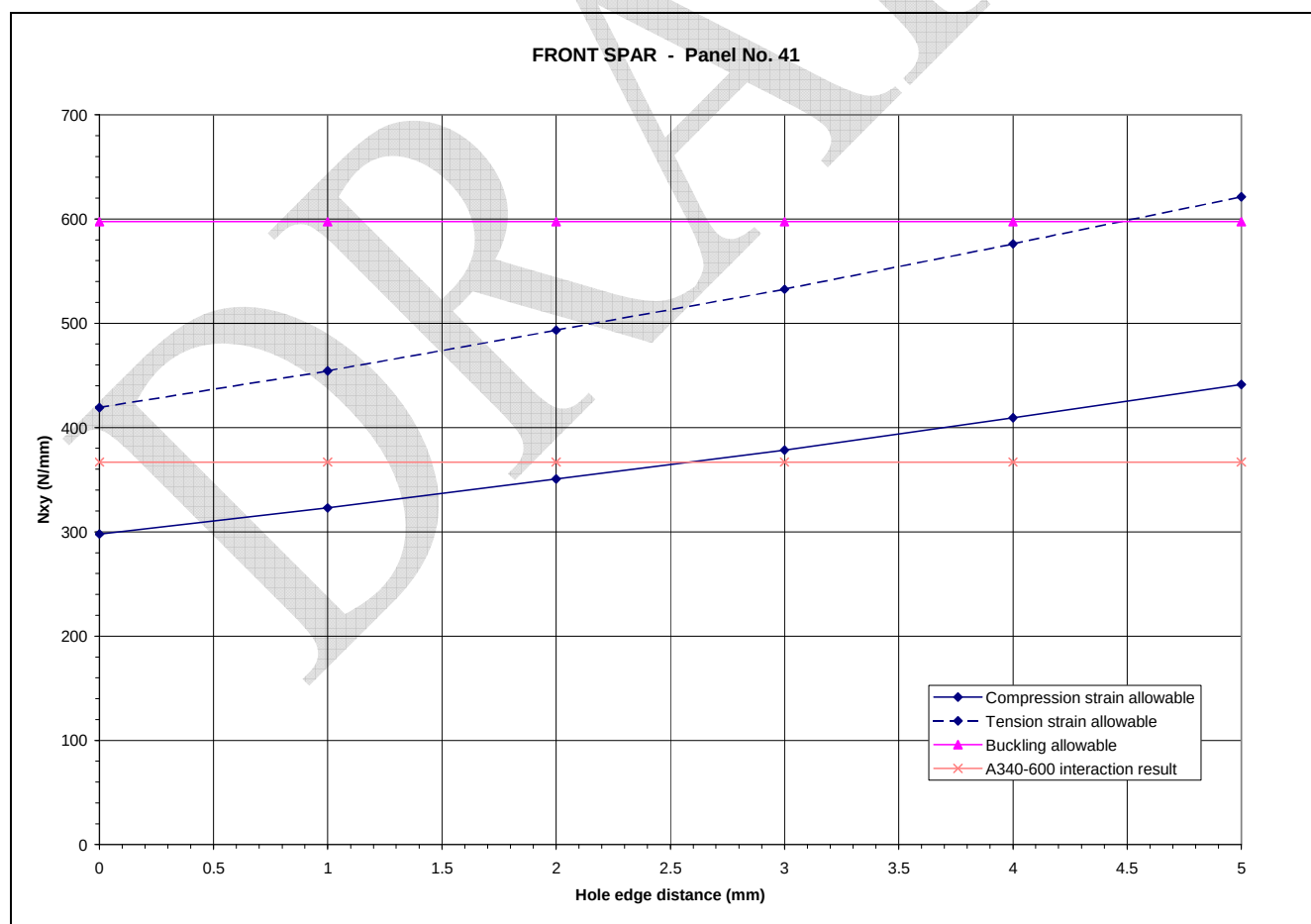


Figure 9-8

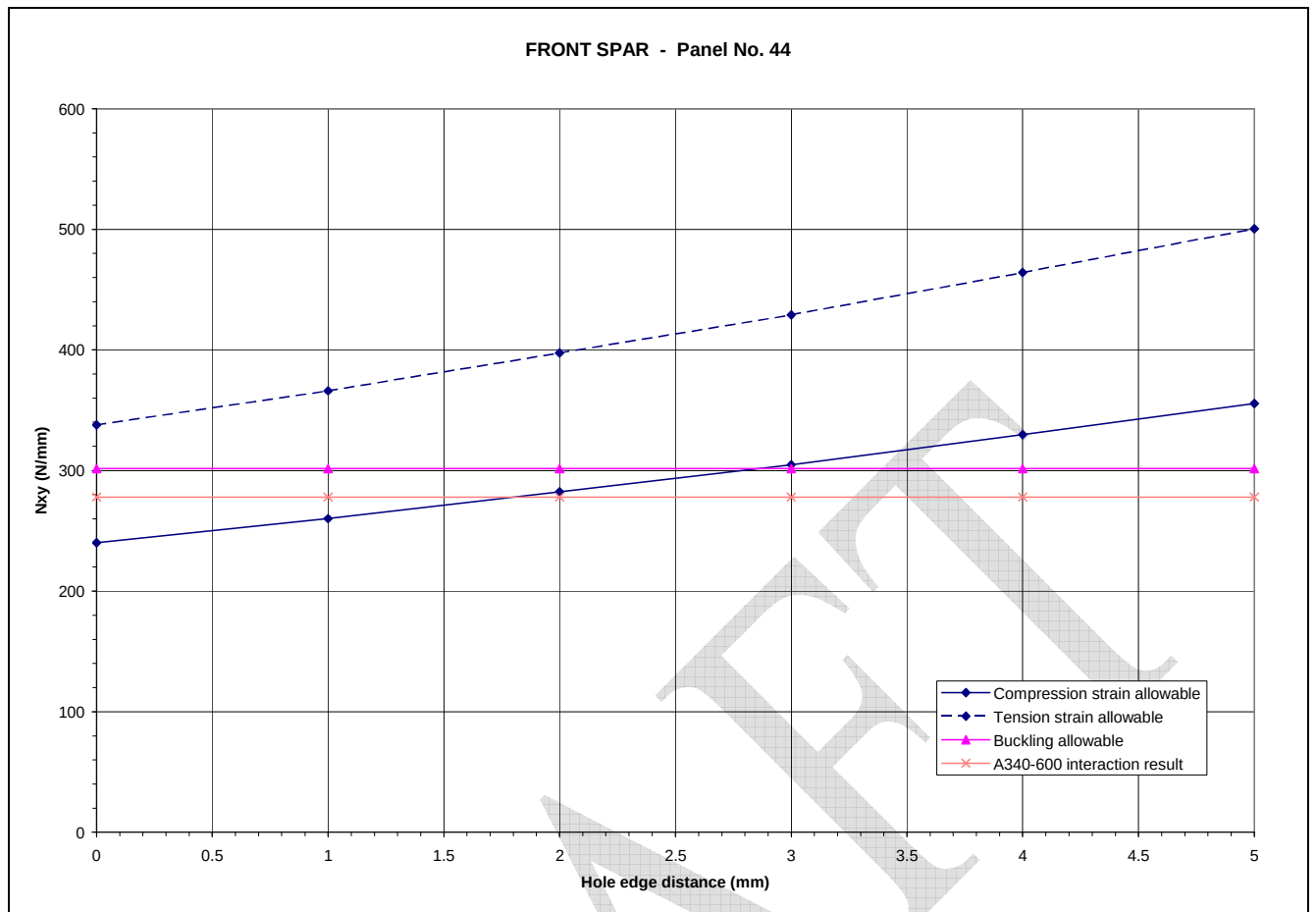


Figure 9-9

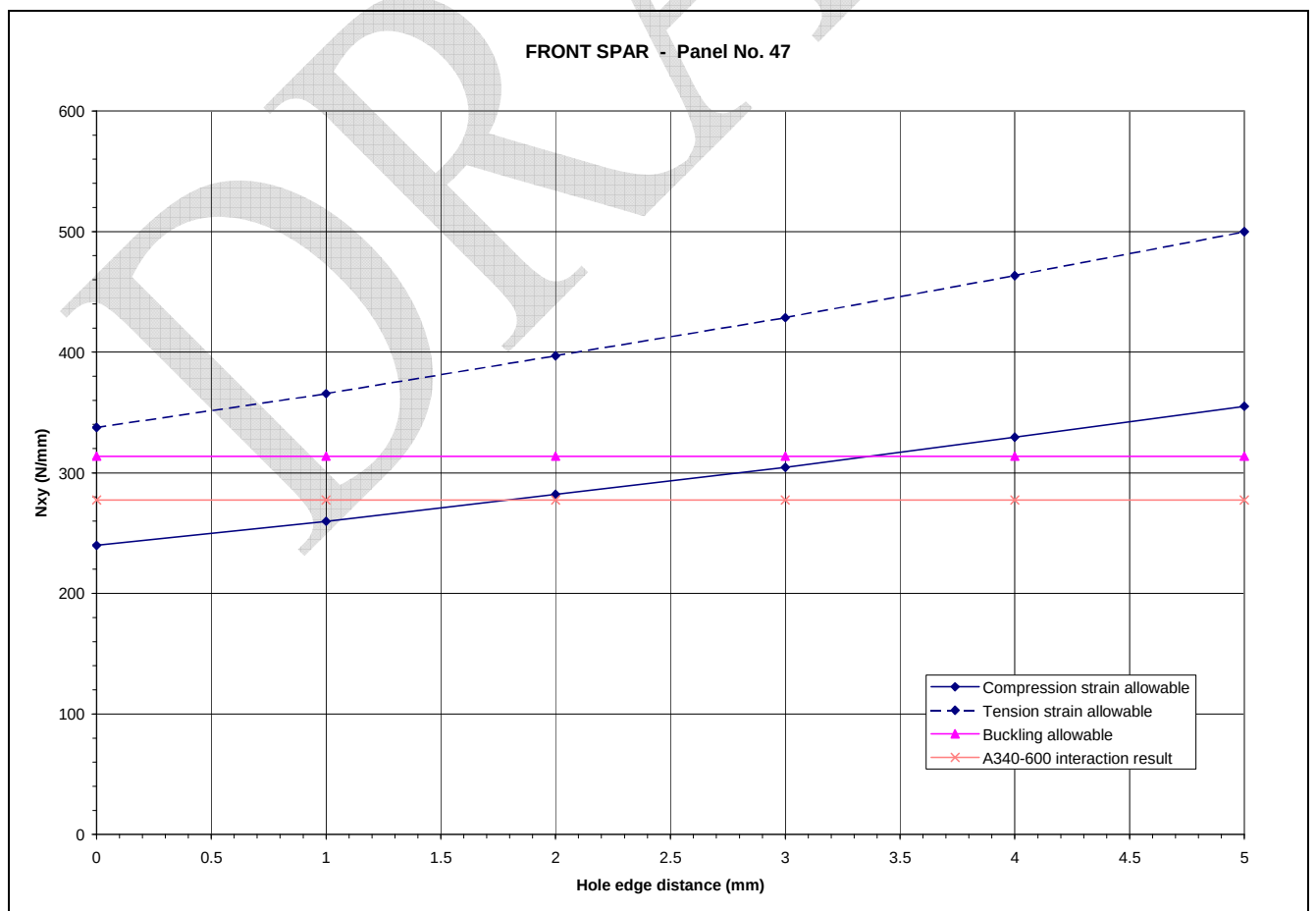


Figure 9-10

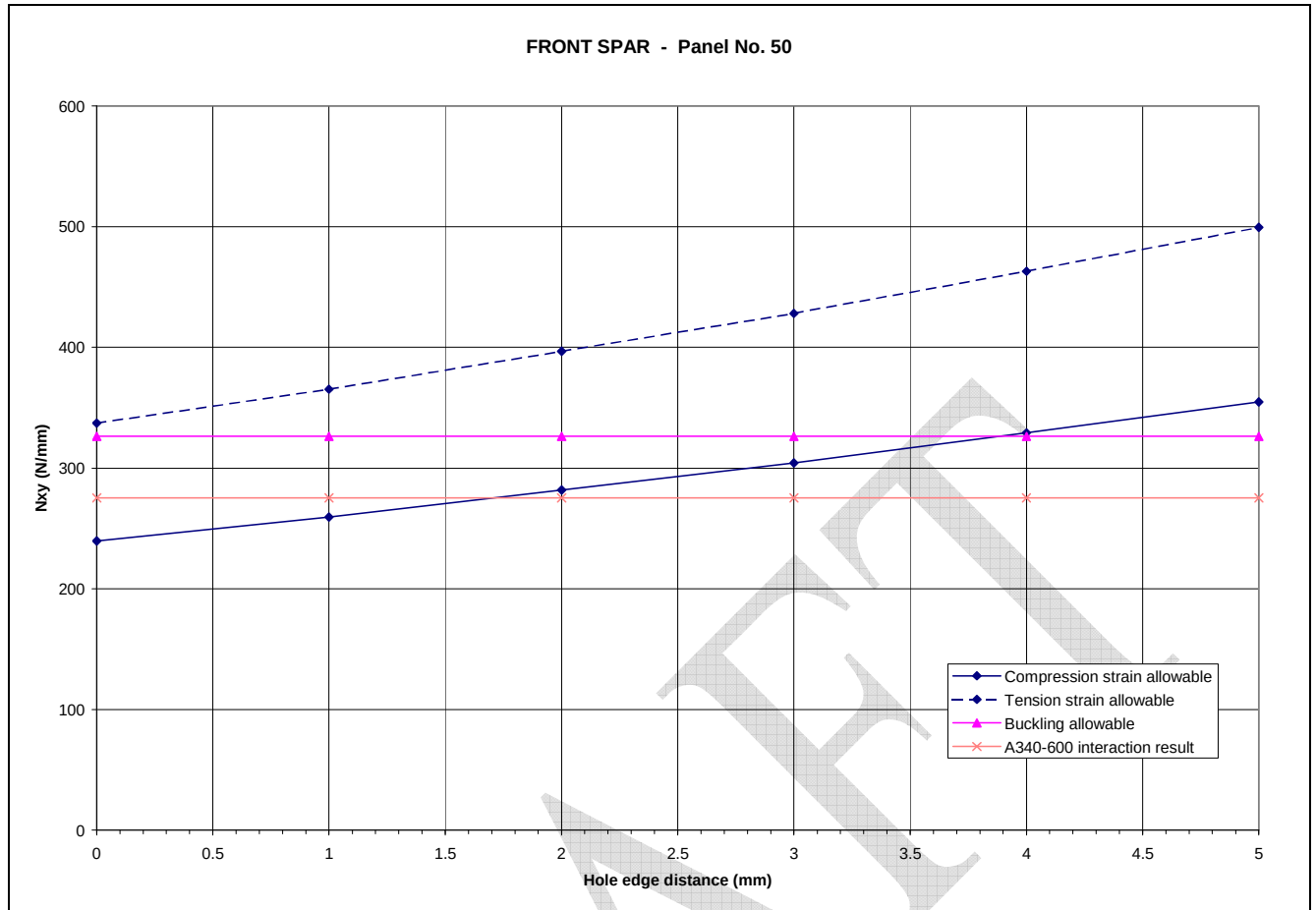


Figure 9-11

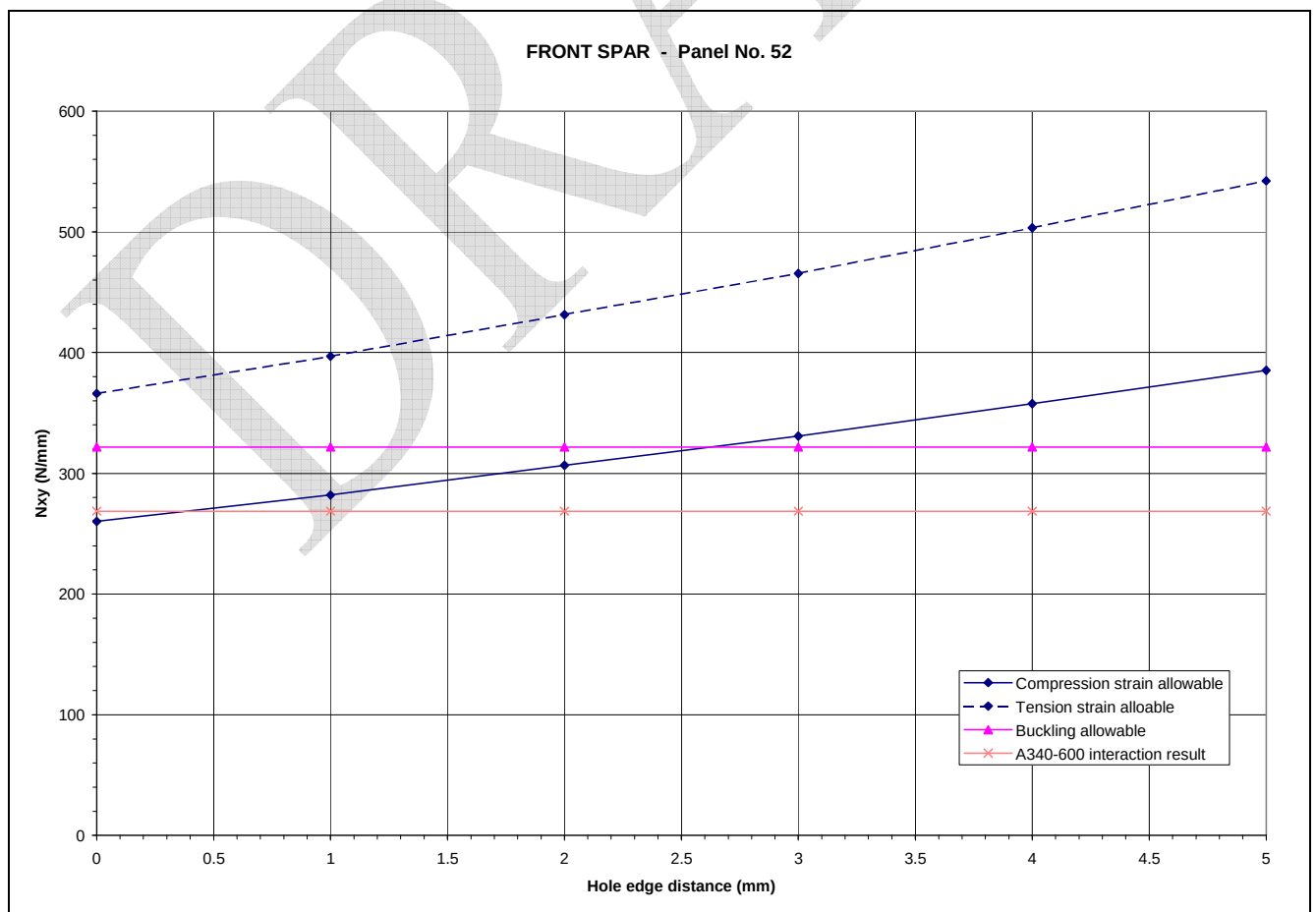


Figure 9-12

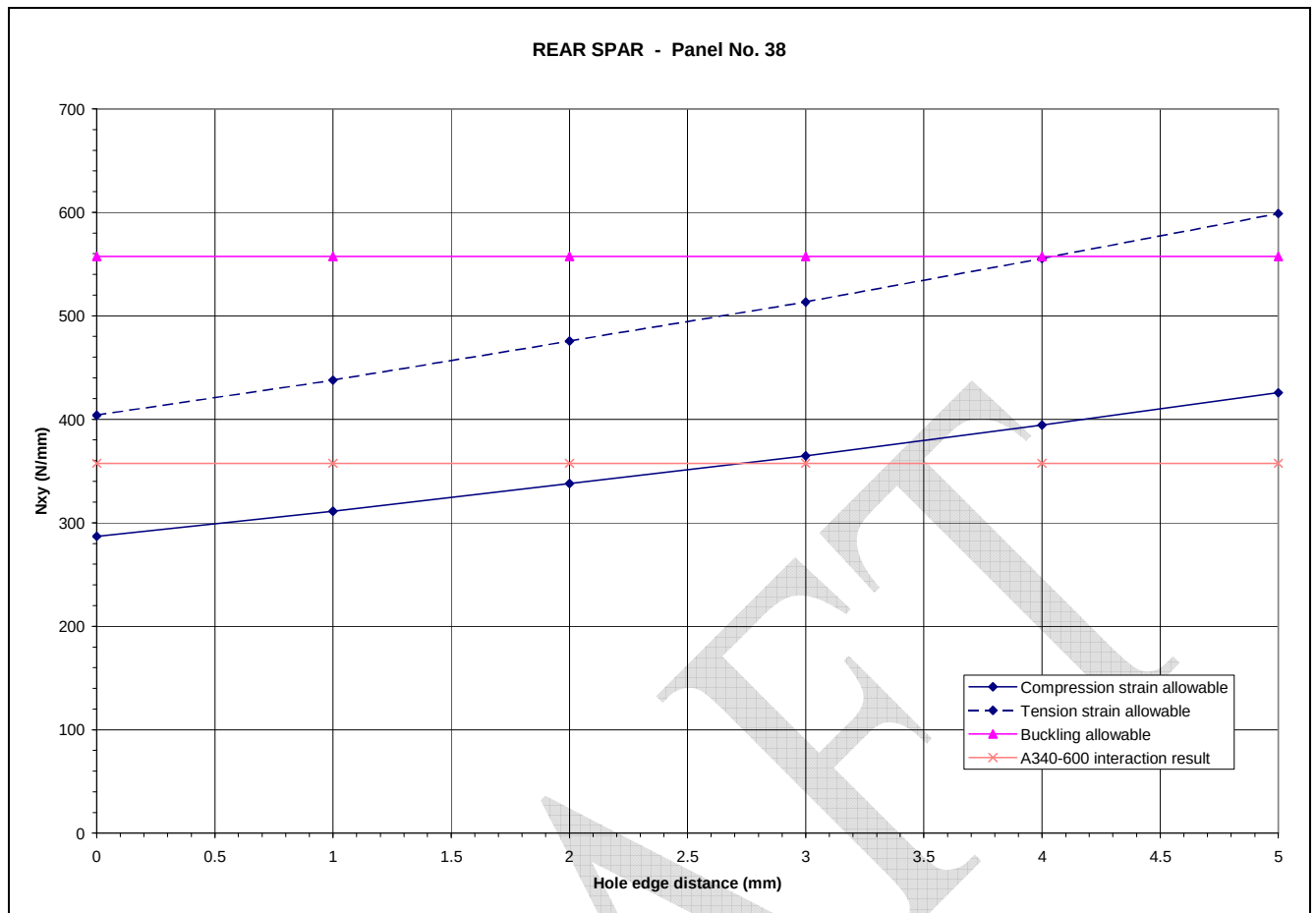


Figure 9-13

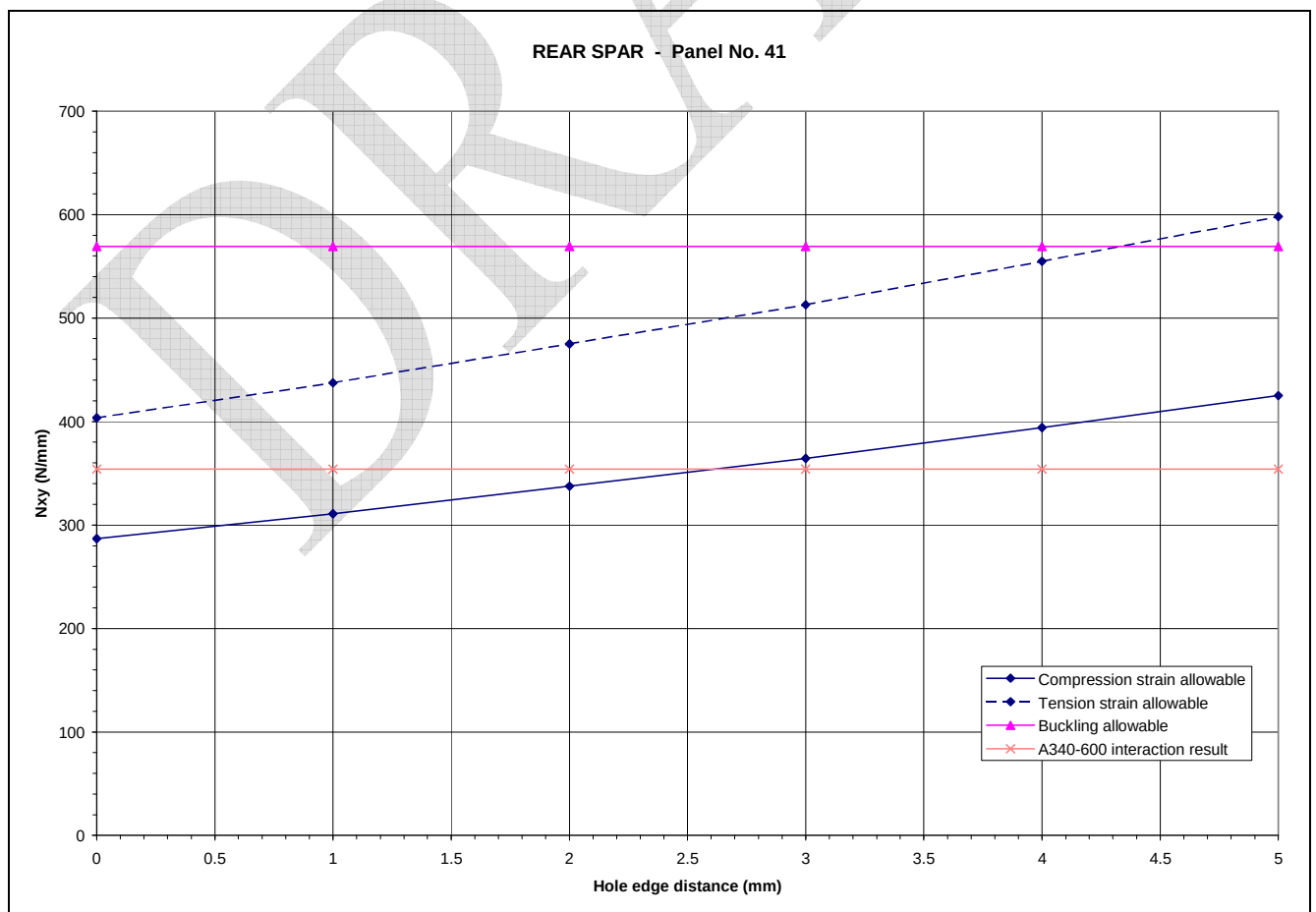


Figure 9-14

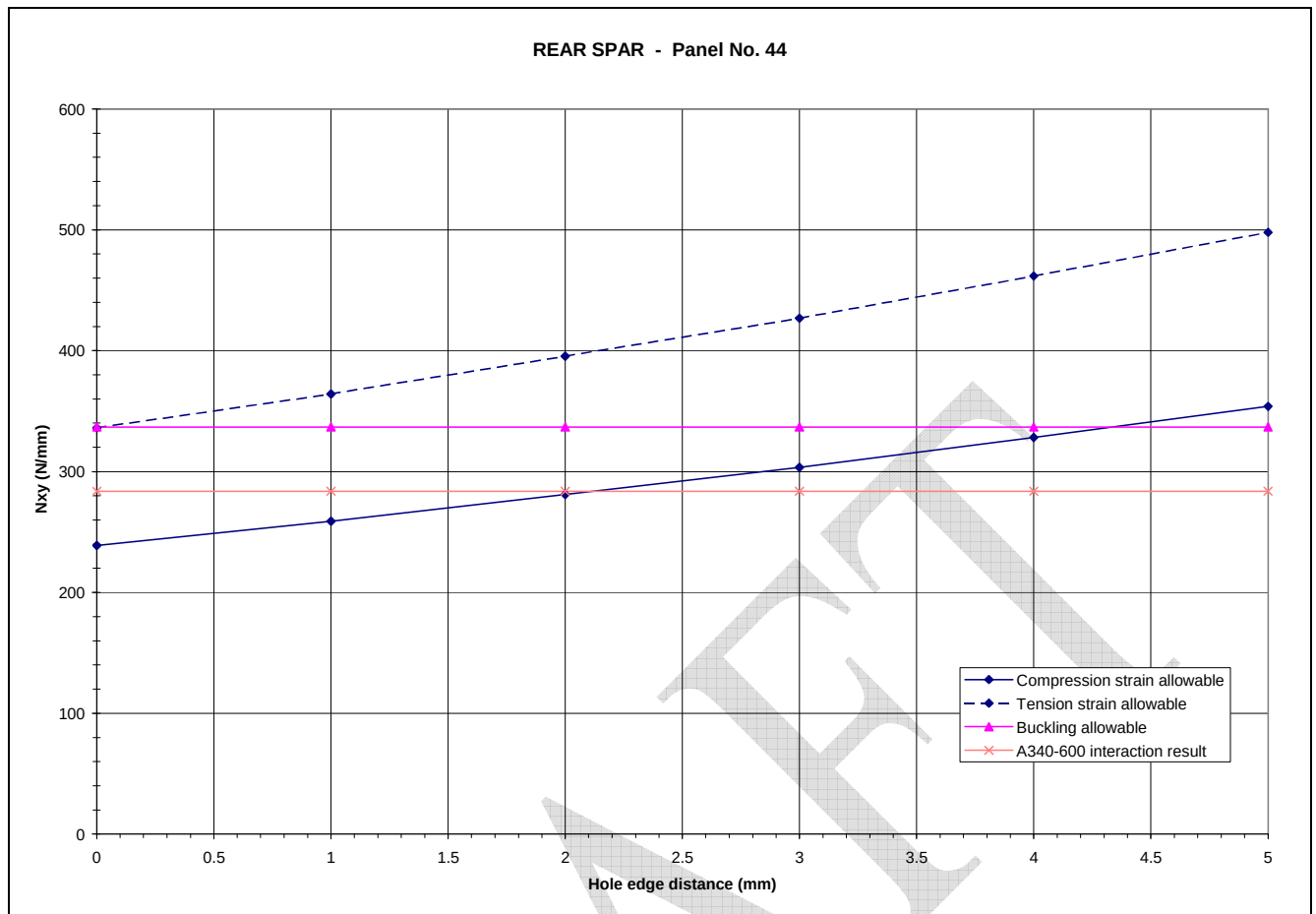


Figure 9-15

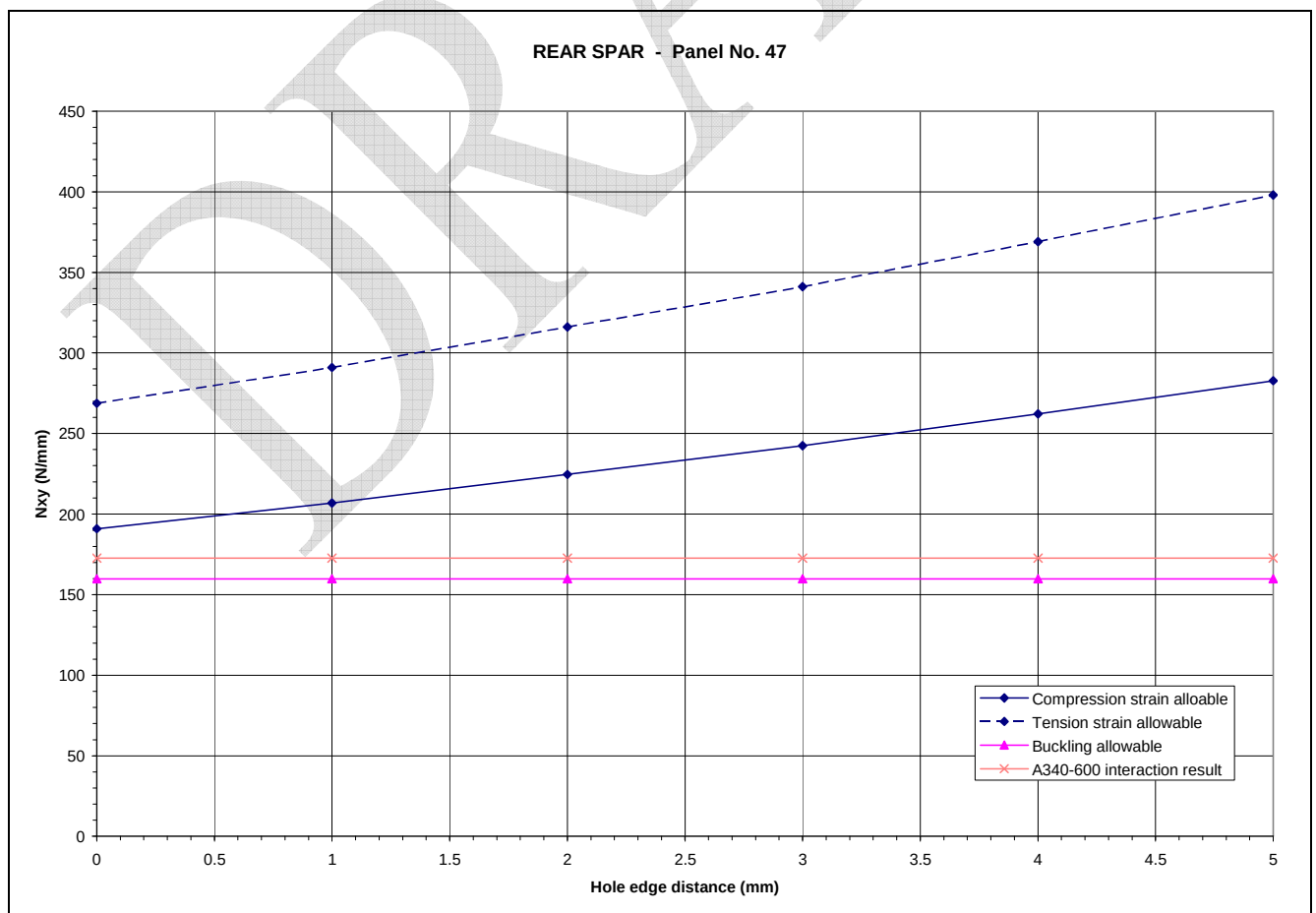


Figure 9-16

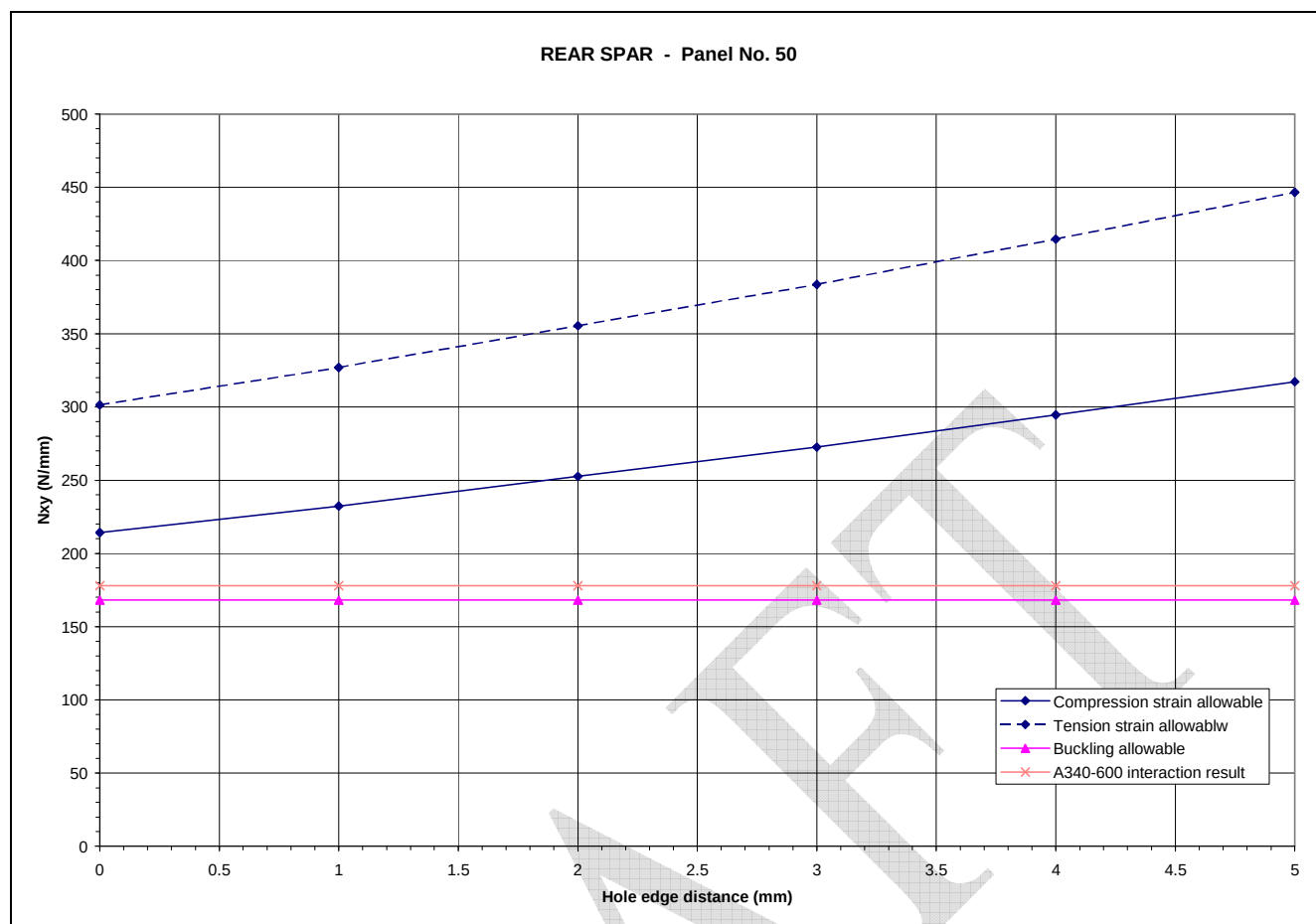


Figure 9-17

Except for panels no. 47 and 50 of the rear spar, all the panels present strength as the critical criterion. When buckling is the critical allowable (Figures 9-16 and 9-17), both methods predict similar failure load, being the new procedure slightly more conservative.

When strength is critical (Figures 9-8 to 9-15), if compression allowable strain ($8100 \mu\epsilon$) is considered at the hole edge, the new procedure results more conservative than the existing one. By the contrary, if tension allowable strain ($11400 \mu\epsilon$) is considered, the most conservative method is the A340-600 procedure.

The behaviour of the method is quite different from the one shown for the A320 tests. The cause is that the A340-600 method considers always the same allowable strain ($7200 \mu\epsilon$), with no dependence on the employed material. This fact makes that one or other method results more conservative depending on the panel material.

9.3.2 Elevator spar

The material employed for the elevator spar is CASA Group V Fabric (AGP 193 – PW / 8552). As in the previous case, H/W 2 conditions are considered. Mechanical properties are shown in Table 9-5.

	Average value	B value
E_{1t} (N/mm ²)	62000	60000
E_{2t} (N/mm ²)	62000	60000
E_{1c} (N/mm ²)	54000	50000
E_{2c} (N/mm ²)	54000	50000
G_{12} (N/mm ²)	3500	2900
μ_{12}	0.05	-
ϵ_{1t} (μ ϵ)	-	9650
ϵ_{2t} (μ ϵ)	-	9650
ϵ_{1c} (μ ϵ)	-	9250
ϵ_{2c} (μ ϵ)	-	9250
γ_{12} (μ ϵ)	-	21700

Table 9-5. Mechanical properties for CASA Group V Fabric (H/W 2 conditions)

The used lay-up is (0/100), which has the following average properties:

- $E_x = 12560$ N/mm²
- $E_y = 12560$ N/mm²
- $G_{xy} = 27620$ N/mm²
- $\mu_{xy} = 0.79$

Table 9-6 shows the geometry of the analysed panels. Only panels with centred circular hole have been considered. Besides, buckling prediction is presented. Buckling shear flow is obtained applying a correction factor to the buckling shear flow of the panel without hole (q_0). In this case, both methods present great agreement. Indeed, the biggest difference is not greater than 5%, (see Δ in Table 9-6).

ELEVATOR		H (mm)	D (mm)	L/H	D/H	q_0 (N/mm)	A340-600	New method	Δ (%)
Panel	L (mm)						q_{bk} (N/mm)	q_{bk} (N/mm)	
4	265.2	250	80	1.06	0.32	131.3	75.9	75.4	0.68
9	250	248.7	80	1.01	0.32	292.7	168.8	165.7	1.90
10	250	244.8	80	1.02	0.33	298.9	171	167.6	2.02
13	250	234.1	80	1.07	0.34	3477.1	1945.3	1897.4	2.52
14	314.6	229.9	80	1.37	0.35	3016.4	1671.5	1719.6	-2.80
15	312	225.5	80	1.38	0.35	3110.4	1705.7	1750.4	-2.56
16	287.2	221.1	80	1.30	0.36	3385.7	1836.5	1842.2	-0.31
17	318	217	80	1.47	0.37	3223.8	1729.8	1784.2	-3.05
18	281.2	213.1	80	1.32	0.38	3604.4	1913.3	1904.4	0.47
24	250	197.8	80	1.26	0.40	539.4	273	260.8	4.68
25	250	194.4	80	1.29	0.41	551.4	275.8	262.8	4.95
30	250	179.3	64.1	1.39	0.36	445.7	243.3	249.7	-2.58
31	250	175.7	64.1	1.42	0.36	457.5	246.9	253.4	-2.55
36	250	159.1	64.1	1.57	0.40	950.6	482.2	494.6	-2.51
37	250	155.9	64.1	1.60	0.41	976.5	488.4	501.1	-2.54
42	250	144.1	64.1	1.73	0.44	501.1	236.3	242.7	-2.65
43	250	140.5	64.1	1.78	0.46	518.7	239.6	246.3	-2.71

Table 9-6. Buckling load prediction for elevator spar panels

Strength results for the new procedure are again presented in graphical way. These figures include the same curves as for the torque box: two strength allowable curves, one corresponding to the compression allowable strain and the other one to the tension allowable strain, the buckling load shown in table 9-6 and the allowable flow resulting from the interaction method of the A340-600 (see Figures 9-18 to 9-34).

For panels no. 4, 9 and 10, buckling is critical. In these cases and for this material, the new method is slightly less conservative. For the rest of the panels, strength is critical. The new procedure is also less conservative in these cases. There are not big differences between considering tension or compression allowable strains (9650 and 9250 $\mu\epsilon$, respectively), since they are rather close.

For all the elevator panels, the interaction procedure results are more conservative. The reason is that CASA Group V Fabric allowable strains are bigger than tape ones, whereas the interaction method keeps the allowable of 7200 $\mu\epsilon$.

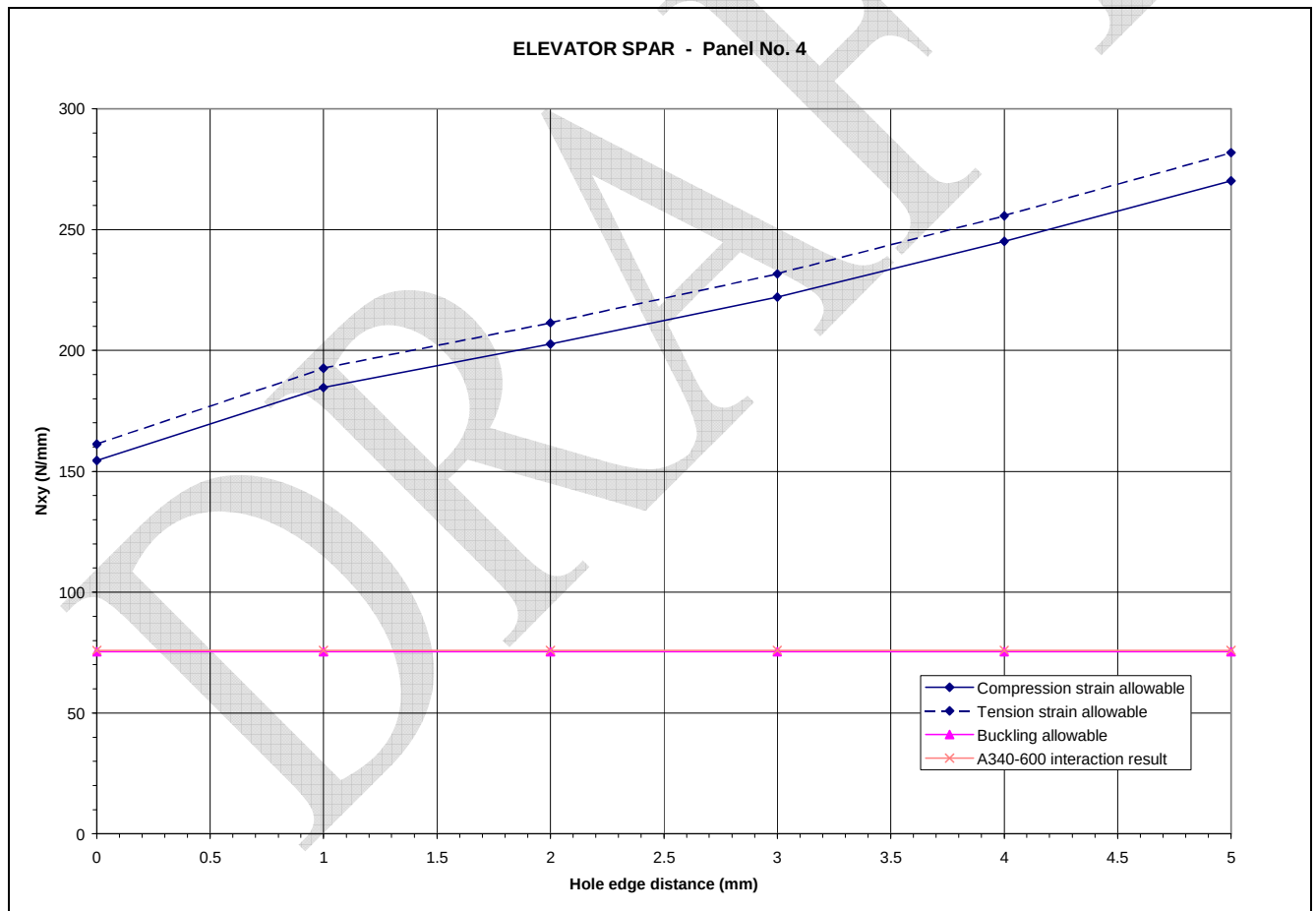


Figure 9-18

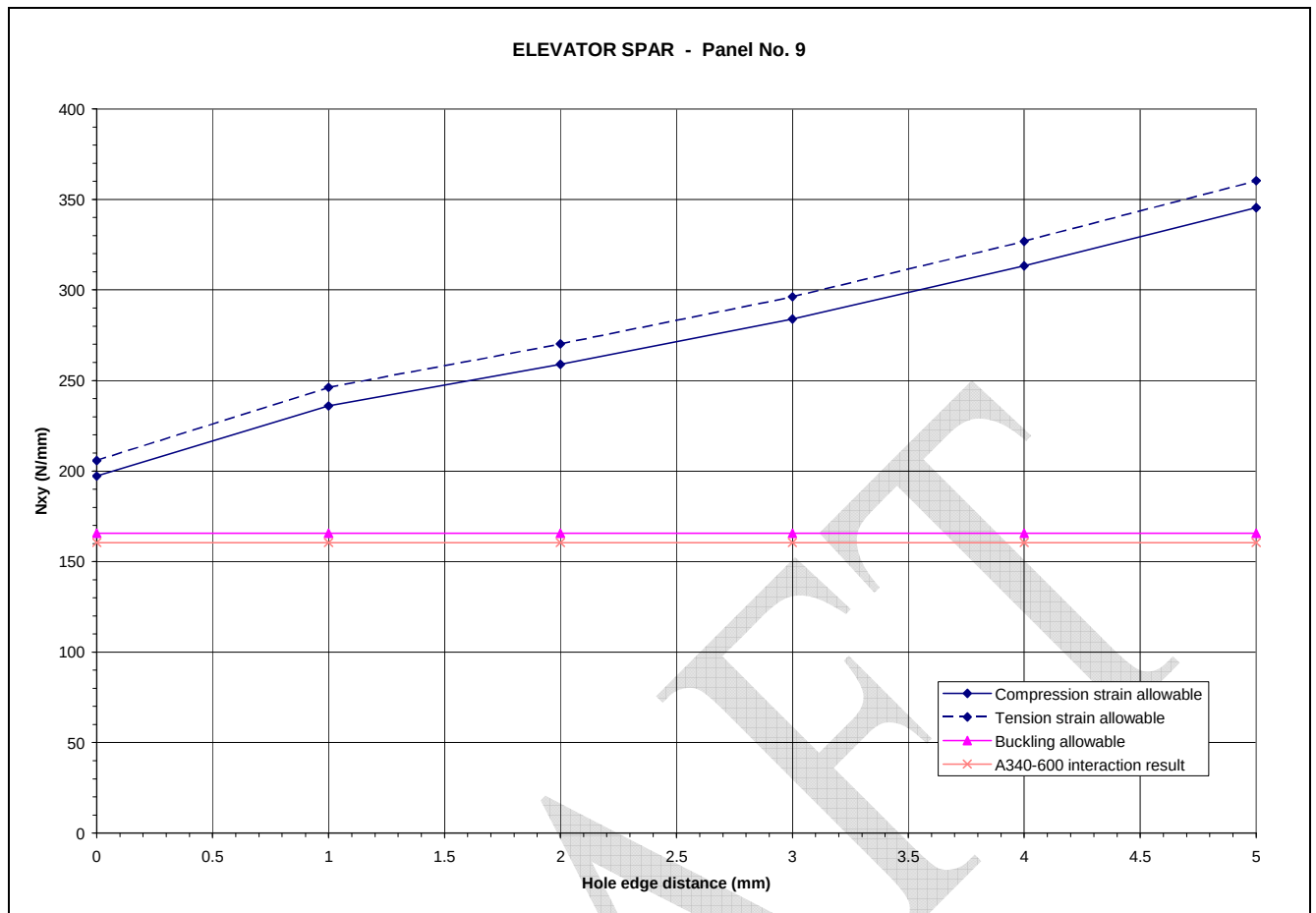


Figure 9-19

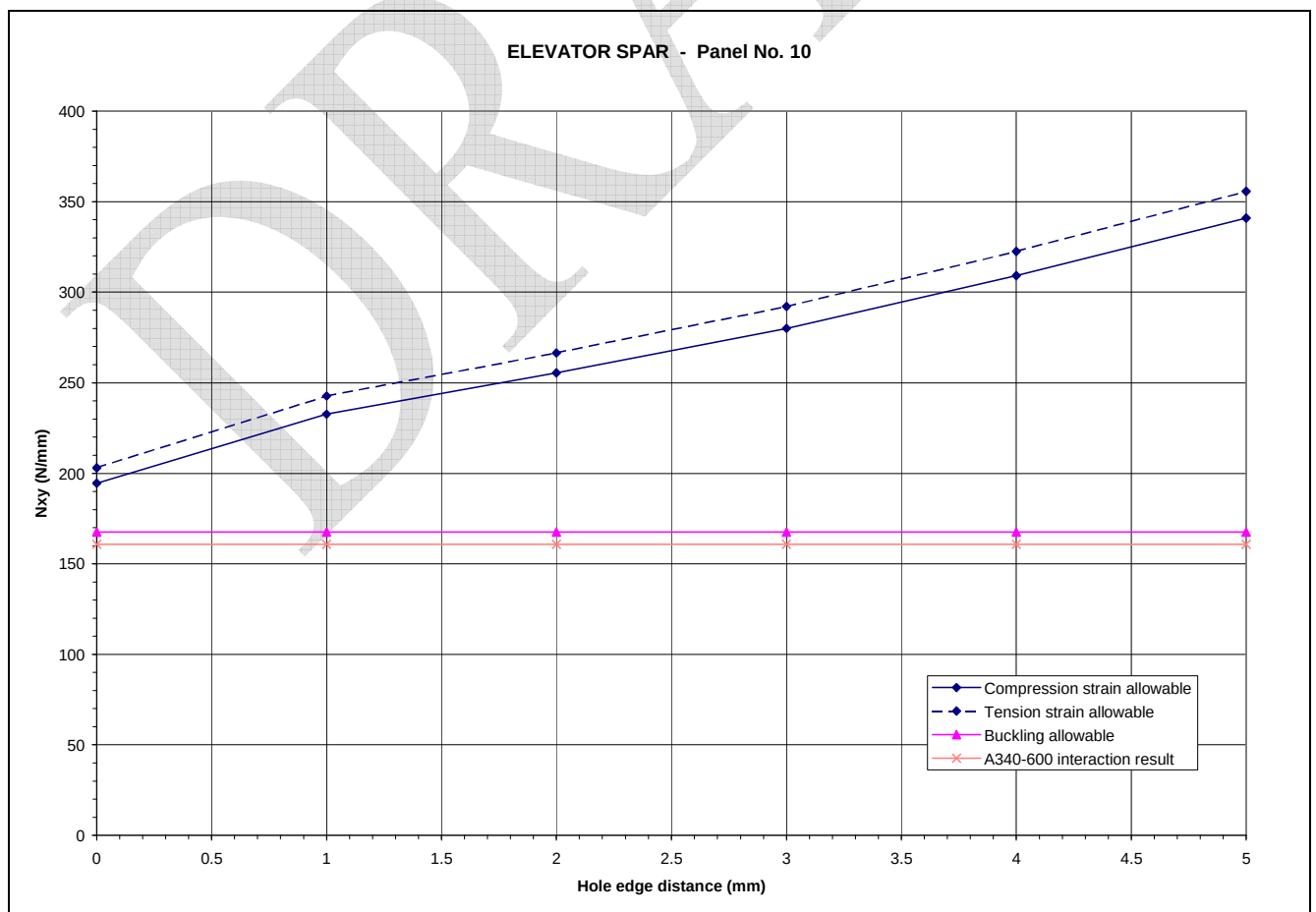


Figure 9-20

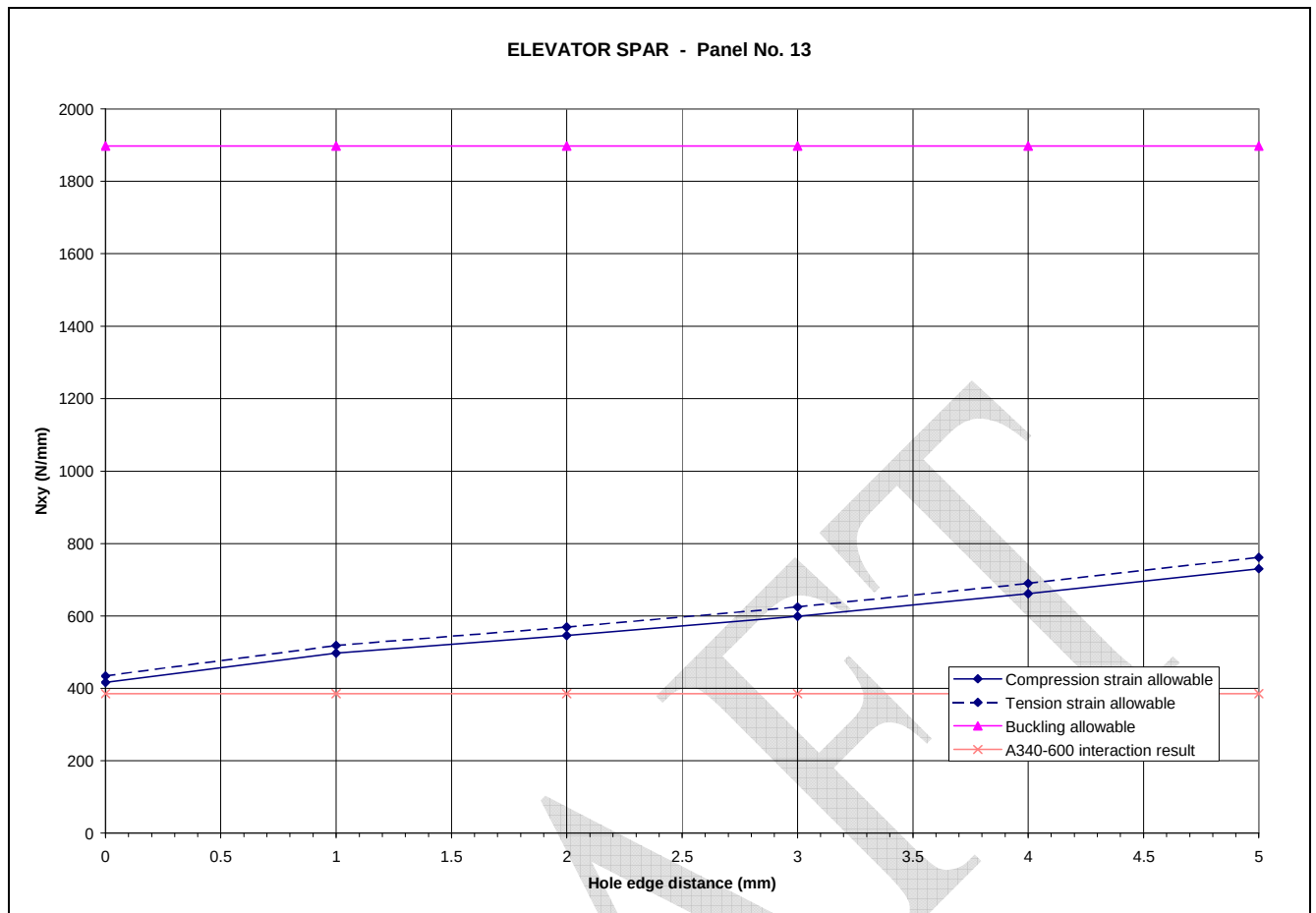


Figure 9-21

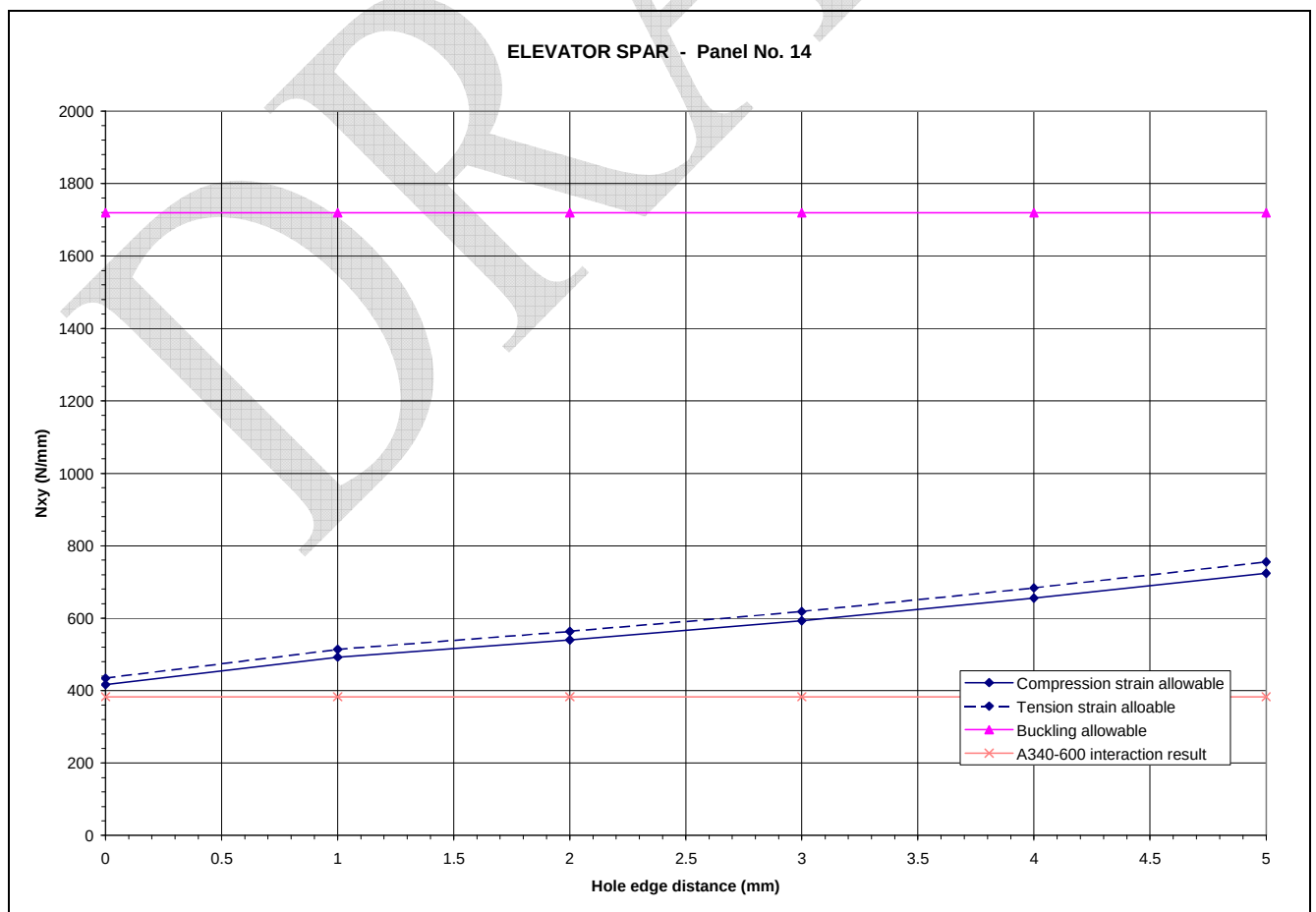


Figure 9-22

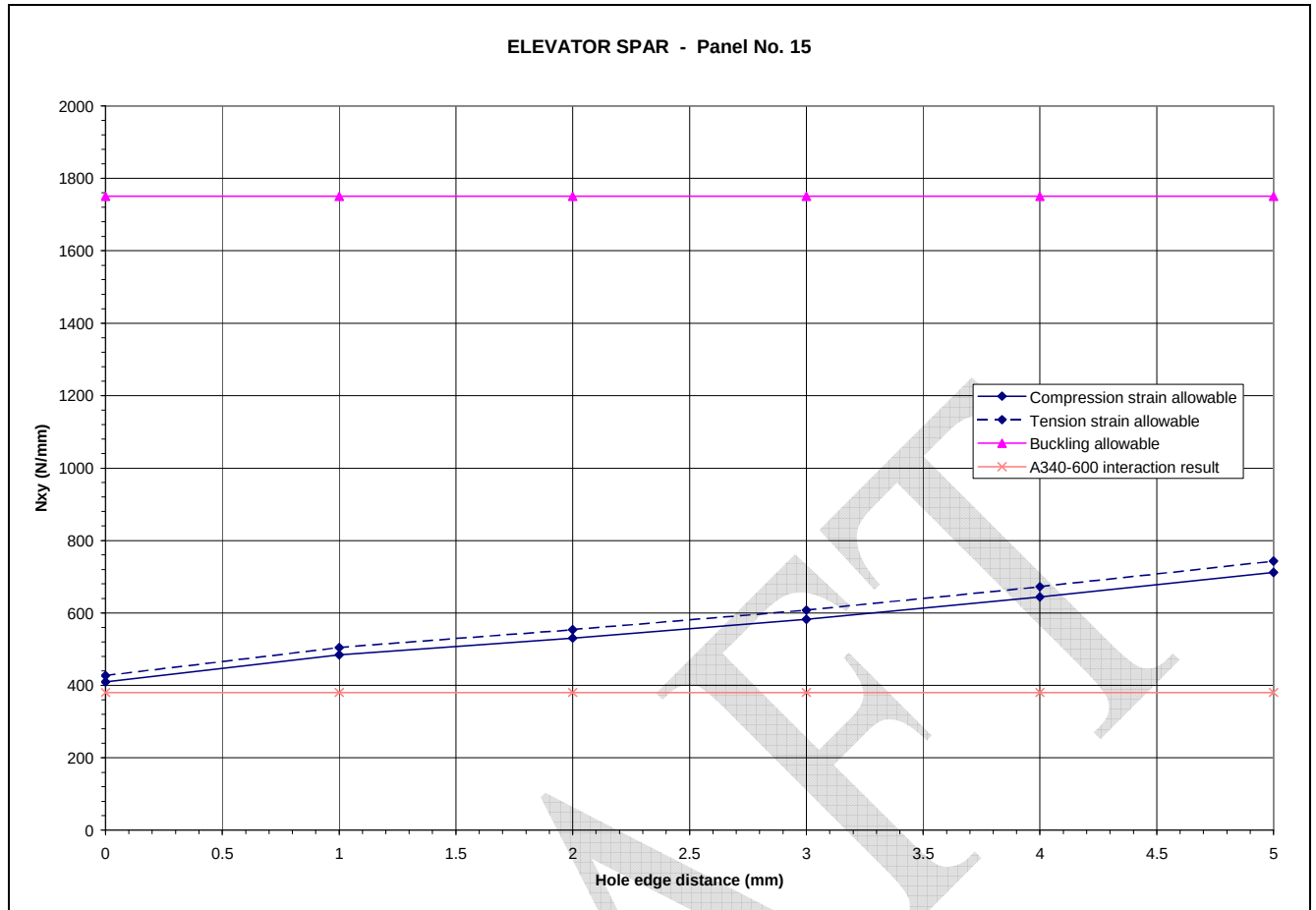


Figure 9-23

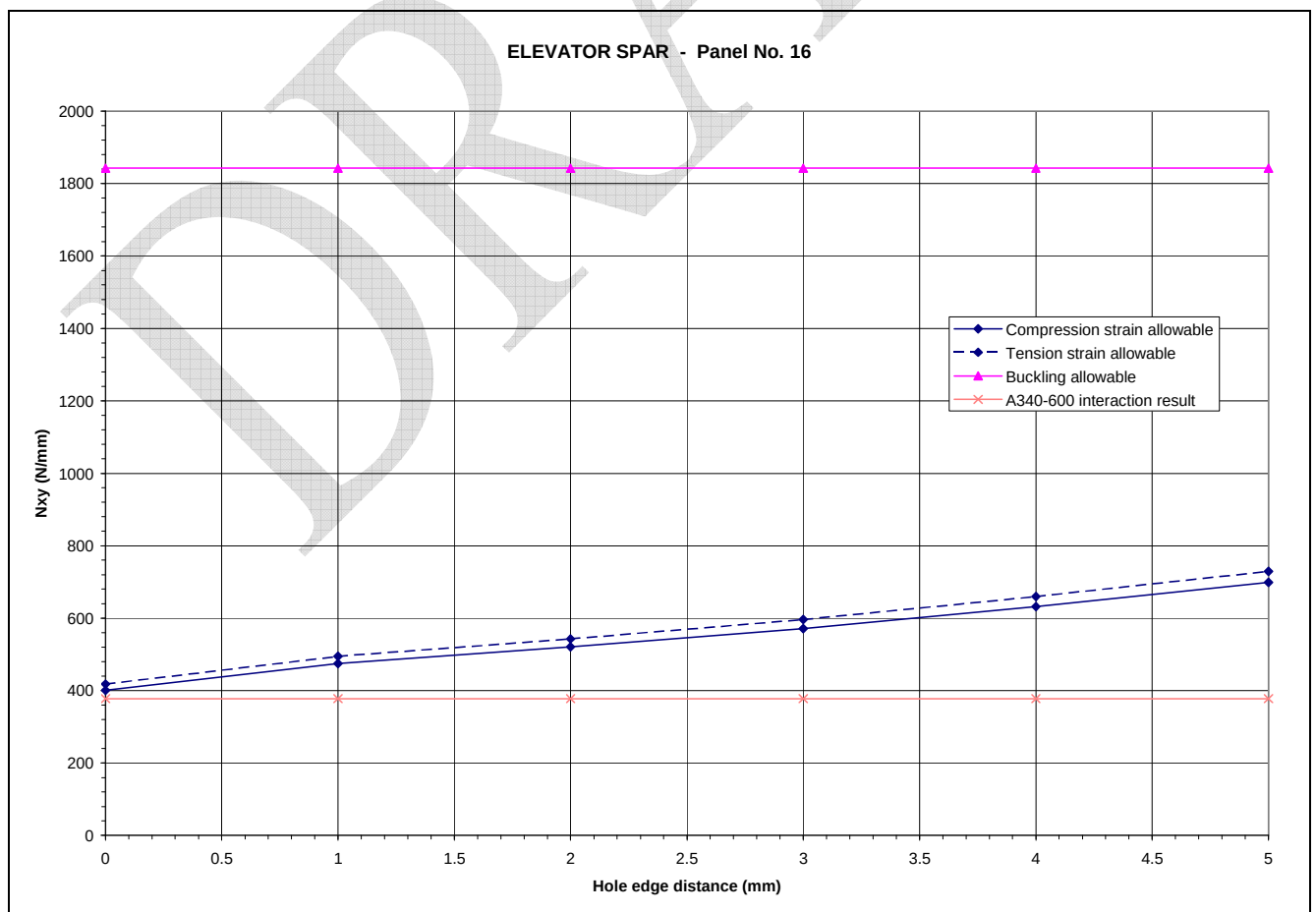


Figure 9-24

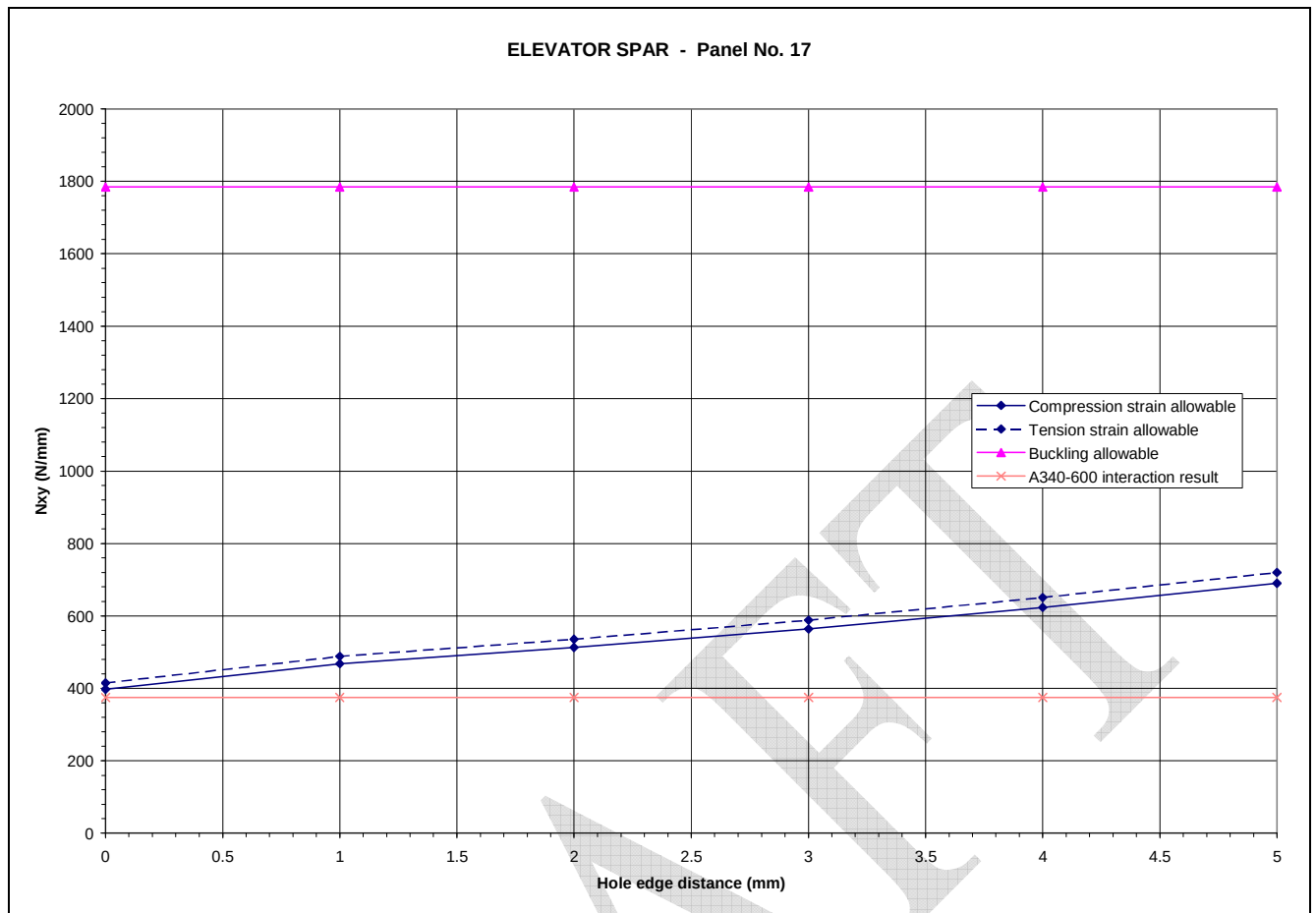


Figure 9-25

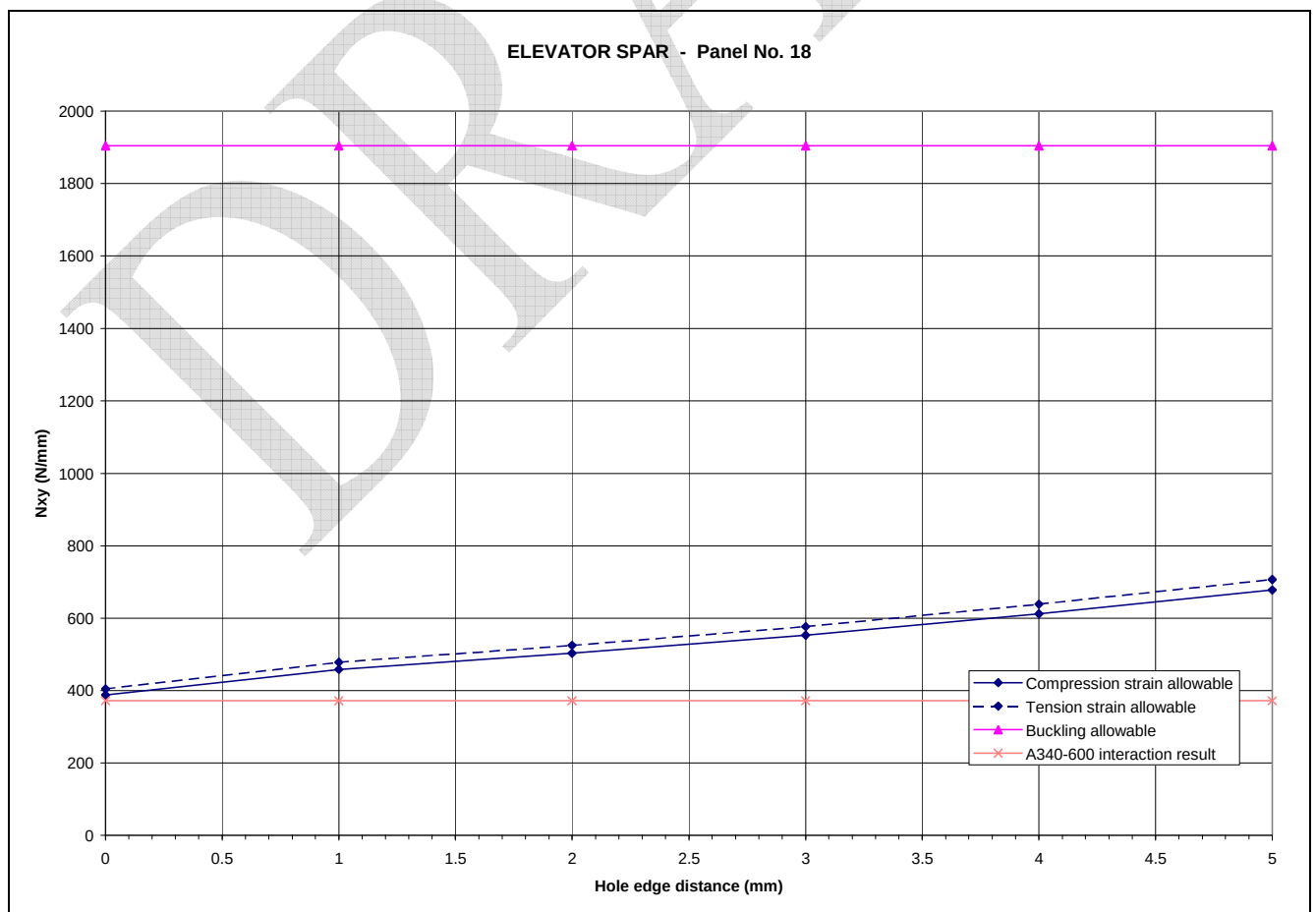


Figure 9-26

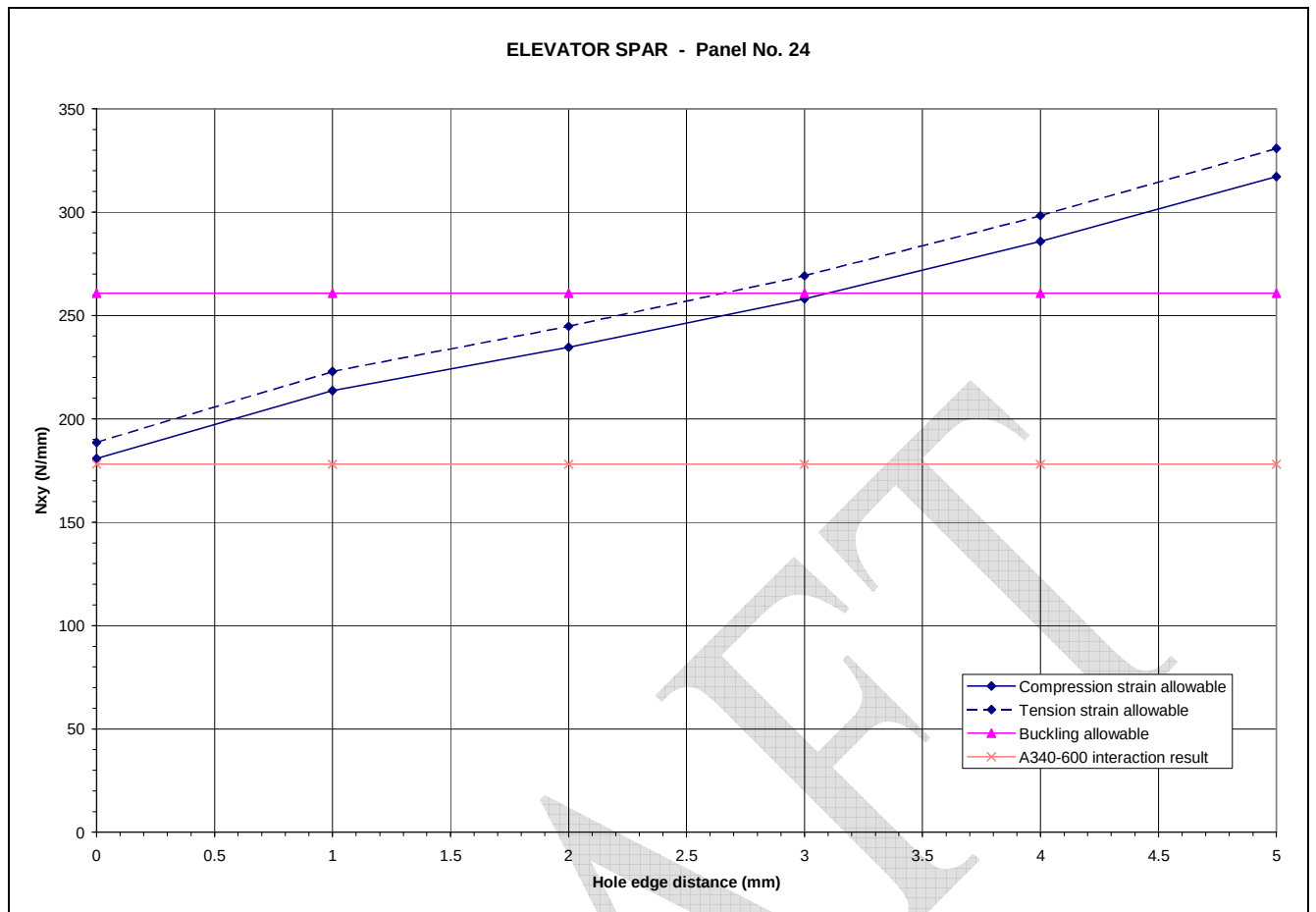


Figure 9-27

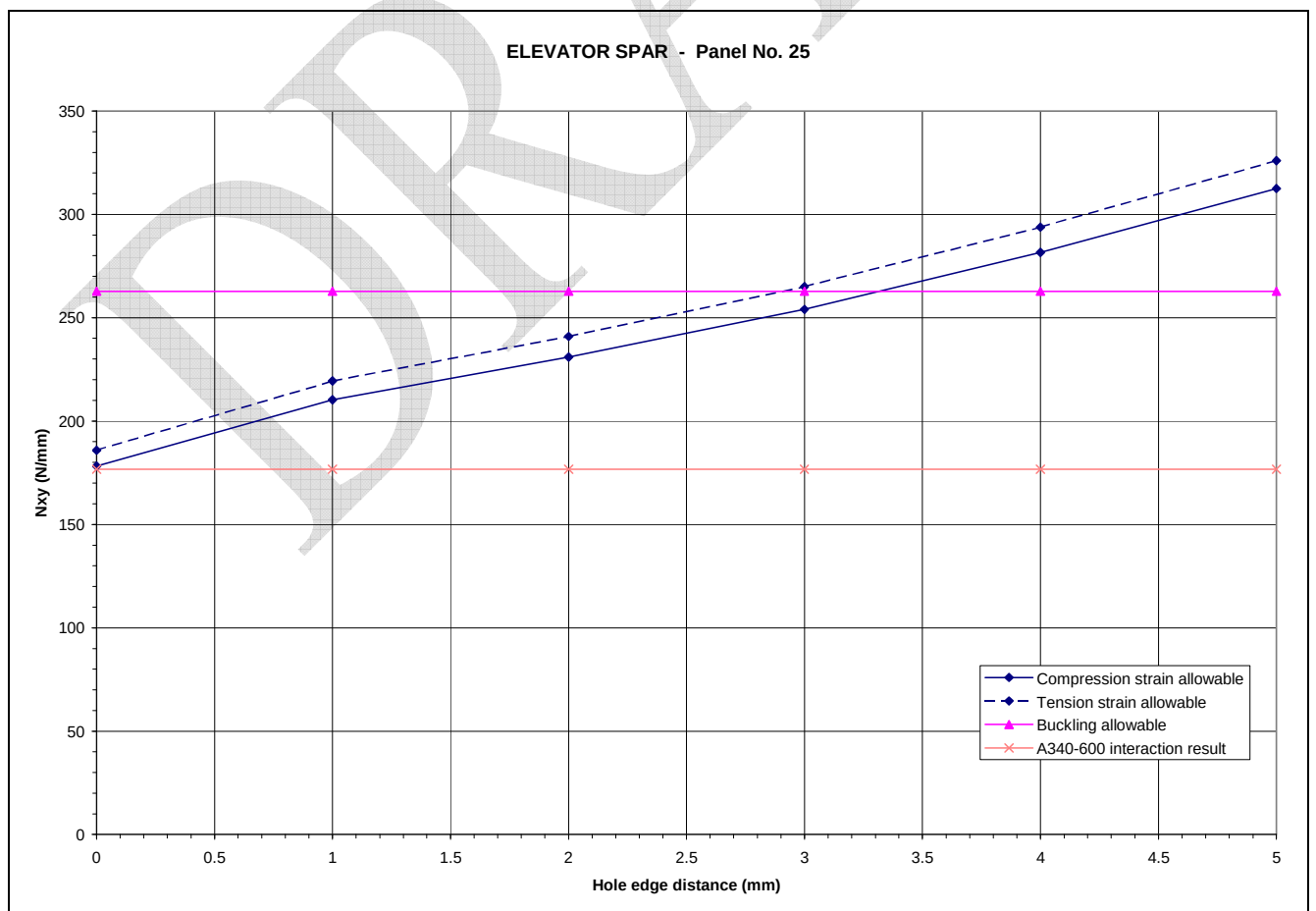


Figure 9-28

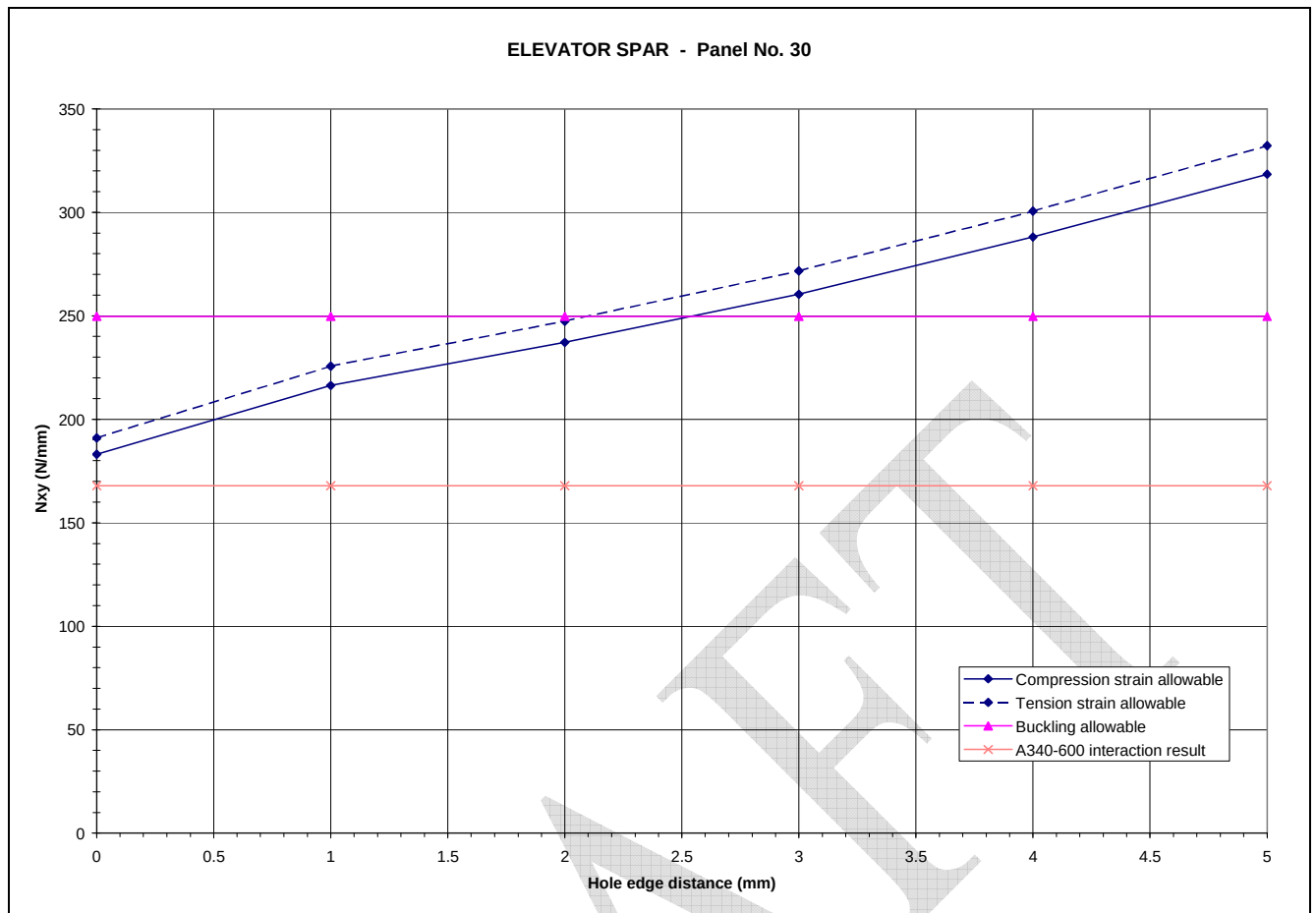


Figure 9-29

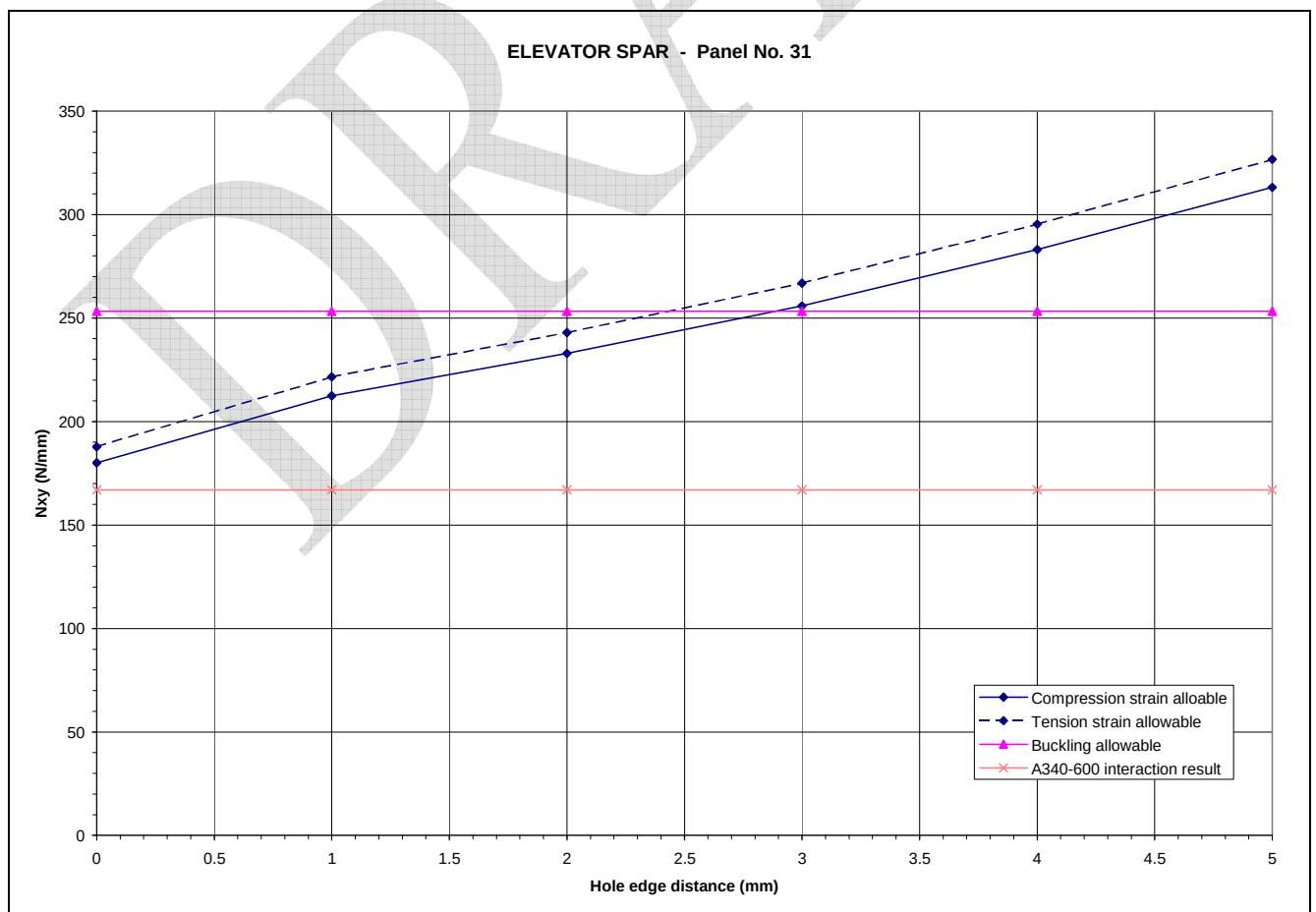


Figure 9-30

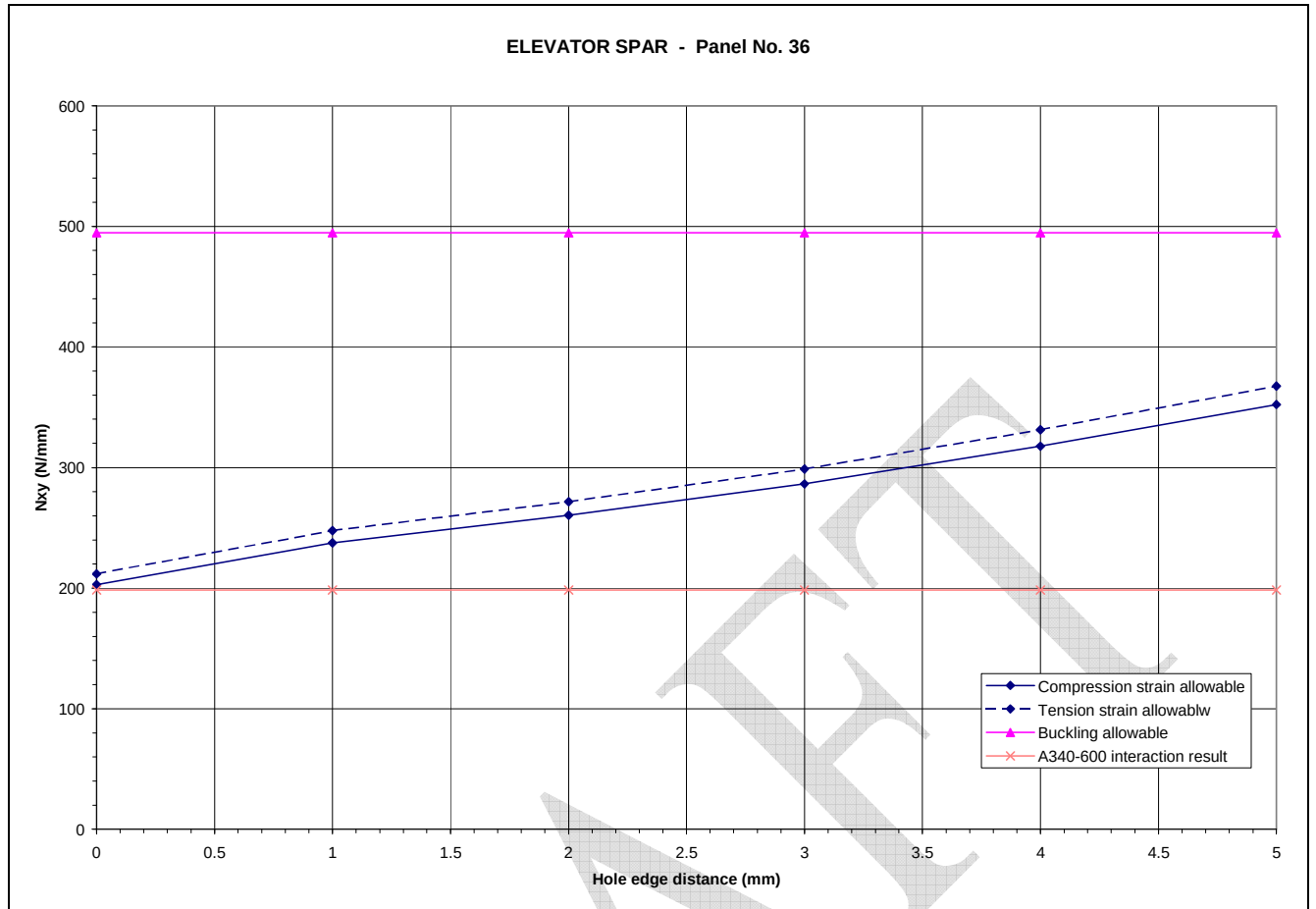


Figure 9-31

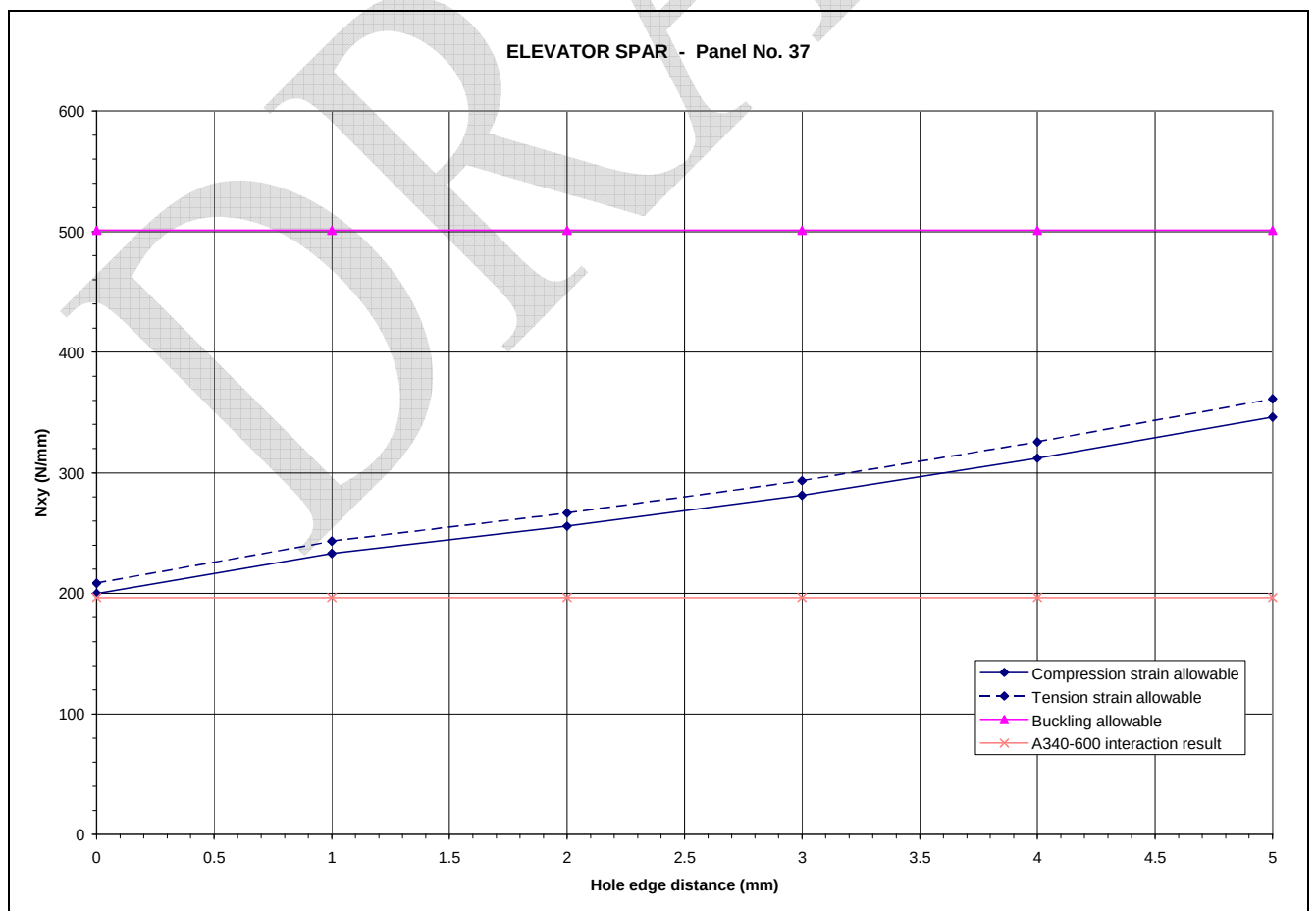


Figure 9-32

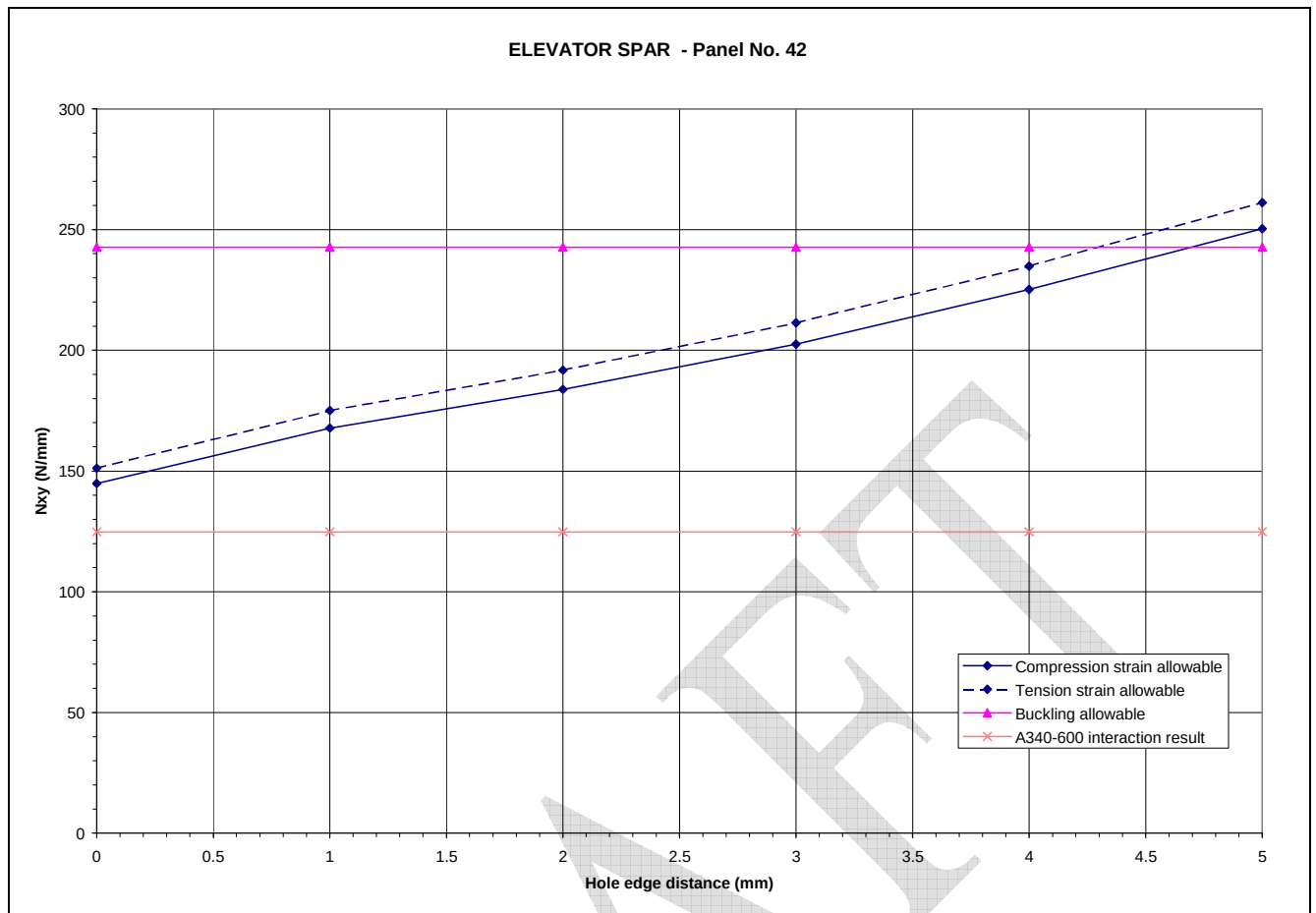


Figure 9-33

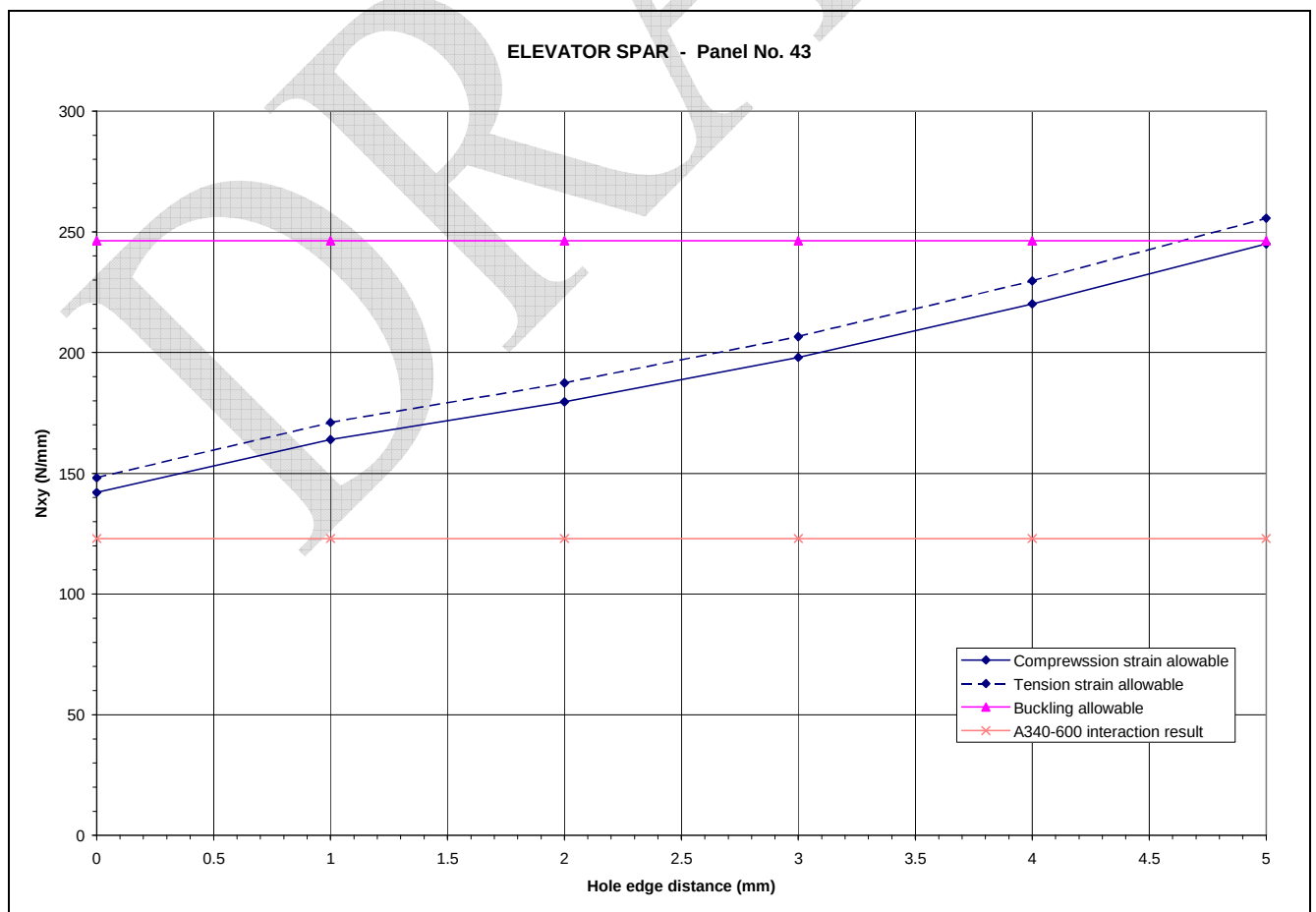


Figure 9-34